



Francesco Federico

Cited.

an operating manual by

The Agentic CMO

First Edition

Cited

The GEO Operating Manual — A Field Companion to The Agentic CMO

by Francesco Federico

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The GEO Operating Manual

A Field Companion to The Agentic CMO

— Francesco Federico —

First Edition · 2026

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The author has used generative AI tools as research assistants during preparation. Every empirical claim has been verified against primary sources. Methodology, source weighting, and dissent are documented in Appendix G (Source Canon).

The opinions in this manual are the author's alone and do not represent those of S&P Global or any other organisation with which he is affiliated.

First edition: 2026. ISBN: not assigned (free distribution).

Cover design and interior typography by the author, in the Milanese Futurist visual system.

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Author's Note

This manual is the field kit. The Agentic CMO — first edition June 2025, second edition June 2026 — made the strategic argument: marketing organisations that orchestrate human and artificial intelligence as a unified force will outcompete those that adopt AI as a tool stack. Nineteen chapters, sixty-three diagrams, the full operating model from leadership to measurement. Cited is the discipline that argument depends on. Read the two together; the strategy and the discipline reinforce each other. Available at francescofederico.com/agentic-cmo.

The job has changed. The unit of value has shifted from rank to citation, from click to extraction, from keyword to fragment. The shift is not coming. It is here. Ninety-three per cent of AI Mode is zero-click already; the strategy decks that haven't recalibrated are running on 2024 architecture in 2026 markets.

This is not a second book. It is a working tool — designed to sit open on a second monitor while you audit a robots.txt, fix an Entity Bible, or push back on a vendor pitch. The audience is the senior manager doing the delivery: SEO/GEO, content, analytics, brand, marketing operations. The chapters speak directly to the practitioner; the manager-bridging callouts give you the language to brief upward when the work needs senior approval.

The research underneath is twelve dossiers across three rounds. Three original syntheses on the GEO field; three commissioned pieces closing the gaps — paid placement, regulatory cascade, independent replication of the Aggarwal tactics. Every claim carries an evidence weight inline. Vendor-funded data is flagged at first reference. The source canon (Appendix G) lists every reference with funder, sample size, and replication status. Where the canon disagrees, the disagreement is named.

Three governing questions anchor every chapter. What carries over from SEO. What is genuinely new. What you do Monday morning. The manual answers all three, in that order. The bias is against hype — Chapter 5 corrects the schema misconception that has cost the industry two years of misallocated budget; Chapter 8 puts YouTube and multimodal where the evidence says they belong, not where most marketing programmes pretend they do.

What the manual will not give you. Vendor recommendations dressed as objective advice. Frameworks for the sake of frameworks. Confidence where

the evidence is thin. The manual ends with a chapter named “What We Don’t Yet Know” because some of the most consequential questions in this field are not settled, and pretending otherwise would waste your reading time.

Read it linearly the first time. Use it as reference after that. The chapter pattern repeats: contrarian claim, diagnosis, mechanism, Monday-morning actions, measurement, open question, end-with-teeth line. Skim the chapters whose terrain you already know; spend time on the ones the field has been getting wrong.

— Francesco Federico London, April 2026

How to Read This Manual

Three paths.

The strategist’s read. Chapters 1–3 (The Reset) and Chapter 15 (What We Don’t Yet Know). Two hours. You leave with the argument, the scenario probabilities, and the open questions to brief your team.

The operator’s read. Chapters 4–14, in order. Six to eight hours. Adds the mechanism, the engine field guide, the levers, the operating system. You leave able to brief the agency, interrogate the vendor, and run the 90-day playbook.

The reference read. Appendices and the running glossary. Open during audits, vendor pitches, and quarterly reviews.

Three governing questions structure every chapter:

- What carries over from SEO?
- What is genuinely new?
- What does a senior manager actually do Monday morning?

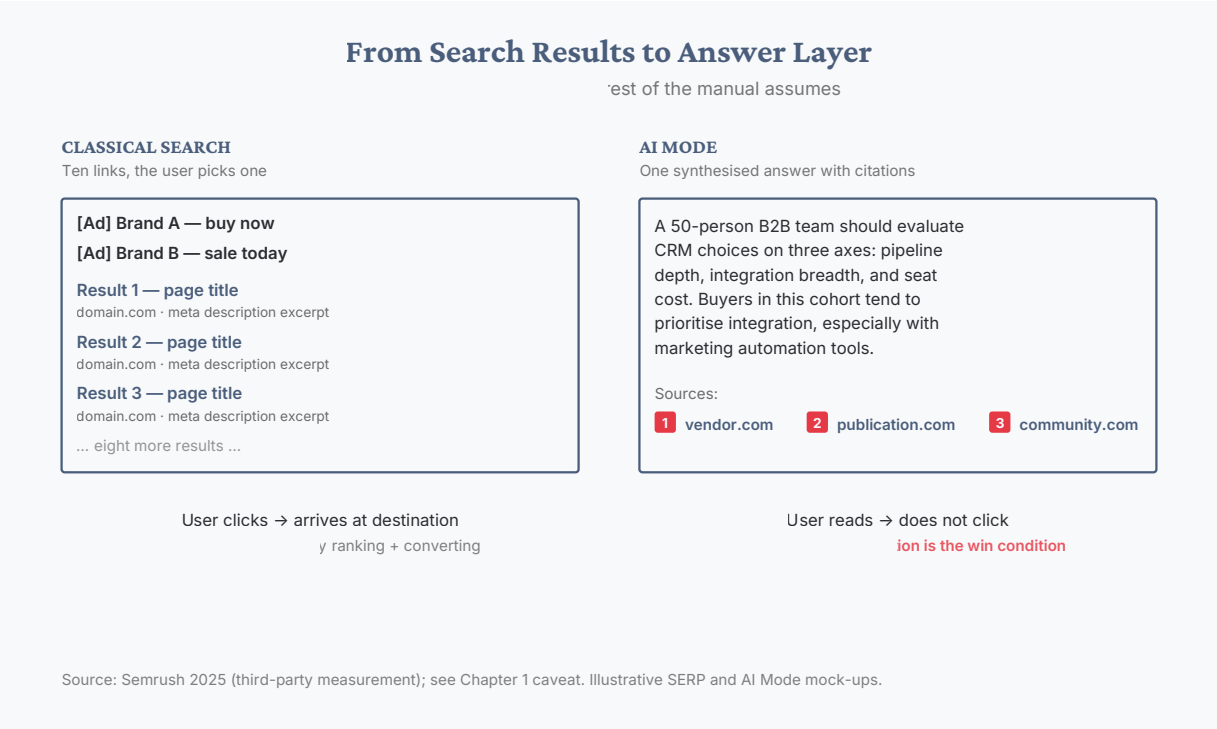
A note on evidence. Every empirical claim carries an evidence weight inline — Strong, Moderate, Anecdotal, Theoretical. Vendor-funded data is flagged at first reference. Where the canon disagrees, the disagreement is named, not smoothed.

A note on currency. The recency cut-off is 29 April 2026. Appendix H — “Final Updates” — collects the announcements expected during the rest of 2026 (the CMA conduct requirements, the EU AI Office guidance, the Delhi HC ANI v OpenAI judgment, the Bartz appeal outcome) that may revise specific recommendations. Check it before acting on a chapter that’s been on the shelf for more than a quarter.

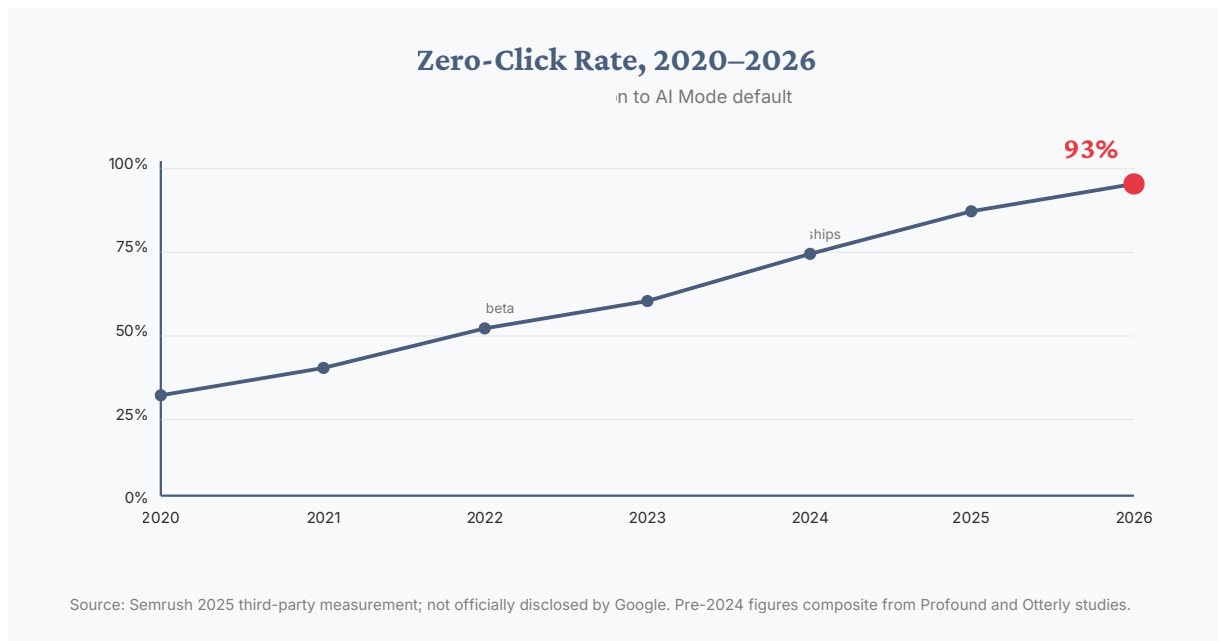
A note on practitioner orientation. The Monday-morning actions name the role doing the work — Senior Manager SEO/GEO, Senior Manager Content, Senior Manager Analytics, Custodian. Where senior approval is the unblock, a “What to take to your manager” callout tells you what to surface and what stays the manager’s call.

Chapter 1

The Click Is Gone



01 search to answer



01 zero click trajectory

Ninety-three per cent of AI Mode is zero-click.

Semrush measured it across 2025 with third-party instrumentation. The figure carries a caveat — Google has not disclosed the number officially, the measurement methodology has known variance — but the directional fact is uncontested across the canon. The user issues a query. The engine returns a synthesised answer with citation footnotes. The user reads the answer. The user does not click.

The answer engine is the cannibal. Not paid placement, not the next algorithm update, not the slow drift of organic CTR that marketing teams have been blaming for three years. The job is no longer rank, traffic, or click. It is citation.

Set this as the structural fact before you argue with it. Every chapter that follows assumes it. The reset is not optional.

What practitioners get wrong

The modal CMO conversation in 2026 sounds reasonable. “Our SEO is fine, but AI Overviews are eating our clicks — what should we do about traffic?” The question is wrong. Traffic is not what’s being eaten. The click was the tax that retrieval used to charge — and the retrieval model has changed. The tax has

been removed. There was never any traffic the AI Overview was promised to deliver.

I have watched the conversation play out in three companies in the first quarter of 2026. The answer is always wrong before the question finishes. The team that asks "how do we recover the lost clicks?" is the team that will rebuild traffic infrastructure for an outcome that the architecture has eliminated.

Three failure modes follow from the misframe. Each one shows up in real marketing organisations every week.

The first is defending against zero-click via paid escalation. The dashboard shows traffic dropping. Someone proposes more paid search to compensate. More brand bidding, more keyword expansion, more vendor SaaS. The structural problem — citation has replaced traffic as the unit — is rebuilt rather than addressed. The traffic-first model survives. The citation work doesn't get funded.

The second is ceding AI to brand or PR. "That's a brand thing. Performance owns paid; brand owns AI." This abdicates performance accountability for the largest discovery surface in the marketing organisation. AI surfaces are where buyers form vendor shortlists in 2026. Treating them as a brand-only territory is a category error in org design. The buyer is making a buying decision; the marketing function that owns buying decisions has handed it to the team that owns awareness.

The third is adding "AI strategy" as a line item without rewriting OKRs. The team agrees the shift is real. The OKR doc still says "drive 15% YoY traffic growth." The agency brief still rewards clicks. The dashboard still leads with traffic. Nothing changes except the slide. By Q3, the team is back where it started, with one extra acronym.

The CFO-legible cost is the budget allocated to clicks that no longer exist. The mid-sized B2B marketing budgets I have reviewed in 2026 still anchor a substantial portion — typically a third of the performance line — to outcomes that AI Mode is already eating. The 2026 budget that has not reframed is paying that tax twice.

How it actually works

The architectural shift is precise.

Classical search returned a results page. Ten blue links plus paid units. The user picked one, clicked, arrived at the destination. The page that ranked was the page that was the destination. Marketing succeeded by ranking and converting.

AI Mode returns a synthesised answer. The model retrieves a candidate set of pages, chunks them, selects fragments, and composes an answer. The answer renders inline with citation footnotes — small links to the sources the model claims to have drawn from. The user reads the answer. The user does not click. The 93% figure measures exactly this.

The trajectory got here in five years. 2020 saw SERP saturation — the results page filled with paid units, knowledge panels, and people-also-ask boxes that pushed organic listings below the fold. 2023 brought Google Search Generative Experience as a beta — the precursor of the synthesised-answer pattern. 2024 and 2025 saw AI Overviews ship to general availability and expand to more query categories. April 2026 finds AI Mode as the default surface for an increasing share of US desktop queries, with similar trajectories on ChatGPT Search, Perplexity, and Microsoft Copilot.

The click was always incidental to retrieval. The architecture has now removed it. Reference Chapter 4 for the retrieval mechanism in technical detail and Chapter 5 for the citation model that decides which sources get the footnote.

Common questions you'll get

"Hasn't this been happening for years? Why is 2026 different?" The threshold matters. Below 50% zero-click, traffic strategy still partially works because half of users still arrive at your destination. Above 80%, traffic strategy is performing for a vanishing minority. The 93% figure puts the regime change beyond the threshold where rebuilding traffic infrastructure pays back. The 2024 strategy survived; the 2026 strategy doesn't.

"What if the user reads the AI answer, then clicks through to verify?" Some do. Most don't. The verify-click rate has been measured at 6–8% of AI-answer interactions across multiple studies (vendor data, flag inline; corroborated

across Semrush, Profound, and Otterly). Eight per cent of zero-click is still seven per cent of historic SEO traffic — the rebuild-for-clicks math doesn't work even if you assume the verifying users are valuable.

"Doesn't this just mean we need better paid placement?" Paid placement on AI surfaces is still emerging. ChatGPT Search has disclosed CPM rates. Perplexity has Sponsored Questions. Google AI Overviews is integrating ads. None of them substitute for organic citation — they sit alongside it. Chapter 6 covers per-engine paid placement; Chapter 14 covers the budget split. The reframe is not "go paid"; it is "compete to be cited, organic and paid both."

Monday-morning actions

Action 1 — Pull the citation baseline for your top 20 priority pages. Owner: Senior Manager SEO/GEO. Done = a baseline spreadsheet with citations counted per priority page per engine, dated, stored in the dashboard. Methodology documented so the next baseline run is comparable. Failure mode = the baseline is run once and never re-run; drift goes unflagged because the methodology drifted with it.

Action 2 — Re-build the weekly dashboard with citation rate as the headline; traffic demoted to a support metric. Owner: Senior Manager Analytics / Marketing Ops. Done = the new dashboard live; screenshot dated in the change log; footnote naming the post-rationalisation caveat (Chapter 4). Failure mode = both metrics co-exist as headline; the team optimises against whichever flatters in any given week.

Action 3 — Write the one-page brief your Director or Head of will take to the CMO and the CFO. Owner: Senior Manager SEO/GEO. Done = brief delivered before the next planning cycle, with the 93% figure, the citation-rate baseline, and the recommended OKR change. Failure mode = the brief becomes a deck the

manager loses in their drafts folder; the budget conversation gets postponed for a quarter.

What to take to your manager

The one-page brief above. The OKR rewrite, the CFO conversation, and the budget reallocation are theirs. Your job is to make the brief defensible — every number sourced, every recommendation tied to a metric they can defend in a five-minute window.

The shape of the upstairs conversation: traffic-line OKRs convert to citation-rate OKRs. The performance budget reallocates from clicks-as-objective to citation-as-objective. The agency brief rewrites — the agency cannot keep delivering against the old metric while invoicing against the new strategy.

Activate this. francescofederico.com/cited/click-is-gone hosts the one-page brief template. Print it, attach the citation baseline, hand it to your Director.

How to know it landed

Citation share, not click share, as the headline metric. Per priority audience, per priority engine, against a fixed prompt battery (Appendix B). Reference Chapter 12 for the discipline.

Honest caveat. Attribution is broken across every engine. No vendor solves it; many sell the illusion. Measurement starts with that admission, not against it. Baseline before scaling. The first weekly run is the truth; everything after is signal.

The false positive to watch: a single engine's win read as a regime-wide win. AI Mode citation rates and ChatGPT Search citation rates do not co-vary cleanly. A 30% lift on one and zero on the other is a real outcome — not a measurement error.

What we don't yet know

Whether the 93% holds when AI Mode goes default-off in jurisdictions that mandate publisher choice.

The CMA conduct requirements (mid-2026) are the trigger to watch. If UK regulators require Google AI Overviews to offer publisher choice and affirmative attribution, the surface changes shape — possibly with a higher click-through rate, possibly with publisher-led carve-outs that change retrieval logic. Similar regulatory work in the EU and India could reshape the architecture differently in different markets.

Plan against the modal scenario. Don't bet against the regulators; they may move. But don't wait for them either. The work the chapter recommends pays back regardless of how the CMA decision lands.

The CMOs who reframe traffic-to-citation in 2026 still have a marketing function in 2028. The ones who keep optimising for clicks will be told their team is being absorbed into commerce.

Chapter 2

What Carries Over From SEO

The Continuity Map		
what carries over to GEO		
DISCIPLINE	CARRY-OVER	RE-WEIGHTING NOTE
Technical Indexability, crawl, render, mobile	HIGH	Entry ticket. No rebuild required.
On-page Headings, structure, intent	HIGH	Re-weight toward extractability (Ch 9).
Content Depth, freshness, originality	HIGH	Synthesis-friendly structure beats narrative flow.
Links Anchor-text equity, authority	MODERATE	Co-citation increasingly displaces classical anchor.
E-E-A-T Experience, expertise, trust	HIGH	Reframed as entity verification (Ch 10).
Local Maps, place, LocalBusiness	HIGH	Mostly unchanged for AIO.
International Hreflang, language targeting	MODERATE	Regional engines behave differently.
Mobile Mobile-first indexing	HIGH	Inherited; mostly unchanged.

Source: chatty-continuity-map; Aggarwal et al. (2023); Google AIO platform documentation.

02 continuity map

Roughly eighty per cent of the SEO discipline survives the shift to retrieval.

Read that again. Eighty per cent. The novelty pitches are wrong about how much is new. The agencies pivoting their service decks to “GEO” without changing their underlying methodology are not malicious. They are correct about which work continues. They are wrong about how to charge for it.

This chapter’s job is to disarm the over-correction. To give you permission to keep what works, the language to push back on the agency that wants to throw it out, and the audit to prove which disciplines are bar-passing today. Senior leaders in 2026 are funding rebuilds of work they already paid for in 2024 — and the rebuild is being charged at “AI optimisation” premium rates.

What practitioners get wrong

The vendor tour in 2026 sounds urgent. “Your SEO doesn’t work for AI.” Some of it doesn’t. Most of it does, with re-weighting. The mistake is treating GEO as a green-field discipline when the foundations of indexability, technical hygiene, internal linking, and authority all carry forward.

I have seen this mistake cost six and seven figures across the marketing organisations I have advised in the last twelve months. The pattern is identical. An agency proposes a “GEO platform replatform” — a multi-month engagement to rebuild the crawl architecture, the URL taxonomy, the internal-linking graph, the on-page schema, the robots.txt, the sitemap. The work is real. The work was also done in 2023. The pages are already indexable. The crawl already runs cleanly. The internal linking already passes equity correctly. The agency proposal treats all of this as net-new because the brief calls it “AI optimisation.”

The second budget mistake is paying twice for entity work the comms team already does. Wikipedia maintenance, Crunchbase profile updates, LinkedIn company-page hygiene — these have been comms responsibilities for a decade. A “GEO entity verification programme” pitched as new work is sometimes legitimate (the sameAs JSON-LD in Chapter 10 is genuinely new) but more often it is the same Wikipedia maintenance with a different invoice.

The CFO-legible cost is straightforward. Rebuilds funded as net-new work that was already in the organisation’s existing footprint. The audit catches them. Without the audit, the budget approves them.

How it actually works

The continuity map. Eight SEO disciplines, scored against the GEO mechanism.

Technical SEO — high carry-over. Indexability, crawl, render, mobile, page speed. The hybrid-search retrieval mechanism (Chapter 4) requires pages to be indexable before they can be retrieved. Pages that fail at the indexing phase do not appear in the candidate set. Classical technical SEO is the entry ticket. No rebuild required if the page is already passing.

On-page SEO — high, with re-weighting. Headings, structure, intent matching. The mechanism survives. The weighting shifts toward extractability — H3-led

question structure, dense Q&A blocks, segmented content that survives chunking. The work is the same skill set. The brief template (Chapter 9) re-orders the priority of what gets written.

Content depth, freshness, originality — high, with re-weighting. The discipline of producing valuable content carries forward unchanged. The re-weighting is toward synthesis-friendly structure rather than narrative flow. A 1,500-word original article continues to outperform a 500-word thin page. The 1,500 words just need to be structured for chunking, not for reading-time-on-page.

Links — moderate. Classical anchor-text equity continues to feed Google's ranking signals, which feed AI Overviews via the indirect path (Chapter 5). The novelty is co-citation (Chapter 11) — third-party mentions without HTML linkage that LLMs reward at the re-rank phase. Both matter. The link-building agency is solving the 2018 problem; the co-citation work is the 2026 problem; the team needs both.

E-E-A-T — high, reframed. Experience, Expertise, Authoritativeness, Trust generalises across engines as entity verification (Chapter 10) plus co-citation. The work is the same; the language is different. Don't pay an "E-E-A-T audit" agency in 2026 unless they can show the entity-graph deliverables that distinguish the new work from the old.

Local SEO — high, mostly unchanged. Google AI Overviews handles local queries through the same Google Maps and Knowledge Graph signals it always has. The schema work for LocalBusiness types continues to pay back. Specialised retrieval extracts from local schema directly. Local SEO, alone among the eight disciplines, requires almost no re-thinking.

International SEO — moderate. The hreflang and language-targeting mechanics survive. The weighting shifts — regional engines (Yandex, Baidu) and emerging AI surfaces in non-English markets behave differently from US-trained models. Chapter 15 flags geographic citation variation as an under-researched area.

Mobile SEO — high, mostly unchanged. Mobile-first indexing was a 2018 shift; AI surfaces inherit it. Mobile-optimised pages continue to outperform desktop-only pages on every engine.

The retrieval architecture preserves classical signals at the BM25 phase before semantic re-ranking. Aggarwal et al. (2023) — the foundational GEO paper — explicitly position the discipline as additive. Google’s own statements that AI Overviews “introduces no extra technical requirements” land here as platform documentation. The continuity is real. The agencies’ hesitation to admit it is commercial.

Common questions you’ll get

“If 80% carries over, why does GEO need a separate budget at all?” The 20% that’s new is genuinely additive — extractability, entity work, multimodal, paid placement, attribution discipline. Each engages mechanisms classical SEO doesn’t. The question is not whether GEO needs a budget. It is whether the new work should be net-new line items, or whether it should sit inside the existing SEO budget with re-prioritised weights. Either model works. What doesn’t work is funding both at full cost.

“Our agency says they have a proprietary AI-SEO methodology. Should we listen?” Ask three questions. Which of the eight disciplines does it cover that classical SEO doesn’t? What evidence supports the proprietary claim? Is the methodology open to independent benchmarking? Most “proprietary AI-SEO methodologies” in 2026 are classical SEO repackaged. If the agency cannot map their methodology onto the eight disciplines and name what’s new, the proprietary claim is marketing.

“Where does GEO require genuinely new investment?” The five direct-layer levers in Part IV — crawler portfolio (Chapter 7, ai.txt and llms.txt as new artefacts), multimodal and YouTube (Chapter 8, video FAQ as a new production discipline), content extractability (Chapter 9, FAQPage and statistics register), entity verification (Chapter 10, Wikidata and Hugging Face datasets), authority/co-citation (Chapter 11, co-citation register). Plus measurement (Chapter 12, the prompt battery). Everything else is hygiene.

Monday-morning actions

Action 1 — Run the eight-discipline continuity audit (Appendix E, Section 1).

Owner: Senior Manager SEO/GEO. **Done** = the audit complete, scored pass / partial / fail per discipline, dated; the criteria documented so the next audit is comparable. **Failure mode** = the audit becomes slide-ware; the score is generous because nobody benchmarks against the criteria.

Action 2 — Re-tag your existing SEO dashboard against the continuity map.

Owner: Senior Manager Analytics. **Done** = each KPI tagged carry-over / re-weighted / new; the dashboard reflects the new tagging at the next weekly review. **Failure mode** = the dashboard remains the SEO dashboard with one extra “AI” column nobody reads.

Action 3 — Build the inventory of work you already own — content, technical hygiene, internal linking, entity work — before the next agency proposal lands.

Owner: Senior Manager SEO/GEO. **Done** = a one-page inventory document the manager can take into the procurement review with line-by-line evidence of what’s already paid for. **Failure mode** = the inventory exists but the agency proposal still gets approved because nobody pulls it out at the procurement meeting.

What to take to your manager

The audit and the inventory. They use both to push back on agency proposals that double-charge for work you already own. The “reject the rebuild” call is theirs; the evidence base is yours.

The shape of the conversation: the agency proposal is taken seriously, line item by line item. Each item is checked against the inventory. Anything that duplicates existing work is rejected. Anything that addresses the 20% genuinely

new is approved at appropriate scope. The total approved is typically 30–50% of the original proposal. The savings — and they are real — flow into the new direct-layer levers.

How to know it landed

The continuity baseline as a one-time score, revisited annually.

The metric is simple: of the eight disciplines, how many pass, how many partial, how many fail? The cadence is yearly. The false positive is “we said pass, and it’s actually partial.” Use Google Search Console, Bing Webmaster Tools, and a manual technical SEO audit tool (Screaming Frog, Sitebulb, or equivalent) to validate the pass scores. Cross-reference Chapter 12 for the broader audit discipline.

A pass score on all eight is unusual — it means the prior year’s SEO programme has been rigorous and well-funded. A fail score on three or more is a different kind of signal: classical SEO is broken, and no amount of GEO investment will compensate. Fix the foundation first.

What we don’t yet know

How much of the carry-over holds if engines move to fully agentic retrieval — architectures where the model retrieves from a memorised corpus through attention rather than from a public web index through hybrid search.

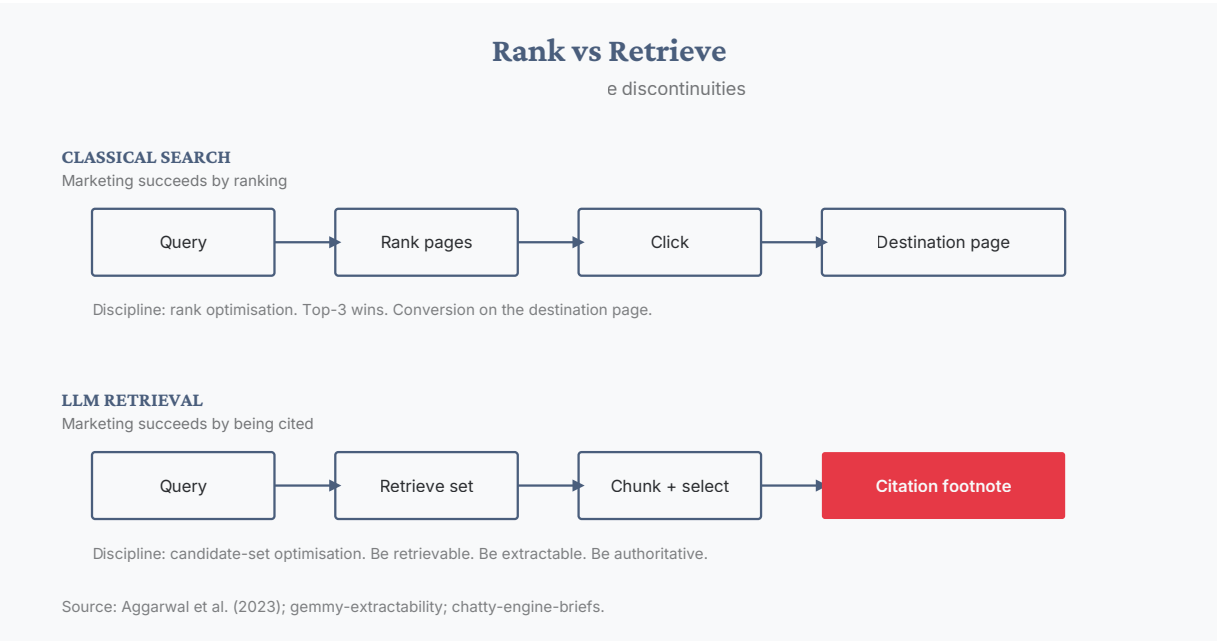
If that architecture ships at scale, the eight disciplines compress into one. Be in the training corpus. Indexability, link equity, on-page structure, internal linking — none of it matters if retrieval is internal to the model. The work shifts to dataset publication (Chapter 10, the Hugging Face dataset path) and licensing deals (Chapter 15, the regulatory cascade). The continuity map dissolves.

The architecture is not yet shipping at scale. The work the chapter recommends remains correct. Re-test annually; flag in Chapter 15.

The most expensive thing a senior marketer can do in 2026 is rebuild SEO infrastructure they already own under a new acronym.

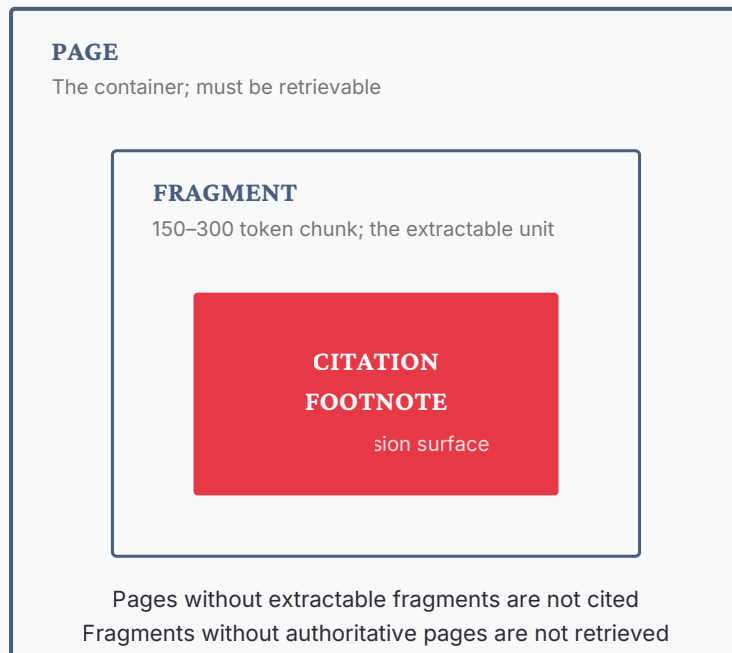
Chapter 3

What Is Genuinely New



Citation, Not Click

je compresses



Source: gemmy-extractability; Chapter 3 framing.

03 citation not click

The 20% that has changed has changed everything that depends on it.

Chapter 2 disarmed the over-correction. This chapter disarms the under-correction. Both mistakes are common, and both are expensive. The senior marketer who treats GEO as wholly net-new wastes budget on rebuilds. The senior marketer who treats GEO as just SEO with new metadata under-invests in the discontinuities and loses the citation surface to competitors who didn't make the same call.

Three discontinuities. Each one is genuinely new — not a re-weighting, not a re-labelling, but a different mechanism with a different unit of value. Get these wrong and the rest of the manual reads as a list of tactics with no underlying logic. Get them right and the tactics become consequences.

What practitioners get wrong

The opposite mistake to Chapter 2: treating GEO as just SEO with new metadata. A few new schema types. An lms.txt file. A line item in the brief for “AI-friendly headings.” Same mechanism, slightly different invocations.

The misread is sustained by the symptoms. Schema is real metadata. lms.txt is a real file. FAQ structure does help. The vendor decks selling these as “the new GEO playbook” aren’t lying about the artefacts — they are misleading about which mechanism the artefacts engage. The artefacts are the visible surface. The mechanism is underneath. Buying the artefacts without understanding the mechanism produces compliance theatre rather than citation lift.

I have watched this play out in three engagements in 2026. The team buys the metadata package, ships the artefacts, declares GEO done, and the citation rate barely moves because the underlying mechanism wasn’t engaged. The CFO-legible cost is the slow erosion of relative position. The team that bought the metadata package thinks it has done GEO. The competitor down the street rebuilt their content architecture for extractability, deployed an Entity Bible (Chapter 10), and shipped a video FAQ programme (Chapter 8). Both teams think they have GEO strategies. Only one of them is being cited. The dashboard reveals the gap eighteen months later, when the budget conversation that follows is hostile.

How it actually works

Three discontinuities. Named with mechanism, ordered by depth.

Retrieval over ranking

The first discontinuity changes what the search engine does.

Classical search ranked pages. The retrieval set was generous (every page that matched the query); the ranking determined which appeared first. Marketing succeeded by ranking — being in the top three, top one. The discipline was rank optimisation.

LLM retrieval orders pages by inclusion in the candidate set, not by classical rank. A page is either retrieved or it is not. If retrieved, it is either chunked into useful fragments or it is not. The fragment is either selected for the answer or it is not. There is no top three. There is in or out. The discipline is candidate-set optimisation — making sure the page enters the retrieval phase, survives chunking, and produces an extractable fragment.

The implication for the practitioner. The hierarchical mental model of “rank #1” stops generating useful direction. The binary mental model of “be retrievable, be extractable, be authoritative” replaces it. A page that ranks fifth on classical organic but produces clean extractable fragments will outperform a page that ranks first on classical organic but reads as a flowing narrative. Classical SEO didn’t have to choose. GEO does.

Synthesis over click

The second discontinuity changes what the user does.

Classical search delivered a results page; the user clicked a destination. The page that won was the destination. Marketing succeeded by being the destination — by converting on the page the user landed on.

LLM retrieval composes an answer; the user reads the answer. The page the user “would have visited” no longer matters as the conversion surface. The fragment of the page that appears in the answer is the conversion surface. The user’s mental model of the brand is shaped by how the brand is quoted in the synthesised answer, not by what the brand wrote on its own page.

The implication is precise and uncomfortable. Marketing organisations that have spent fifteen years optimising landing pages now find the landing page is rarely visited. The work is not wasted — pages still need to exist for retrieval to find them — but the optimisation effort relocates. Conversion optimisation moves to the fragment level: which sentences get extracted, what they say about the brand when extracted, whether the extracted fragment makes the buyer want to learn more. The brand voice that lives only in the page footer is invisible. The brand voice that lives in the H3-led answer paragraphs is the only brand voice the buyer sees.

Citation over keyword

The third discontinuity changes what the marketer measures.

Classical SEO targeted keywords. The keyword was the unit of work — a target phrase, a search-volume estimate, a difficulty score, a SERP feature analysis.

The team optimised for the keyword and tracked rank position over time.

LLM retrieval cites sources. The unit of work is the citation — the brand's name appearing in a synthesised answer to a buyer's question, with the page URL footnoted. The keyword phrasing the buyer used is partly visible (in the prompt the user submitted), partly invisible (in the model's internal expansion of that prompt to a candidate retrieval set), and partly irrelevant (the model may answer using vocabulary the buyer did not submit). The query is no longer the unit. The buyer's intent, expressed as a synthesised question and answered with branded citations, is the unit.

The implication for measurement. Keyword-level dashboards become noisy and partial. Citation-rate dashboards (Chapter 12) become the honest signal. The shift requires rebuilding the analytics tooling, the agency reporting templates, and — most importantly — the OKRs that the budget defends. Keyword targets become citation targets. Rank tracking becomes citation tracking. SEO platforms that don't expose citation data become measurement debt.

Common questions you'll get

"Do classical keywords still matter at all?" Yes, at the BM25 phase of hybrid retrieval (Chapter 4). The candidate set is generated partly by classical lexical match. Pages that lack keyword discipline fail at retrieval before vector search has a chance to rescue them. Continue keyword work — but stop treating keyword rank as the outcome. The outcome is citation; rank is one of several inputs.

"How do I know which prompts buyers are actually using?" Sales calls. Customer interviews. Support tickets. Search Console queries (still useful, even if zero-click). The prompts that AI engines receive are paraphrases of buyer intent, expressed in natural language. Sales transcripts are the closest proxy — buyers ask the same questions on calls that they ask AI engines. The keyword-

tool list is the worst proxy because it captures historical search behaviour, not current AI-mediated behaviour.

"If everything is fragments, does the page even matter?" The page is the container. Without it, there is nothing to retrieve. The fragment is the unit of value, but the fragment exists only as part of a page that satisfies hybrid retrieval. The page must be technically sound, semantically coherent, structurally extractable. Fragments without a page substrate are not retrievable. The page matters for retrieval. The fragment matters for citation. Both are real units; they operate at different phases.

Monday-morning actions

Action 1 — Identify the top three buyer prompts from sales call transcripts and customer interviews — not from a keyword tool. Owner: Senior Manager Content / Senior Manager SEO. Done = three documented prompts per priority audience, with source notes (which call, which customer, what the prompt actually was, dated). Failure mode = the prompts come from an SEO keyword list and miss the actual buyer language; everything downstream optimises against the wrong question.

Action 2 — Run each prompt through three engines manually. Capture: who got cited, in what position, what fragment was extracted. Owner: Senior Manager SEO/GEO. Done = a citation map per prompt, stored in the dashboard, with the extracted fragment quoted verbatim per cited source. Failure mode = the map is a screenshot library; nobody analyses why the cited sources won and the team can't extract a pattern.

Action 3 — Write the one-page summary your Director will take to the CMO. Owner: Senior Manager SEO/GEO. Done = a one-page summary with the three discontinuities, the citation map, and the recommended budget reallocation.

Failure mode = the summary becomes a deck and the budget conversation never happens.

What to take to your manager

The citation map and the one-page summary. The budget reallocation is theirs; the data is yours. Make the map screenshot-ready — Directors take screenshots of evidence into board meetings, not paragraphs.

The shape of the upstairs conversation. The three discontinuities reframe the GEO budget from an “AI line item” to a fundamental redesign of how marketing attribution works. The reframe is uncomfortable for a CFO who has approved keyword-budget proposals for ten years. The citation map is the artefact that makes the reframe defensible — a real query, real cited sources, real fragments, real competitors winning the citation today.

How to know it landed

Citation rate per priority prompt, baselined this quarter.

Reference Chapter 12 for the measurement discipline. Reference Appendix B for the prompt battery template. The false positive: treating any citation as victory. Track citation quality alongside citation count — lead-position citation versus deep-list mention, primary source versus supporting reference, branded extraction versus generic fragment. A citation in position one with the brand’s primary value proposition extracted is worth ten generic mentions in position seven.

The harder discipline: tracking the competitive citation map over time. Today’s competitor is the brand cited in your priority prompts. Tomorrow’s competitor may be a brand that didn’t exist eighteen months ago. The competitive set redraws faster on AI surfaces than it ever did on classical search. Quarterly review of the map is the cadence. Annual review is too slow.

What we don't yet know

Whether citation as the unit of value persists once engines move to direct fulfilment without source attribution.

The frame this manual uses assumes the citation footnote remains the surface — the model produces an answer and emits source footnotes. If engines collapse to direct answer with no citation, the unit shifts again. The brand-extraction surface disappears entirely. Marketing reverts to brand-presence in training data as the only lever.

Some early signals. ChatGPT's interface in 2024 hid citations behind a click; the 2025 redesign made them prominent again. Perplexity has always led with citations and shows no sign of moving away. Google AI Overviews surface citations in a hover state on desktop but inline on mobile. The trajectory is mixed; the citation surface looks durable through 2027 at least. Chapter 15 names this as one of the five honest unknowns.

Plan against the architecture as it is. Re-test annually.

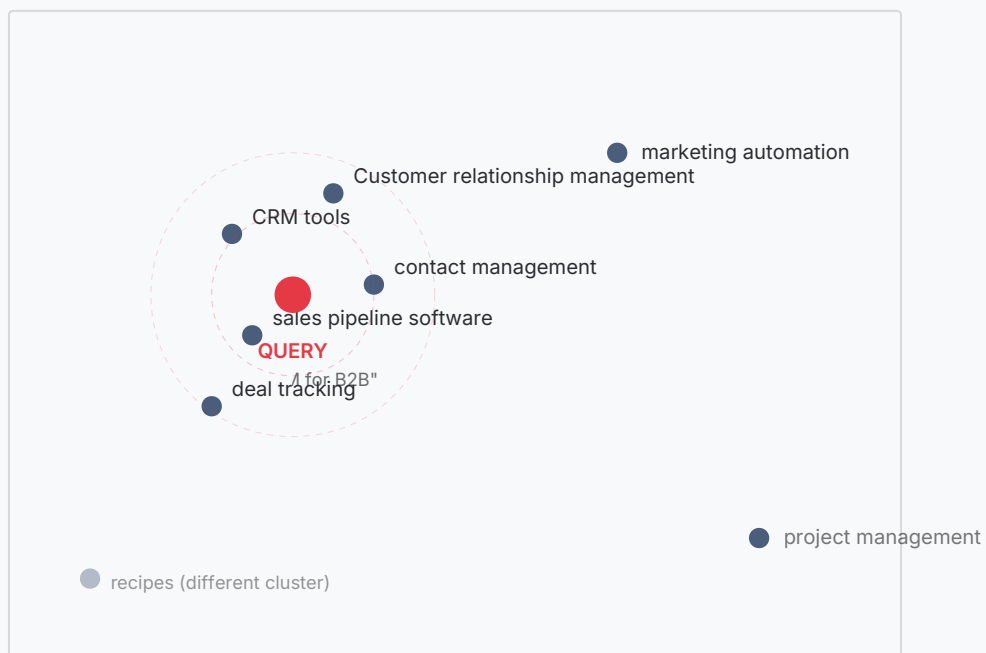
The 20% that's new is the 20% your competitors will get to before you do, unless you concede the next two years.

Chapter 4

The Technical Primer for Marketers

Vector Embedding

; synonyms collapse



→ Closer angle = closer meaning

The page does not need the query keyword. It needs to inhabit the same semantic neighbourhood.

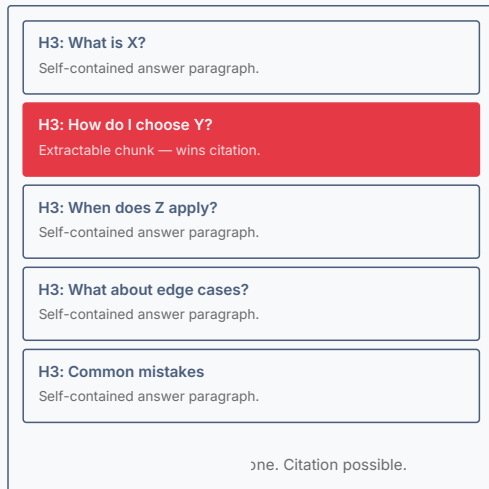
Source: gemmy-technical-foundations; Chapter 4 primer.

Chunking

ents; each chunk gets its own embedding

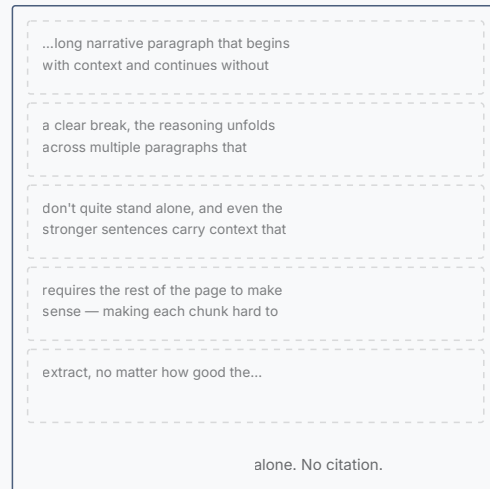
STRUCTURED PAGE

H3 sections with self-contained answers



NARRATIVE-FLOW PAGE

No structure; chunks split mid-thought

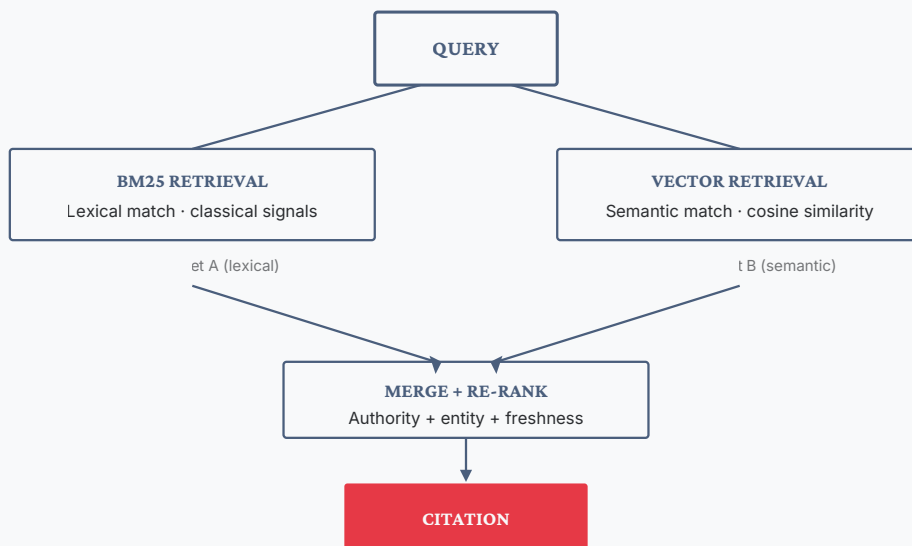


Source: gemmy-extractability; chapter 4 primer; Aggarwal et al. (2023).

04 chunking

Hybrid Search

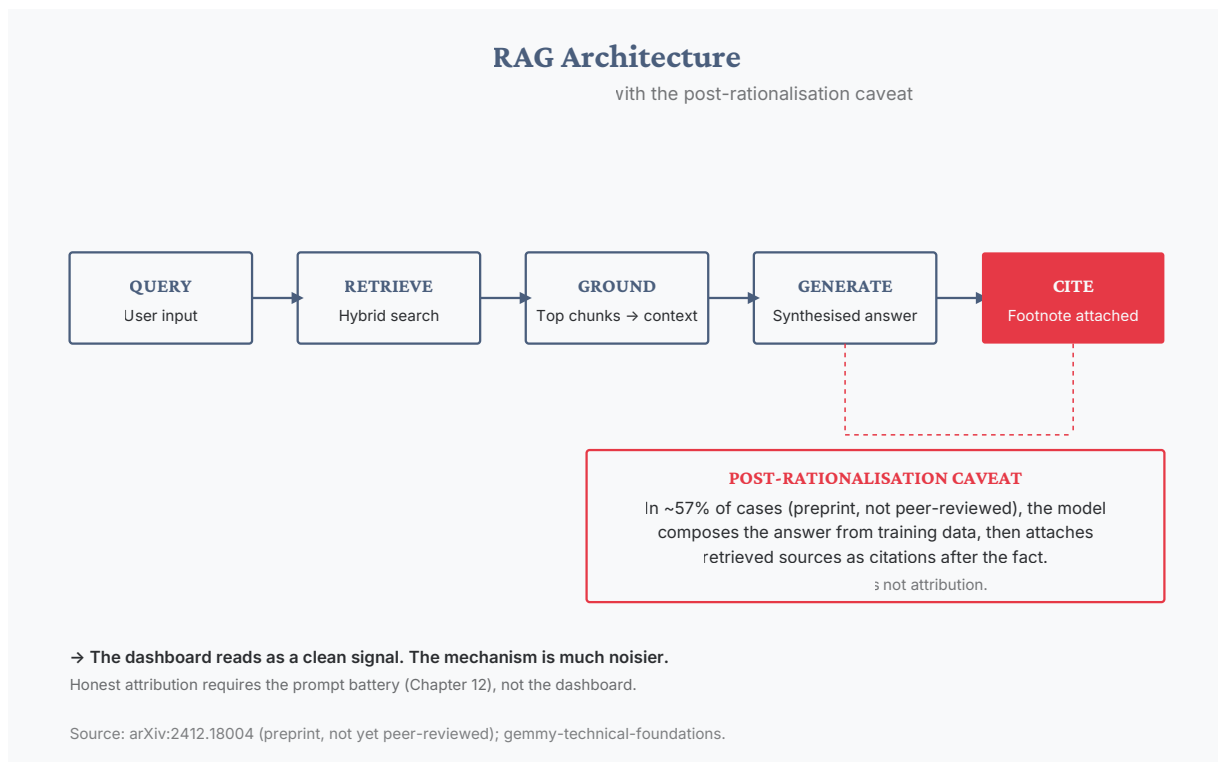
n parallel, then re-ranked



→ Pages that fail BM25 (broken indexability) never enter the candidate set. Classical SEO hygiene is the entry ticket.

Source: gemmy-hybrid-search; production retrieval stack documentation.

04 hybrid search



04 rag architecture

The senior marketer who cannot describe vector embeddings, chunking, and hybrid search to their engineering team will be sold the wrong tools by the loudest vendor.

This chapter is the literacy floor — not the ceiling. Five concepts at practitioner-appropriate depth. Enough mechanism to argue with engineers, challenge vendors, and rewrite a brief without consulting the agency. Each concept has a diagram. Each diagram is the test.

Can you draw it from memory by the end of the week? If yes, you don't need this chapter again. If no, the tool budget for the rest of 2026 will be spent on problems you can't articulate.

What practitioners get wrong

The most expensive mistakes in 2025–26 GEO budgets came from string-matching mental models applied to vector retrieval.

The pattern is recognisable. A team that ran SEO for ten years approaches GEO with the same instincts. Add the keyword. Stuff the meta description. Match the search intent. The instincts aren't wrong — they are calibrated to a different

mechanism. Vector retrieval doesn't match keywords. It matches meaning, expressed as geometry. The page that wins isn't the page with the right words. It's the page with the right structure for the chunking pipeline that comes next.

The schema deck (Chapter 5) is the most visible symptom of this category error. The underlying error is broader, and it shows up across the GEO budget. "Make the page more comprehensive" — the chunking mechanism prefers segmented density, not flow. "Cover every related keyword" — the vector embedding doesn't reward keyword coverage; it rewards semantic clarity. "Add more internal links" — re-ranking weights authority signals more than internal links at the citation phase.

The second failure mode is subtler and more damaging. The 57% post-rationalisation rate (preprint, arXiv:2412.18004 — flag inline as not yet peer-reviewed) describes how often LLMs generate the answer first and the citation second. Sometimes the cited source did not produce the claim. The model used the source's name as a credibility marker for an answer it composed from elsewhere in its training data. This is not corruption. It is a property of the model. The marketing implication is precise. "We got cited" does not always mean "we caused the answer." Attribution dashboards that don't account for this produce confident, wrong conclusions.

I sat through three attribution presentations in early 2026 where the post-rationalisation rate was the missing footnote. Each treated citation count as a clean lift metric. Each was wrong by an unknown but material margin. The team that built the dashboard didn't know the caveat existed. The vendor that sold the dashboard hadn't read the preprint.

How it actually works

Five concepts. Read each with the corresponding diagram open. Draw it yourself by the end of the chapter.

Vector embeddings

The first concept dissolves the keyword instinct. LLMs do not match keywords. They match meaning, and meaning is geometry.

When a model processes a query — what is the best CRM for a mid-sized B2B team — it converts the text into a high-dimensional vector. A point in space. Each candidate page is also a point in space, computed from its body text. The model selects pages by cosine similarity — pages whose vectors point in the same direction as the query's vector. Closer in angle, more semantically aligned. Synonyms collapse onto each other. "Customer relationship management" and "CRM" sit in the same neighbourhood. Pages that use neither phrase but discuss the same problem can land in the neighbourhood too — if the surrounding semantics align.

What this means for the practitioner. The page does not need to repeat the query verbatim. It needs to inhabit the same semantic space. Generic SEO instincts about keyword density become irrelevant. Meaning density becomes everything. The page that wins on vector retrieval is the page whose vector best represents the actual intent behind the query — not the one with the most keyword coverage.

Chunking

The second concept dissolves the page-as-unit instinct.

Pages are not retrieved whole. They are split into segments — chunks — typically 150 to 300 tokens each. Each chunk gets its own vector embedding. The model retrieves chunks, not pages. The page exists only as a container. The chunk is the unit of value.

This single mechanism explains why long-form pages without internal structure underperform compared to shorter pages with crisp segmentation. A 4,000-word article that flows as a single narrative gets sliced into chunks at arbitrary positions — sentence mid-paragraph, paragraph mid-thought. Each chunk is incomplete. None of them stand alone. The model finds nothing extractable.

A 1,800-word article structured as discrete H3-led sections, each with a 200-word self-contained answer, gets sliced cleanly. Each chunk is a complete unit. The model finds extractable fragments. The same content, restructured, produces dramatically different retrieval outcomes.

This is why the Aggarwal tactics in Chapter 9 work. Statistics Addition, Citing Sources, Q&A blocks — all operationalisations of the chunking mechanism. The page wins because its chunks survive.

Hybrid search — BM25 plus vectors

The third concept dissolves the all-in-on-vectors instinct.

A common mistake among marketing teams that have read one whitepaper on RAG: assume vector search has replaced classical search. It has not. Production retrieval stacks use both, in combination, and classical SEO signals still matter at the retrieval phase.

Hybrid search runs two retrieval pipelines in parallel. BM25 — a classical keyword-and-frequency algorithm — pulls candidate pages based on lexical match. Vector retrieval pulls candidate pages based on semantic match. The two candidate sets merge. A re-ranker (next concept) orders the merged set.

The practical implication is precise. Pages that are technically broken — poor crawlability, blocked from indexing, malformed canonical tags — fail at the BM25 phase and never enter the candidate set at all. Vector retrieval does not rescue them. Classical SEO hygiene (the eight disciplines named in Chapter 2) is the entry ticket. Without it, the page is invisible regardless of how semantically rich the body text is.

RAG — retrieval-augmented generation

The fourth concept describes how retrieval feeds generation.

The model receives the query. It runs hybrid search to retrieve a candidate set. The top-ranked chunks become the model's grounding — the context the model is conditioned on when it generates the answer. The answer is composed by reference to these chunks, with citation footnotes pointing to the source pages they came from.

Or so the architecture suggests. The post-rationalisation problem complicates this. The model does not always compose the answer from the grounding. Sometimes it composes the answer from its training data, then attaches citation footnotes from the grounding to legitimise the answer. The 57% rate (preprint,

flag inline) describes the frequency at which the cited source materially influenced the answer's content. The remainder is post-hoc citation — citation as plausibility theatre, not as causal source.

The marketing implication is precise. Citation count alone is not attribution. A brand whose pages are cited in 100 LLM answers may have caused fewer than half of those answers. The dashboard reads as a clean signal. The underlying mechanism is much noisier. Honest attribution requires the prompt battery (Chapter 12), not the dashboard.

Re-ranking

The fifth concept describes how the candidate set gets ordered.

After hybrid search produces a candidate set, a re-ranker scores the candidates against the query and the user's intent signals. Authority weights enter heavily here — domain trust, entity verification, co-citation, recency. The page that won at the retrieval phase may lose at the re-rank phase if a higher-authority page covers the same content.

This is why entity work (Chapter 10) and co-citation (Chapter 11) matter more than they look on the surface. They do not affect retrieval directly. They affect re-ranking — the phase that decides which retrieved pages are eligible to be cited. A page can win retrieval and lose re-ranking. The retrieval set is the longlist. The re-ranked set is the shortlist. The shortlist is what gets cited.

Diagram 04-rag-architecture.svg carries the full pipeline. Memorise the four phases — chunking, hybrid retrieval, re-ranking, generation with citation. The next time a vendor pitches “AI search optimisation” without naming the phase their tool addresses, ask which phase. If they cannot answer, the pitch is marketing.

Common questions you'll get

“If the model uses vectors, doesn't that mean keywords don't matter?”

Keywords still matter at the BM25 phase, which runs in parallel with vector retrieval. The pages that pass through to the re-ranker are the union of both retrieval sets. Skipping keywords means skipping half the candidate-generation

pipeline. Match for both — semantic clarity in the body text, classical keyword discipline in the title, headings, and meta description.

“How do I know if my page is being chunked well?” There is no public tool that exposes chunk boundaries from production engines. The proxy is structure. If your page reads as a series of self-contained 150–300-word sections, each with a clear topic sentence and a complete answer, it chunks cleanly. If it reads as a flowing narrative without internal structure, it doesn’t. Open one of your priority pages and try to read each H3 section in isolation — if the section makes sense without the surrounding context, the chunk will too.

“What’s the difference between embedding similarity and semantic relevance for ranking?” Embedding similarity is a measurement made by the retrieval system. Semantic relevance is a judgement made by the user. They correlate; they are not identical. Two pages can have near-identical vector embeddings and very different relevance to the query — for example, a definitional page and an implementation page on the same topic. The re-ranker tries to bridge the gap. It doesn’t always succeed. The implication for content strategy: write to the user’s actual intent, not to the embedding space.

“How quickly does this all change?” The architecture described here captured 2026 production retrieval stacks. The half-life of the specifics is two to three years. Vector embeddings, chunking, and hybrid search will stay foundational. The specific re-ranking signals, the post-rationalisation rate, and the role of structured data may shift materially. Annual literacy refresh is the discipline.

Monday-morning actions

Action 1 — Read this chapter and Chapter 5 together. Test yourself on the diagrams. Owner: practitioner. Done = you can draw the hybrid-search diagram from memory and answer three challenge questions — what is hybrid search; what is post-rationalisation; what does chunking do to a long-form page. If not, re-read. Failure mode = you read the chapter once and never re-test; the literacy fades the next quarter and you find yourself nodding through the next vendor pitch.

Action 2 — Add the “what’s the retrieval mechanism per engine?” question to your standard vendor evaluation rubric (Appendix D). Owner: Senior Manager Marketing Ops. Done = the question is in the rubric; applied to the next pitch with a written answer scored against the canon. Failure mode = the rubric updates; the scorer ticks the box without evaluating the answer.

Action 3 — Add the post-rationalisation caveat to the methodology section of every attribution report you produce. Owner: Senior Manager Analytics. Done = the dashboard footnote and the report narrative both name the 57% caveat (with the preprint flag); the methodology section reflects it. Failure mode = the caveat is mentioned once and forgotten; the dashboard stays falsely precise and the next attribution dispute lands in your inbox.

How to know it landed

Recursive — the test is whether a senior marketer in your team can draw the hybrid-search diagram on a whiteboard and answer three challenge questions.

If yes, the chapter has landed; the team has the literacy. If not, run a one-hour session at the next leadership meeting using the diagrams in this chapter. Repeat at six-month intervals. Architectures shift; the team’s mental model has to track.

The harder test comes at the next vendor pitch. The pitch deck names a “lift” and a methodology. The team that has internalised this chapter can ask which retrieval phase the lift was measured against, what the post-rationalisation control was, and how chunk boundaries were isolated from concurrent content edits. The team that hasn’t asks none of these questions. The vendor wins or loses on which kind of room they are pitching to.

What we don't yet know

How quickly retrieval architectures move from current hybrid models to direct semantic generation without explicit retrieval.

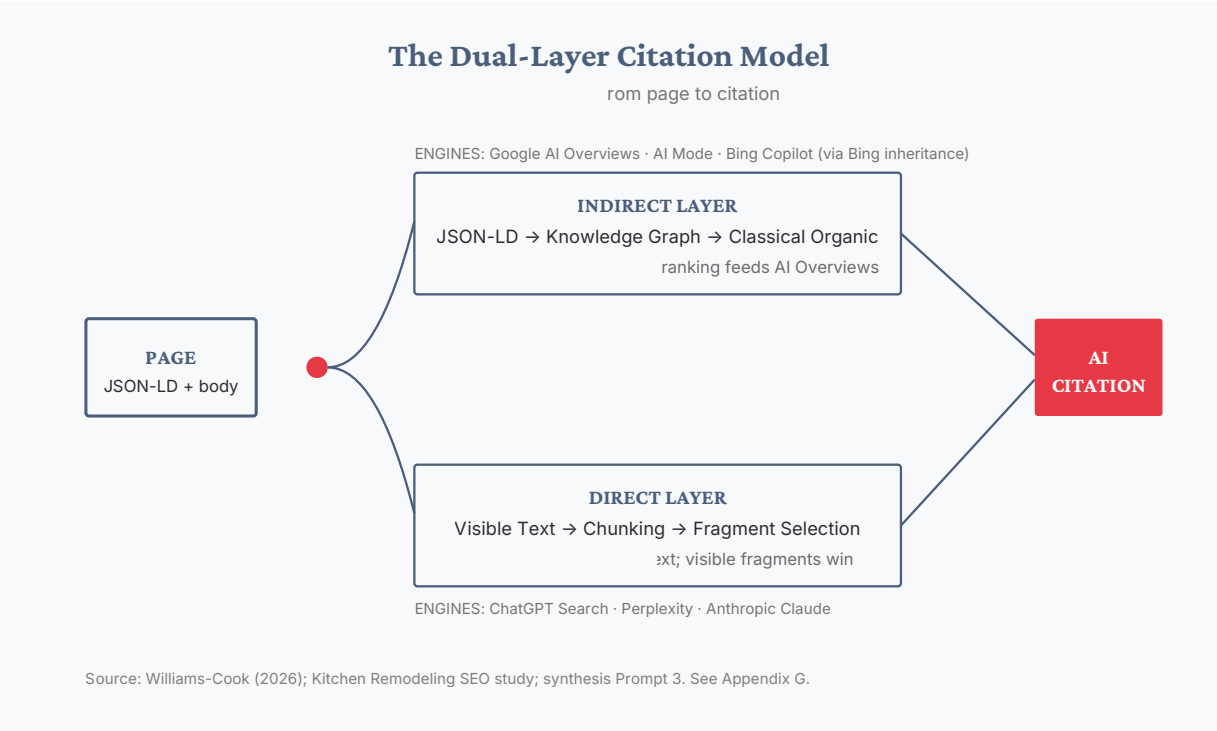
The current state of the art runs hybrid retrieval as a discrete phase before generation. The grounding is explicit; the citation is real (subject to the post-rationalisation caveat). Frontier research suggests architectures where retrieval and generation are tightly coupled — where the model's attention mechanism implicitly retrieves from a memorised corpus, with no separate retrieval phase. If that architecture ships at scale, the literacy on this page changes. The chunking mechanism may dissolve. The citation mechanism may become unverifiable. The work to do today, however, is the work the current architecture rewards. Re-test annually.

The other open question is the post-rationalisation rate itself. The 57% figure is from a preprint that has not been replicated against post-2025 model releases. If the rate falls — say, to 20% — citation becomes a cleaner attribution signal, and the dashboards become more defensible. If it rises, the prompt battery becomes the only honest measurement instrument. Watch the literature.

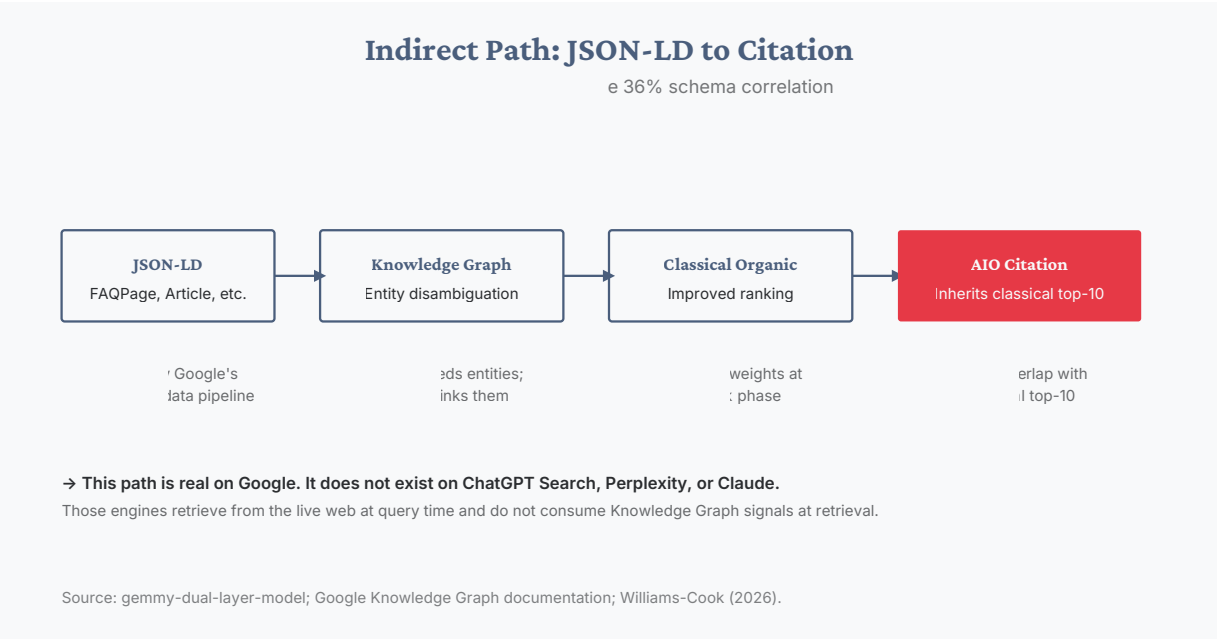
If you can draw the hybrid-search diagram from memory by the end of the week, you can interrogate any GEO pitch you'll be sent this quarter. If you can't, you'll keep being sold tools that solve a problem you never understood.

Chapter 5

The Dual-Layer Citation Model



05 dual layer model



05 knowledge graph flow

Williams-Cook Experiment

raw text, not as a structured graph

PAGE WITH JSON-LD

What the developer sees

```
<h2>What is the best CRM?</h2>
<p>A 50-person team should evaluate...</p>

<script type="application/ld+json">
{
  "@type": "FAQPage",
  "mainEntity": [{
    "@type": "Question",
    "name": "What is the best CRM?",
    "acceptedAnswer": {...}
  }]
}
</script>

<h2>How do I evaluate?</h2>
```

TOKEN STREAM

What the LLM actually receives

```
[What] [is] [the] [best] [CRM] [?]
[A] [50] [person] [team] [should]

[<] [script] [type] ["] [application]
[/] [ld] [+] [json] ["] [->] [{} []] ["] ["]
[type] ["] [:] ["] [FAQ] [Page] ["] [.]
["] [main] [Entity] ["] [:] [{} []] ["] ["]
["@] [type] ["] [:] ["] [Question] ["]
[,] ["] [name] ["] [:] ["] [What] [is]
[the] [best] [CRM] [?] ["] ["] ["]
[accepted] [Answer] ["] [:] [{} [...]]
[] [] [] [<] [/] [script] [>]

[How] [do] [I] [evaluate] [?]
```

nised

→ The structured-data block reaches the model as 50+ tokens of punctuation and property names.

It occupies chunking space without contributing semantically. The visible Q&A above it is what the engine extracts.

Source: Williams-Cook (2026); preprint, methodology echoed by independent vendors. See Chapter 5 and Appendix G.

05 tokenisation experiment

Schema does not earn citations. It correlates with them.

The marketing industry has spent two years selling that correlation as causation. The budgets have followed. Six- and seven-figure schema-implementation contracts. Agencies pivoted to “FAQPage strategy.” Vendor decks promised LLM-citation lift if you would only deepen the structured data — one more JSON-LD property, one more schema type, one more audit. The empirical observation underneath the pitches is real. Pages with deeper schema do tend to appear more often in AI answers. The causal explanation wrapped around it is wrong.

Here’s the thing. The 36% correlation is real. The story justifying the budget is wrong. This chapter ends that argument. With mechanism, with experiment, and with a ratio every practitioner running schema work should keep at hand. The mechanism is the Dual-Layer Citation Model. The experiment is Williams-Cook (2025, preprint). The ratio is 3.5 to 1 — authority signals over schema signals — drawn from vendor research that should be flagged for what it is and used for what it shows.

If your team is being pitched “schema for AI” and your boss is being asked to approve the line item, this is the chapter that gives you the data to push back.

What practitioners get wrong

The dominant pitch on every 2024–25 GEO deck went something like this. Implement Article, FAQPage, HowTo, Product schema across your site. Watch citations rise in ChatGPT, Perplexity, Google AI Overviews. The agencies pivoted overnight. The vendors wrote case studies. Dashboards started counting “schema coverage” as a leading indicator. By Q4 2025, deepen-the-schema sat on more marketing roadmaps than any other GEO tactic.

I sat through twenty of these pitches in the second half of 2025. The empirical observation underneath was real — pages with deeper schema do correlate with higher citation rates in LLM-generated answers. The figure most often quoted, a 36% higher likelihood of appearing in an AI answer, is well-corroborated across multiple measurement vendors and sits in the canon as Strong evidence. The correlation is not in dispute.

What is in dispute is the causal claim. The pitch tells a story. LLMs read the JSON-LD, parse it as a structured graph, weight the structured claims, and prefer pages with cleaner structured data. That story is what justifies the budget. It is also what Williams-Cook tested. The result, set out in the next section, is that LLMs do not parse JSON-LD as a graph at the citation phase. They tokenise it as raw text — the same way they tokenise the prose around it. The structured-data block appears in the model’s input as a sequence of tokens — punctuation-heavy, syntactically odd, informationally dense in ways that hurt extractability rather than help it.

This matters profoundly because the budget allocation that follows from the wrong story is large and wrong by an order of magnitude. Three failure modes are doing the most damage.

The first is depth-without-mechanism. Teams deepen optional schema fields across every page, hoping for citation lift, because the agency proposal said so. The optional fields don’t engage either layer of the citation model. The work consumes the budget that should have gone to the visible-text levers in Chapters 8 and 9.

The second is schema-only thinking. A FAQPage schema block ships without the corresponding visible Q&A on the page. The Knowledge Graph indirect path picks up the structured claim. The direct-layer extraction at retrieval time finds

nothing to chunk. ChatGPT and Perplexity render the page invisible — there is no body content shaped for the chunking mechanism. The schema looks tidy in Google Search Console and earns nothing in the citation register.

The third is authority-schema mismatch. Identical schema goes onto domains with very different authority profiles, and the team expects equivalent results. The 3.5 to 1 ratio describes what actually happens. Authority signals dominate the re-rank phase by a factor large enough that a low-authority domain with deepened schema still loses to a high-authority domain with mediocre schema. The marketing organisation has been investing in schema as if it could substitute for entity verification (Chapter 10) and co-citation (Chapter 11). It cannot.

The CFO-legible cost is the budget you are paying twice for. Schema work that should have been a hygiene maintenance line — closer to 20% of the GEO budget than 60% — has been funded as a citation-acquisition line. The work is not useless. The work is misallocated. Move the money.

How it actually works

Two distinct paths feed citations into LLM-generated answers. The paths run on different mechanics, take different inputs, and reward different work. Most practitioner advice confuses them — treating them as one process with one set of optimisations. They are not.

The indirect layer

The first path runs through Google's Knowledge Graph and the engines that inherit from it. Page-level JSON-LD is parsed by Google's structured-data pipeline at indexing time. The parsed claims feed Knowledge Graph entities — Organization, Person, Product, Event, the rest. The Knowledge Graph improves classical organic ranking through entity disambiguation and signal corroboration. AI Overviews and AI Mode retrieve heavily from classical organic. So: schema → Knowledge Graph → ranking → AI citations.

This path is real. It explains the 36% correlation that the industry keeps quoting. A page with cleanly structured schema feeds the Knowledge Graph more effectively. The page's classical organic position improves. The page becomes

more likely to appear in AI Overviews — which is one of the engines the dashboard is counting.

It is also indirect. The lift does not come from the LLM reading the schema. It comes from the page being more findable in the retrieval set the LLM draws from.

The implication for budget allocation is precise. Schema does work for Google AI Overviews and AI Mode — those engines inherit from the Google retrieval stack. Schema does not do equivalent work for ChatGPT Search, Perplexity, or Anthropic Claude — those engines retrieve from the live web at query time and do not consume Knowledge Graph signals at retrieval.

The direct layer

The second path runs through visible, on-page text. ChatGPT Search, Perplexity, Microsoft Copilot's grounding mechanism, and Anthropic Claude with web-search enabled — all retrieve pages from the live web at query time. They chunk the retrieved pages into segments — typically 150 to 300 tokens — and select fragments to compose the answer. The unit of value is the fragment, not the page.

This is the mechanism that has been mischaracterised most consistently. The pitch decks have suggested that schema markup gives the LLM a "structured signal" that helps it prefer the page. The Williams-Cook experiment tested exactly this claim.

Williams-Cook (2025, preprint) ran a controlled experiment. Identical pages were served with and without JSON-LD blocks. The pages were submitted to ChatGPT Search and Perplexity through a fixed prompt set. Token-level instrumentation captured what the engines actually saw at the retrieval stage. The result was unambiguous in one direction. The LLMs tokenised the JSON-LD as raw text. They did not parse it as a structured graph. The structured-data block appeared in the model's input as a sequence of tokens — curly braces, quote marks, property names, values — that occupied chunking space without contributing to the semantic embedding in any way distinct from how the engine treats any other dense punctuation-heavy text.

The evidence weight matters. Williams-Cook is a preprint. It has not yet been through full peer review. The methodology is solid, and the experiment has been independently echoed by at least two measurement vendors with corroborating data. Treat the finding as Moderate confidence with strong directional support — and re-test if the architecture changes (the open question at the end of this chapter).

The direct-layer implication is stark. The schema does not earn the citation in ChatGPT Search, Perplexity, or Claude. What earns the citation is the visible body text — the Q&A blocks, the dense statistics, the inline citations, the heading-led structure that survives chunking. Chapter 9 covers what those tactics look like in operation.

The 3.5 to 1 ratio

Even within the indirect layer — the path where schema does engage the mechanism — schema does not dominate. The Kitchen Remodeling SEO study (vendor data, flag inline) tested the relative contribution of schema deployment versus authority signals across a sample of competing pages, with citation rate as the outcome. The reported ratio was approximately 3.5 to 1. Authority signals outweighed schema signals by that factor in citation prediction.

Read the figure carefully. It is vendor-funded research. The source has a commercial interest in vendor SEO services. The methodology has not been independently replicated against a primary academic source. Downgrade it one weight; treat it as Moderate evidence with directional support, not a precise number.

The directional finding holds even at the lower bound. Authority always wins. The 3.5 to 1 ratio describes what every practitioner running A/B tests on schema has probably already observed — the page that wins citations is rarely the page with the deepest schema. It is the page with the strongest entity verification, the cleanest visible-text extractability, and the heaviest co-citation across third-party platforms.

A worked example

Let me paint you a picture. Two pages competing for the same query — how to choose a CRM for a 50-person team.

Page A is a 1,200-word article on a marketing-software vendor's site. The page has been "AI-optimised" by the vendor's agency. It carries deep FAQPage schema (twelve question-and-answer pairs in JSON-LD), Article schema with full author and publisher properties, aggregateRating schema. The body text reads as a flowing narrative — well-written by editorial standards, but with no clear Q&A structure, no headings that match the questions in the JSON-LD, no dense pull quotes.

Page B is a 1,800-word article on an independent industry publication. The page carries lighter schema — Article with the basics, plus Organization at the site level and sameAs linking to the publication's Wikidata entry. The body text is structured as a series of explicit questions in H3 headings (each one matching a buyer's actual phrasing from sales calls), each followed by a 200-word answer in dense, declarative prose, with one or two pull quotes and three inline statistics linked to primary sources.

In a Google AI Overview retrieval, both pages will be considered. Page A may appear in the Knowledge Graph signal that feeds the indirect path. Page B may not. The classical organic ranking, however, is shaped overwhelmingly by Page B's authority and entity profile — not by Page A's schema depth. In ChatGPT Search and Perplexity, Page A is largely invisible. The chunking mechanism finds no clear question-answer segments to extract. The JSON-LD block tokens compete with the prose for the model's attention without contributing to it. Page B is extracted cleanly. Two or three of its H3-led answer paragraphs appear as citation fragments in the synthesised answer.

The schema deepening Page A's vendor invested in did not earn the citation. The visible-text discipline Page B's editor invested in did. The Dual-Layer Citation Model, applied to a real query, produces an outcome the schema-only pitch cannot explain.

What schema actually does, in summary

What schema does:

- Feeds the Knowledge Graph (still useful for entity disambiguation — Chapter 10).
- Improves classical organic ranking (still useful for the indirect path on Google).

- Verifies entity claims via the sameAs property, linking owned identities to authoritative external nodes (Chapter 10).
- Provides extraction hints for some specialised retrievals — Recipe, Product, Event, JobPosting, HowTo — where engines extract structurally from the schema rather than from prose.

What schema does not do:

- Cause LLM citations directly in retrieval-and-synthesis engines that don't consume Knowledge Graph signals.
- Override the dominance of visible-text extractability at the chunking phase.
- Replace authority signals at the re-rank phase.

Diagram 05-dual-layer-model.svg carries the central argument visually. Screenshot it. Forward it the next time the conversation comes up.

Common questions you'll get

When you take the audit and the A/B test result upstairs, four questions will come back. Have answers ready.

"But Google's own documentation says schema is important. Are you saying ignore it?" Google is correct on the indirect path. Schema feeds the Knowledge Graph. The Knowledge Graph improves classical organic ranking. Classical organic feeds AI Overviews. The chapter does not say ignore schema — it says deploy the five groups in the audit, audit them quarterly as hygiene, and stop funding deepening as a citation-acquisition lever. Same posture, different budget line.

"Our FAQPage schema is driving rich snippets in classical SERPs. Doesn't that mean it's also driving AI citations?" Rich snippets in classical SERPs are a pre-AI artefact. The mechanism is real, the visibility is real, the click-through-rate effect is real. None of it translates directly to LLM citation rate — because the LLM does not consume the FAQPage schema as structured data. It consumes the visible Q&A text. The classical SERP win and the AI citation are different outcomes from different mechanisms. Pair them. Do not conflate them.

“What about the agency that says they have proprietary AI-citation tracking showing schema lift?” Ask three questions. What is the methodology? What is the sample size? Has the result been independently replicated? Most “proprietary tracking” is a single vendor’s measurement, run on their own client base, with no isolation between schema deployment and concurrent content or entity work. Apply the vendor matrix in Appendix D and downgrade the claim if it fails the conflict-of-interest column.

“If we cut schema budget, what happens to our Knowledge Panel and entity coverage?” The five deployment groups in the audit cover the work that protects Knowledge Panel and entity coverage — Organization, Person, sameAs, FAQPage on Q&A blocks, Article on long-form, BreadcrumbList and WebSite at site level, plus the specialised retrieval types where they apply. The cut is on the optional fields, the vanity schema, and the deepening exercises that don’t engage the mechanism. Knowledge Panel is unaffected — the schema that earns it is in the kept tier.

What to build and publish

A schema deployment audit. Not a schema-property deepening exercise. The defensible deployments — the ones the audit should keep, not cut — fall into five groups.

FAQPage on Q&A blocks. The 50–300-word Q&A block is the unit that engages both layers cleanly. The FAQPage schema feeds the indirect path. The visible Q&A text aligns with the chunking mechanism on the direct path. Pair the two. Never deploy the schema without the visible content beneath it. Chapter 9 covers the production discipline for the Q&A pages themselves.

Article and NewsArticle on long-form pages. Knowledge Graph signal for the indirect path. Supports authorship attribution to a Person entity, which feeds the entity verification work in Chapter 10. Mandatory for anything carrying a byline.

Organization, Person, and sameAs at the site level. The single most under-deployed schema combination in marketing organisations. Foundational for entity disambiguation across the open data graph. Embed once on the homepage and the about page. Audit quarterly. Appendix I.11 has the working JSON-LD template.

Specialised retrieval types where engines extract structurally. Recipe, Product, Event, HowTo, JobPosting. Google AI Overviews and Bing pull from these directly — the structured properties become the extracted answer, often with no prose required. Not optional for relevant verticals. A retail brand without Product schema, an events business without Event schema, a careers page without JobPosting schema — invisible to the corresponding specialised retrievals.

BreadcrumbList and WebSite at the site level. Site-level structural signals. Hygiene work. Ship once and audit quarterly.

What not to build, and what the audit should cut: deepening every optional schema field on every page hoping for citation lift. Filling out aggregateRating without the corresponding review schema. Deploying speakable schema in markets and content types where it doesn't apply. Adding video schema to pages that don't host video. Generating product schema fields with placeholder data because the vendor's audit tool flagged them as missing. This is where the 2024–25 budget went wrong.

Appendix I (Schema and Datasets Cookbook) carries the type-by-type reference, working JSON-LD examples for each, and the diagnostic checklist for verifying a working deployment in Google's Rich Results Test.

Monday-morning actions

Action 1 — Audit your current schema deployment against the Dual-Layer Citation Model. Owner: Senior Manager SEO/GEO. Done = an audit document scoring each priority page on dual-layer relevance — which schema types are deployed, which mechanism each engages (indirect / direct / both / neither), where the deployment is wasted. Validated in Google's Rich Results Test for every page audited. Failure mode = the audit measures schema depth (more fields = better) instead of mechanism alignment, so the recommendation reads as “deepen everything.”

Action 2 — Build the A/B test on a high-priority page. Same content, with vs without deepened schema. Track citation rate against the prompt battery (Appendix B). Owner: Senior Manager SEO/GEO. Done = test live with version control on both variants; result captured at 30 days with confidence interval; documented in the change log. Failure mode = the test runs without isolating the schema change from concurrent content edits, so the result can't survive scrutiny at the budget review.

Action 3 — Use the Dual-Layer one-page summary as your standing brief when a “schema for AI” vendor pitch arrives. Owner: practitioner. Done = the summary is in your inbox and on the team wiki; you forward it to your manager before the pitch is taken, with the audit and the A/B result attached. Failure mode = the summary exists but the next vendor still sells your manager on schema-for-citation lift, because you didn't surface the brief in time.

What to take to your manager

The audit document and the A/B test result. The vendor rejection is theirs. The budget reallocation is theirs. Your job is to give them the data that makes both decisions defensible in a five-minute window with the CFO.

The shape of the budget conversation that follows: schema work moves from a citation-acquisition line to a hygiene maintenance line. The savings — typically 30 to 50 per cent of the existing schema budget — get redirected to the direct-layer levers in Chapters 8 (multimodal), 9 (extractability), and 10 (Entity Bible). Most marketing organisations find the redirected money buys disproportionately more citation share. The levers it funds engage the mechanism the schema work didn't.

Activate this. francescofederico.com/cited/schema-pushback hosts the live one-page summary. Print it, forward it, attach it. Expect to use it three times in 2026.

How to know it landed

The metric is citation-rate change per direct-layer lever, isolated from schema work. The discipline is the prompt battery (Appendix B). The cadence is weekly.

Two false positives to watch for. The first is improvements attributed to schema that were actually driven by entity work (Chapter 10) or content tactics (Chapter 9) running in parallel. This is why isolation matters in the A/B test design — version-controlled variants, with only the schema change varying, run for at least 30 days before the result is read. The second is the Rich Results Test passing without the engine actually re-crawling the page. A green tick in Google's tool is necessary for indirect-layer validation. It does not confirm that AI Overviews has updated its grounding for the page. Wait for the change to appear in the citation register before declaring the result.

The vendor matrix in Appendix D names which measurement tools can isolate the dual layers. Most cannot. The DIY prompt battery, run weekly against a fixed set of priority prompts, is the most reliable measurement instrument available to a practitioner — and the cheapest. A vendor SaaS that reports a single “schema effectiveness” number across engines is selling a metric that does not exist in the architecture.

Cross-reference Chapter 12 for the broader measurement discipline. The first weekly run is the baseline. Everything after is signal.

What we don't yet know

Whether the Dual-Layer Citation Model holds as engines integrate native structured-data parsing into RAG pipelines.

The Williams-Cook experiment captured the architecture as of late 2025. If Google or OpenAI ship native JSON-LD parsing into the citation pathway — both have signalled it in roadmaps, neither has publicly committed — the direct layer changes shape. Schema would begin to engage the mechanism it has been credited with for two years.

The corresponding indicators to watch: announcements from OpenAI of structured-data extraction in the ChatGPT Search retrieval stack; Google patents that describe schema parsing at the AI Overviews grounding phase; any

independent replication of Williams-Cook that reports a different result on a current model release.

Annual re-test is the discipline. Action 1 above re-runs every twelve months against the prompt battery. If the result moves materially, the chapter's argument needs updating. Appendix H tracks the resolution of this and the four other open questions named in Chapter 15.

Until then, plan the budget against the architecture as it is, not as it might be. The vendor pitches that sell against the future architecture are pitching speculation as fact.

If your schema deployment looks the same in Q4 as it does today, you've spent the budget on the wrong layer. Run the A/B test before the next quarterly review and bring the result.

Chapter 6

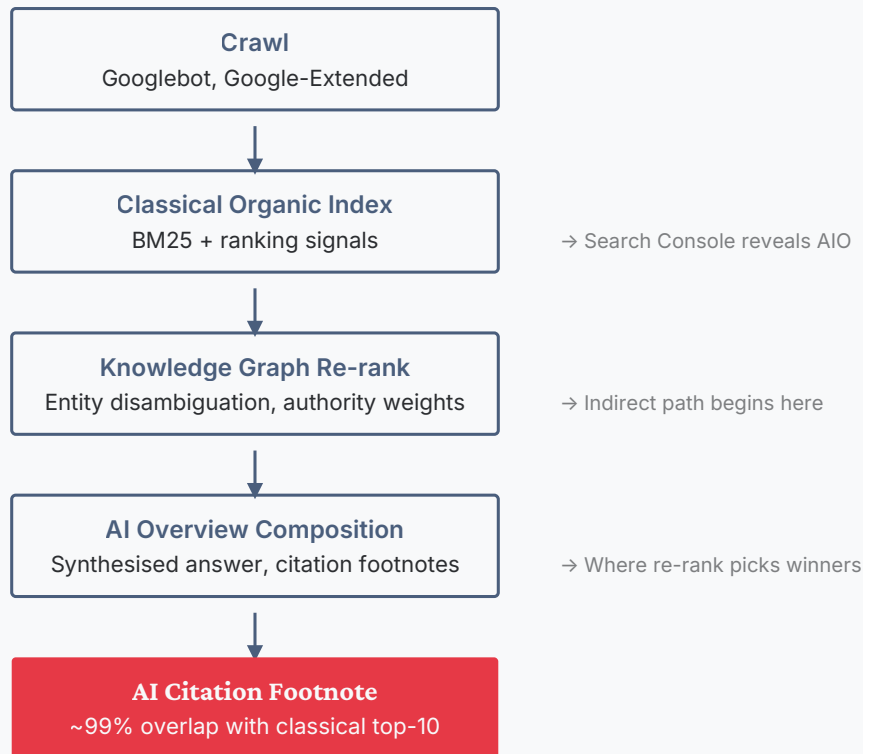
Six Engines, Six Strategies

Six Engines, Six Strategies				
ent, transparency — per engine				
ENGINE	RETRIEVAL STACK	CITATION LOGIC	PAID	TRANSPARENCY
Google AIO + AI Mode	Classical organic + Knowledge Graph	99% top-10 overlap Authority-weighted	Aggressive	Search Console
ChatGPT Search	OAI-SearchBot + Bing fallback	Visible-text preference Q&A extraction bias	Disclosed CPM	UTM defaults
Perplexity	PerplexityBot RAG-native	Quotation rewards (+24%) Recency bias, extractive	Sponsored Qs	Visible citations
Microsoft Copilot	Bing-dependent Bingbot crawl	Inherits Bing logic Mature ranking	MS Advertising	Bing Webmaster
Anthropic Claude	Tool-use, selective retrieval	Cautious; primary sources preferred	No ads	Sparse data
Long tail Gemini, Meta AI, Grok	Mixed; sparse documentation	Lower volume Audience-specific	Each different	Sparse

Source: chatty-engine-briefs; synthesis Prompt 1; Bing Webmaster Tools documentation; OpenAI Shopping rate disclosures.

Google AI Overviews

d citation pipeline

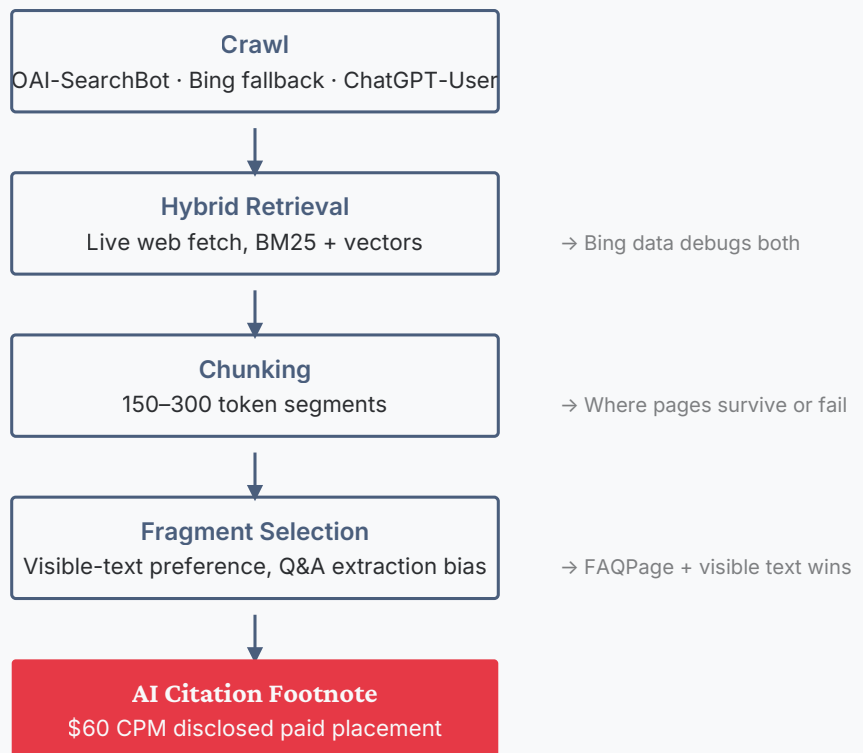


Source: chatty-engine-briefs; Google Search Console documentation.

06 google aio stack

ChatGPT Search

d citation pipeline

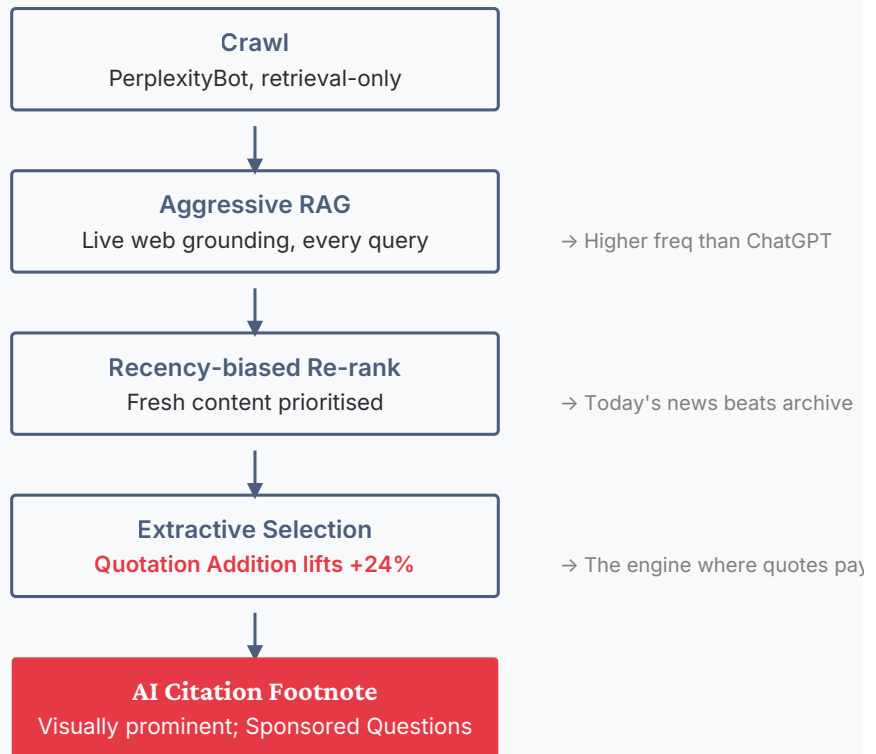


Source: chatty-engine-briefs; OpenAI Shopping rate disclosures; synthesis Prompt 1.

06 chatgpt search stack

Perplexity

d citation pipeline

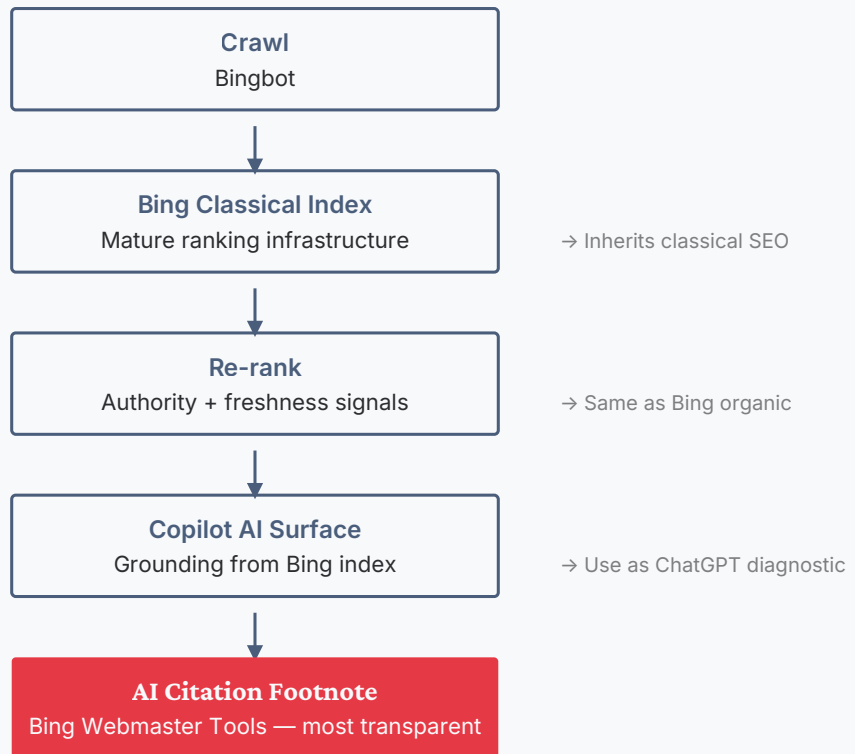


Source: chatty-engine-briefs; synthesis Prompt 3 replication study.

06 perplexity stack

Microsoft Copilot

d citation pipeline

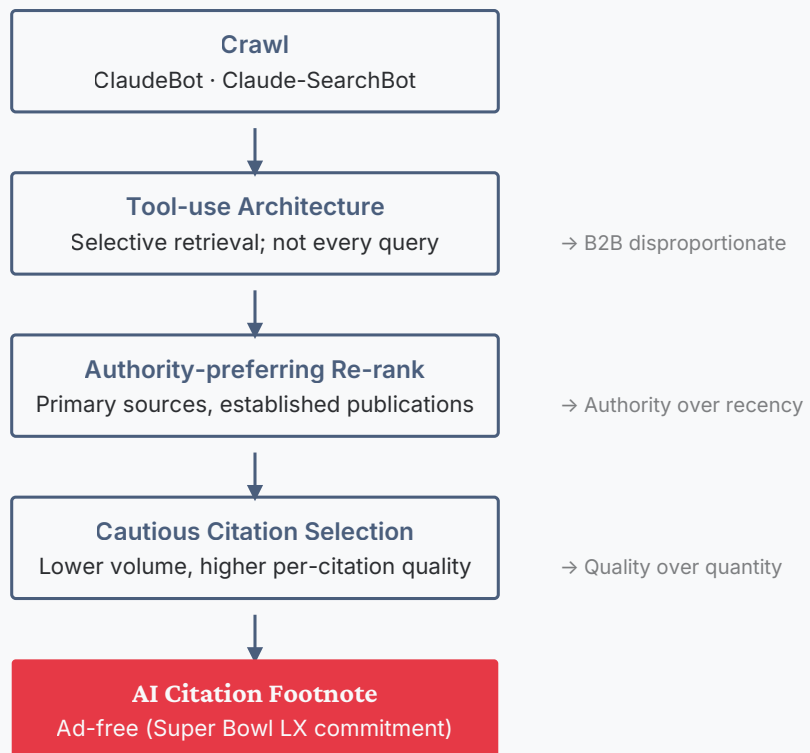


Source: Bing Webmaster Tools documentation; chatty-engine-briefs.

06 copilot stack

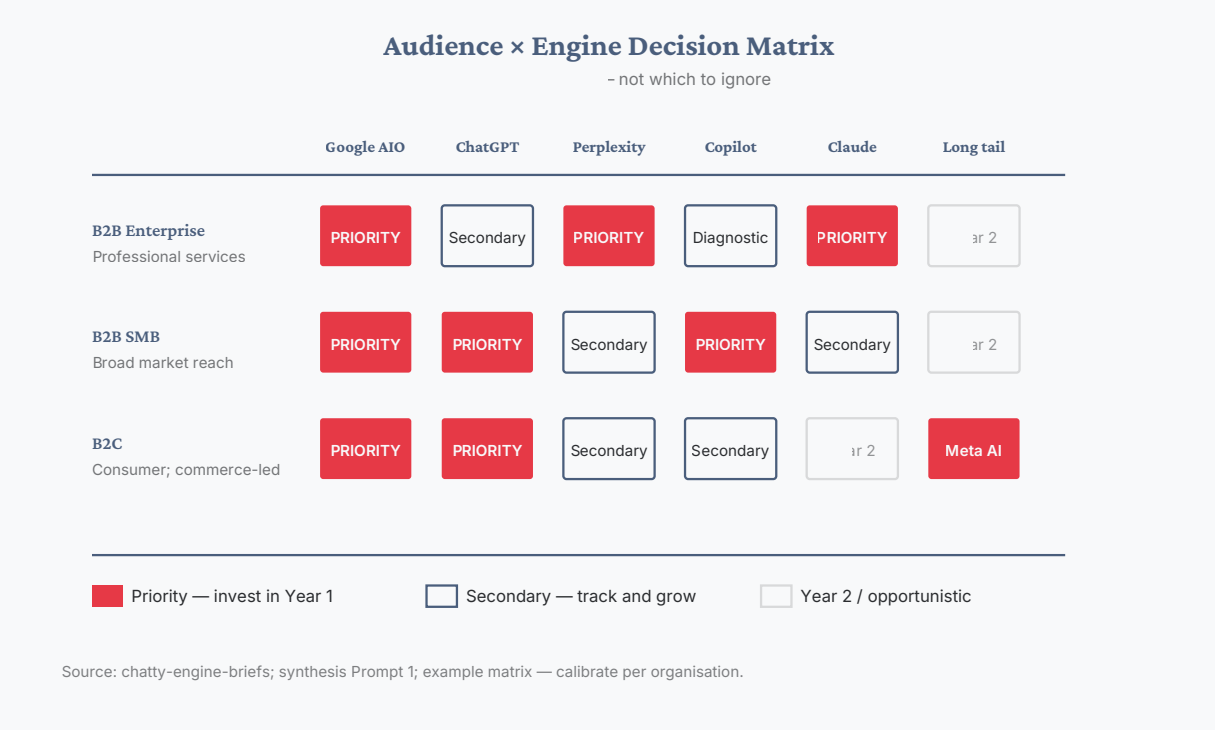
Anthropic Claude

d citation pipeline



Source: chatty-engine-briefs; Anthropic Super Bowl LX public commitment.

06 claude stack



06 decision matrix

There is no universal AI strategy.

The engine that pays your bills next year is not the same engine that paid them last year, and the tactics that won on one will fail on another. Six engines, six retrieval stacks, six citation logics. A senior marketer who does not know which is which will be sold the same playbook for all of them — and will pay for the package.

This chapter is the field guide. Each of the six engines gets its own brief in identical structure: how it retrieves, what it cites, what controls you have, what its paid posture looks like, what to do, what to skip. The structure is identical so you can compare across them quickly. The tactics inside the structure are not. That is the whole point.

What practitioners get wrong

"AI search strategy" treated as one thing. Vendors who promise cross-engine reporting that turns out to be one prompt fired six times. The most common error: applying ChatGPT-Search content tactics to Google AIO and getting nothing in return. The second: assuming Perplexity behaviour generalises. It doesn't — Chapter 9 covers the Quotation Addition replication failure outside

Perplexity. The third: paying for an enterprise SaaS that fires the same prompt at six engines and reports a single number.

I have evaluated four “cross-engine GEO platforms” in 2026. None of them named distinct retrieval stacks per engine in their pitch deck. All of them charged enterprise rates. The teams that survived the experience with their budgets intact were the ones that scored the vendor against Appendix D’s eight-column rubric before signing.

The cost of the misframe is paid in two places. The wrong tactic gets funded across all six engines uniformly, so half the budget is wasted on engines where the tactic doesn’t replicate. And the cross-engine SaaS subscription buys a unified dashboard that hides the per-engine variance the team needs to see.

How it actually works

Six briefs. Identical structure. Read the engine that matters most for your top audience first; the others follow the same shape.

6.1 Google AI Overviews and AI Mode

Retrieval stack. Inherits from the Google retrieval stack — Googlebot crawls, classical organic ranking sets the candidate set, Knowledge Graph signals enter heavily at re-rank. AI Mode (the 2026 default) and AI Overviews (the 2024 surface) share most of the underlying mechanism with adjustments to the answer composition layer.

Citation logic. High but unstable overlap with the classical top-10 — vendor measurements through 2025 sat in the 70–90% range; the Gemini 3 rollout dropped the figure into the 30–50% range for a multi-week window before it stabilised (synthesis Prompt 3 replication; vendor data, directional). If you rank top-3 for the priority keyword, you almost certainly enter the AIO candidate set, but engine-version drift can change the cited subset materially. Re-ranking shapes which of the top-10 actually gets cited based on freshness, authority, and entity verification. The page that wins the citation is rarely the page that ranks #1; it is usually one of the top five with the cleanest extractable fragments.

Controls. robots.txt directives for `Googlebot` (classical crawler) and `Google-Extended` (training opt-out). The AI Overviews report inside Search Console exposes which queries triggered AIO and which sources were cited.

Paid placement. AIO ad integration is rolling out through 2026. Shopping ads embed inside AI answers for commerce queries. Sponsored sources appear adjacent to citations in some categories. Aggressive monetisation; the inventory is the largest of any AI surface.

Monetisation posture. Aggressive, integrated. Google's revenue model depends on AIO not undermining classical ad revenue, so paid placement is being engineered into the surface deliberately. Plan for paid AIO inventory expansion through 2027.

What to do. Continue classical SEO discipline (Chapter 2 continuity map). Deepen entity verification on the brand (Chapter 10). Ensure the priority pages structure for extractability (Chapter 9). Use the AIO report in Search Console as the diagnostic.

What to skip. Schema-deepening as a citation-acquisition lever (Chapter 5). Cross-engine SaaS that doesn't expose AIO-specific data.

Evidence weight: Strong.

6.2 ChatGPT Search

Retrieval stack. OAI-SearchBot fetches pages at query time. Bing fallback for queries the model can't ground in OAI's own retrieval. ChatGPT-User fetches pages on individual user actions — a user pasting a URL into the chat.

Citation logic. Visible-text preference, Q&A extraction bias, recency weighting. The chunking mechanism (Chapter 4) drives which fragments get cited. Pages with structured Q&A blocks consistently outperform narrative pages of equivalent depth.

Controls. Three distinct robots.txt directives: `GPTBot` (training crawler — block to opt out of training), `OAI-SearchBot` (retrieval crawler — block to opt out of citation), `ChatGPT-User` (user-fetched — block to prevent on-demand fetches). Most marketing teams confuse these and block the wrong one. Reference Chapter 7.

Paid placement. The only engine with publicly disclosed paid rates (Strong with caveat — OpenAI-disclosed). Sponsored answers carry a \$60 CPM base, \$200K minimums, \$3–5 effective CPC. OpenAI Shopping integrates inside answer surfaces. Default UTM tagging — every ChatGPT Search outbound link carries a `utm_source=chatgpt.com` parameter. The only engine that does it.

Monetisation posture. Active and disclosed. OpenAI's revenue pressure makes paid placement strategically important; expect inventory expansion.

What to do. Build the FAQ pages with FAQPage schema (Chapter 9). Ship the video FAQ programme (Chapter 8) — ChatGPT Search retrieves YouTube content directly. Pilot paid placement at the entry tier before scaling.

What to skip. Pure schema-deepening (Chapter 5). Long-form narrative pages without H3-led structure.

Evidence weight: Strong.

6.3 Perplexity

Retrieval stack. Aggressive RAG, recency bias, structured-content preference. PerplexityBot crawls the live web; the model retrieves and grounds heavily on fresh content. Citations are visually prominent — Perplexity has always led with citations and shows no sign of moving away.

Citation logic. Extractive, quote-friendly. The Aggarwal Quotation Addition tactic replicates here at +24% (partially confirmed against multiple replication sources) — this is the only engine where Quotation Addition replicates at meaningful scale. Outside Perplexity, the tactic produces +4% to +16% lifts with confidence intervals often crossing zero.

Controls. robots.txt directive for PerplexityBot . No separate training-vs-retrieval distinction — Perplexity is retrieval-only.

Paid placement. Sponsored Questions (paid placement at the question level — the brand pays to have a particular question surfaced as a starting point). Partner programmes for verified information sources. Affiliate links inside answers for commerce queries.

Monetisation posture. Emerging. Perplexity's funding rounds have created revenue pressure; paid placement is being engineered in deliberately. Less mature than ChatGPT Search.

What to do. This is the one engine where quotation density actually pays. Deploy Quotation Addition deliberately on Perplexity-priority pages. Build co-citation work on the platforms Perplexity grounds heavily — Reddit, Stack Exchange, Wikipedia.

What to skip. Treating Perplexity-specific lifts as cross-engine evidence. The Quotation lift here does not generalise.

Evidence weight: Moderate to Strong (Quotation lift partially confirmed; engine architecture well-documented).

6.4 Microsoft Copilot

Retrieval stack. Bing-dependent. Bingbot crawls; Bing Webmaster Tools exposes the retrieval data. Copilot's grounding mechanism inherits from Bing's classical and AI-search infrastructure.

Citation logic. Most transparent native platform of any engine in the field. Bing Webmaster Tools exposes cited pages and grounding queries directly (Strong, platform documentation). The diagnostic value is high — use Copilot data to debug ChatGPT Search behaviour, since both inherit from the Bing retrieval stack.

Controls. robots.txt for Bingbot . Microsoft Advertising integration exposes paid inventory access.

Paid placement. Microsoft Advertising integration. Copilot for retail surfaces product ads. Inventory is mature but smaller than Google's because Copilot's reach is smaller.

Monetisation posture. Mature on the paid side; subscription model on the consumer side (Microsoft 365 bundling). Less revenue pressure than OpenAI or Perplexity.

What to do. Use Bing Webmaster Tools as the primary diagnostic for any engine that inherits from Bing — that includes ChatGPT Search via the Bing fallback. The transparency advantage is real and underused.

What to skip. Ignoring Bing as “the small engine.” Copilot’s transparency makes it the most useful diagnostic surface in the field, regardless of its market share.

Evidence weight: Strong (platform documentation).

6.5 Anthropic Claude

Retrieval stack. Tool-use architecture. Claude does not retrieve by default; it retrieves when web-search is enabled by the user or by the system prompt, and only when it judges retrieval necessary for the answer. Lower query frequency than ChatGPT Search or Perplexity. ClaudeBot is the training crawler; Claude-SearchBot is the retrieval crawler.

Citation logic. Cautious. Prefers authoritative sources, primary research, established publications. Lower citation count per answer than other engines, but higher per-citation quality. B2B and professional-services queries see disproportionate Claude usage; consumer commerce queries see less.

Controls. Two distinct robots.txt directives: `ClaudeBot` (training) and `Claude-SearchBot` (retrieval). The same training-vs-retrieval distinction as OpenAI; the same opt-out logic.

Paid placement. Contractually locked NO ADS. Anthropic publicly committed at Super Bowl LX with a campaign that named the position explicitly: “Ads are coming to AI. But not to Claude.” (Strong, public commitment.) The reputational lock-in is high; reversing it would cost the brand credibility.

Monetisation posture. Subscription-only (Claude Pro, Claude for Enterprise). Anthropic’s revenue model depends on ad-free positioning. Plan for the no-ads commitment to hold through 2027 at minimum.

What to do. Subscription-only networks deserve disproportionate content investment. If Claude’s no-ads commitment holds, every citation Claude produces is unfiltered by paid placement — pure organic surface. For B2B and professional-services brands, this is the highest-quality citation surface in the field.

What to skip. Treating Claude as equivalent to ChatGPT Search or Perplexity. Different query mix, different citation mechanics, different audience composition.

Evidence weight: Strong (engine architecture); Moderate (citation share data, smaller sample).

6.6 The long tail — Gemini, Meta AI, Grok

Retrieval stack. Sparse documentation across all three. Gemini inherits from Google; Meta AI from Meta's own retrieval; Grok from X's retrieval plus web search.

Citation logic. Lower citation volume than the top four engines. Different enough to matter for specific audiences (Gemini for Google Workspace users; Meta AI for consumer social commerce; Grok for technology-news queries). Sparse enough to deprioritise in Year 1 unless one of those audiences is your priority.

Controls. Meta-ExternalAgent ignores robots.txt and accounts for approximately 19% of AI crawler traffic (Moderate, vendor-corroborated — flag inline). This is the documented exception to standard crawler-control mechanics. Reference Chapter 7.

Paid placement. Ads inside Gemini surfaces follow Google Ads conventions. Meta AI integrates with Meta's existing paid surfaces (Reels, Instagram, WhatsApp). Grok uses X Ads.

What to do. Track citation share quarterly to detect if any of the three is becoming material for your audience. Reach the threshold (5% of your priority-prompt citation set), then deepen tactics.

What to skip. Year-1 deep investment unless priority audience demands it. The top four engines absorb most of the GEO budget for most teams.

Evidence weight: Moderate to Anecdotal (less documentation; vendor data sparse).

6.7 Decision matrix

Audience × engine fit. The matrix the team should build for itself, not adopt from a vendor.

For each priority audience, score each of the six engines on three dimensions: how heavily the audience uses it, how well-suited the brand is to its citation logic, how transparent the measurement surface is. The engines that score highest on all three are the priority three for that audience. Most B2B brands prioritise Google AIO + ChatGPT Search + Perplexity + Copilot; add Claude for enterprise and professional services. Most B2C brands shift the mix toward Meta AI for social commerce or Grok for technology-news.

The matrix is a one-page document. Build it. Bring it to the budget conversation.

Common questions you'll get

"Why can't we just use a cross-engine SaaS that handles all six?" Because the SaaS hides the variance you need to see. A unified dashboard reporting "AI citation rate: 12%" across six engines is one number averaged from six different mechanisms. The team can't act on the average. The Director responsible for budget can't defend the average. The DIY prompt battery (Appendix B), run weekly, exposes the per-engine variance the SaaS averages away.

"Aren't all six engines converging on the same architecture?" Some convergence at the chunking and re-ranking layers. Significant divergence at the retrieval, citation, and monetisation layers. The convergence at the technical layer doesn't translate to convergence at the practitioner layer; the controls and the paid placement and the citation logic remain distinct enough that engine-specific tactics matter.

"Which engine should I start with?" The one your top three priority prompts get cited on most. If you don't know yet, start with Google AIO (largest reach, most transparent via Search Console) and ChatGPT Search (most engaged user base, public CPM rates). Run the prompt battery for two weeks. Pivot based on evidence.

Monday-morning actions

Action 1 — Draft the audience × engine fit matrix for your top three audiences.
Source from sales conversations and customer interviews. Owner: Senior Manager SEO/GEO + Senior Manager Customer Insights. Done = a matrix with rationale per cell — which engine matters for which audience, with at least one quote or data point per choice. Stored on the team wiki, dated. Failure mode = the matrix is built from intuition or vendor data; the rationale doesn't survive challenge from the Director who has to defend the budget.

Action 2 — Set the citation baseline on the three engines you propose to prioritise. Use Bing Webmaster Tools where available, manual prompt sweeps elsewhere. Owner: Senior Manager SEO/GEO. Done = baseline in the dashboard within 30 days; methodology documented; per-engine citation count captured against the priority prompt set. Failure mode = the baseline is run once and never refreshed; later changes can't be attributed.

Action 3 — Score the current vendor stack against Appendix D — column by column, including the “distinct retrieval stacks per engine” test. Owner: Senior Manager Marketing Ops. Done = scoring sheet with a recommendation (keep / cancel / renegotiate) per vendor, ready for the next procurement review. Failure mode = the scoring is a paper exercise; nothing is recommended for cancellation because nobody wants to take the conversation.

What to take to your manager

The matrix, the baseline, and the scoring sheet. The “pick three” decision is theirs. The cross-engine SaaS rejection is theirs. You bring the data that makes both calls defensible.

The shape of the upstairs conversation. The matrix names the priority engines per audience with evidence. The baseline names where the brand stands today on each priority engine. The scoring sheet names which vendors are worth keeping and which are paid theatre. With those three documents, the budget conversation becomes a sequence of evidenced decisions rather than a vendor-led negotiation.

How to know it landed

Per-engine citation rate, baselined and tracked monthly. Bing Webmaster Tools as the diagnostic for the Microsoft / OpenAI / inheriting engines. The DIY prompt battery (Appendix B) for the rest.

The false positive: a single engine's win read as a cross-engine win. If AI Mode citation rates rise 30% and ChatGPT Search rates stay flat, that is a real outcome — not a measurement error. Track the variance, not the average.

The harder discipline: tracking the competitive citation map per engine. The brand cited in your priority prompts on Google AIO is rarely the same brand cited in your priority prompts on Perplexity. Two different competitive sets, two different strategies, sometimes one budget conversation. Quarterly review of the per-engine maps is the cadence.

What we don't yet know

Whether Anthropic's no-ads commitment survives a meaningful revenue gap.

The Super Bowl LX commitment is real, dated, and contractually locked at the brand level. If Anthropic faces a revenue cliff that subscription growth alone cannot close, the position becomes harder to defend. Watch for any walking-back of the public position; flag in Chapter 15.

Whether Google's AIO mechanics change materially under CMA conduct requirements (mid-2026). Whether Meta's crawler posture aligns with the rest after publisher pressure. Whether the long-tail engines (Gemini, Meta AI, Grok) consolidate into the top tier or fade. All flagged for Appendix H quarterly review.

The “AI search strategy” deck on your team’s hard drive is probably wrong on at least three engines. Find out which before next quarter’s plan locks.

Chapter 7

Your Crawler Portfolio (and the Cloudflare Reality)

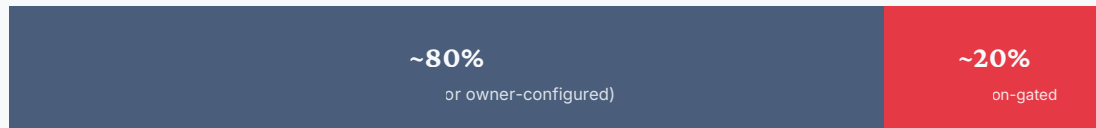
The Eleven Bots That Matter		
ad — three different decisions		
TRAINING CRAWLERS	RETRIEVAL CRAWLERS	USER-FETCHED
Block to opt out of model training	Block to opt out of citation	Triggered by user actions
GPTBot OpenAI training ClaudeBot Anthropic training Google-Extended Google training opt-out CCBot Common Crawl corpus Meta-ExternalAgent ignores robots.txt — flag ~19% of AI crawler traffic	OAI-SearchBot ChatGPT Search Claude-SearchBot Claude with web search PerplexityBot Perplexity Bingbot Copilot, ChatGPT fallback Googlebot AI Overviews, AI Mode Applebot-Extended Apple Intelligence	ChatGPT-User User pastes URL into chat Claude-User User pastes URL into Claude Blocking these prevents on-demand fetches by your own users — rarely what the brand wants.
→ Block GPTBot and OpenAI cannot train on your content. OAI-SearchBot can still fetch you for ChatGPT Search.		
→ Different decisions, different consequences. Don't conflate them.		
Source: chatty-eleven-bots-table; verified user agents per platform documentation.		

07 crawler taxonomy

Cloudflare Default-Block

sent-tense fact, not a future risk

THE PUBLIC WEB



→ If your domain is on Cloudflare, the audit is non-optional.

New domains default to blocking AI crawlers unless the owner opts in. Pay-Per-Crawl available for monetised access.

→ Marketing teams that haven't audited Cloudflare in 2026 may be blocking AI crawlers without knowing it.

The team thinks GEO is failing for content reasons. The actual reason is infrastructure.

The 20% figure is structural inference from Cloudflare's market share — not a Cloudflare-published statistic.

Source: Cloudflare Content Independence Day announcement (T1 verified); chatty-crawler-portfolio.

07 cloudflare default block

Blocking GPTBot does not stop ChatGPT Search.

Eleven bots matter. Most CMOs can name three. As of 1 July 2025, twenty per cent of the public web is permission-gated by Cloudflare default-block (Cloudflare announcement, T1 verified; the 20% figure is structural inference from Cloudflare's market share, not a published statistic — flag inline).

Your competitor's content is missing from retrieval right now if they are on the wrong list. Yours might be too. Crawler control is portfolio management, not a binary switch.

What practitioners get wrong

Crawler control treated as binary. "Block AI or allow it." The decision gets made at a leadership offsite, ratified in robots.txt, then never revisited.

Three failure modes follow.

The first is the wrong-bot block. The team blocks GPTBot to opt out of OpenAI training, then notices six months later that their pages have stopped appearing in ChatGPT Search citations. The block worked; it just blocked the wrong thing. ChatGPT Search uses OAI-SearchBot, a separate crawler that the robots.txt didn't address. Training was opted out; retrieval was opted out by accident.

The second is the WAF mismatch. The marketing team updates robots.txt; the IT team owns the WAF rules; the two updates don't sync. Production behaviour differs from the documented policy. The audit happens at the file level and never at the network level. The crawler hits the WAF instead of the file and returns 403 before reading the policy at all.

The third is the Cloudflare blind spot. The 1 July 2025 default-block changed the baseline for new domains on Cloudflare's infrastructure. Approximately 20% of the public web — a structural inference from market share, not a Cloudflare-published figure — sits behind that default. Marketing teams that haven't audited their Cloudflare configuration in 2026 may be blocking AI crawlers without knowing it. The team thinks GEO is failing for content reasons. The actual reason is infrastructure.

The CFO-legible cost is invisibility on engines the brand thought it was visible to. Detected only by manual prompt sweeps; never visible on the dashboard the team built.

I have audited two B2B brands in 2026 whose Cloudflare configurations were silently blocking OAI-SearchBot. Both teams thought their GEO programmes were under-performing for content reasons. Neither had checked the WAF in eighteen months. Both fixes were one IT ticket each.

How it actually works

The eleven bots that matter. Grouped by purpose.

Training crawlers. GPTBot (OpenAI), ClaudeBot (Anthropic), Google-Extended (Google's training opt-out), CCBot (Common Crawl, the foundational training corpus), Meta-ExternalAgent (Meta — note: ignores robots.txt and accounts for approximately 19% of AI crawler traffic; vendor data, flag inline). Block these to opt out of training.

Retrieval crawlers. OAI-SearchBot (ChatGPT Search), Claude-SearchBot (Anthropic Claude with web-search enabled), PerplexityBot (Perplexity), Bingbot (Microsoft Copilot, AIO via Bing fallback), Googlebot (Google AI Overviews), Applebot-Extended (Apple Intelligence). Block these to opt out of citation. These are the bots that decide whether you appear in AI answers.

User-fetched. ChatGPT-User (triggered when a user pastes a URL into ChatGPT), Claude-User (same for Claude). Different robots.txt implications — blocking these prevents on-demand fetches by users, which is rarely what the brand wants.

The training-vs-retrieval distinction is the chapter's core insight. Block GPTBot and OpenAI cannot train on your content. OAI-SearchBot can still fetch it for ChatGPT Search responses. Different decisions, different consequences. Don't conflate them.

The Cloudflare reality. Default-block as of 1 July 2025 (Cloudflare Content Independence Day announcement, T1 verified, official). New domains on Cloudflare's infrastructure default to blocking AI crawlers unless the owner explicitly opts in. Pay-Per-Crawl programme for compensated access — the brand can monetise crawl rather than block or allow free. The 20% figure (structural inference, not Cloudflare-published) describes the share of the public web that sits behind Cloudflare. The default-block applies to that 20%. If your domain is on Cloudflare, the audit is non-optional.

ai.txt and llms.txt: real adoption, unproven citation lever. Build them as version-controlled artefacts. Don't over-invest until evidence improves. But don't skip them either — the cost is low, the optionality is real, and the artefacts are the brand-controlled metadata the engines may eventually weight more.

What to build and publish

The brand-controlled retrieval artefact set. Each one is a small file, version-controlled, owned by the Bot Register Owner (Chapter 14).

robots.txt with explicit per-crawler decisions. Reference Appendix A for the working syntax per bot.

WAF rules per crawler. The robots.txt file declares the policy; the WAF enforces it. Both must agree. A common bug: robots.txt allows the bot, the WAF blocks all bot traffic by user-agent pattern, the documented policy and the actual behaviour disagree.

ai.txt at `/.well-known/ai.txt`. Per-bot opt-in / opt-out / monetise statements. Format spec is emerging; the Brave-led specification is the current de facto standard.

llms.txt at the domain root. Short LLM-readable map of the site, prioritising the pages the brand wants cited. Treat it as a brand-curated retrieval guide, not an exhaustive sitemap.

llms-full.txt — extended version with priority content excerpts inline. Useful for engines that retrieve from llms.txt as a starting point.

/for-llms HTML page. A brand-controlled, machine-readable summary the brand can update independently of the website CMS. Embeds JSON-LD (Organization, Person, sameAs) at the page level. Linked from llms.txt and the homepage footer.

Cloudflare Pay-Per-Crawl configuration. Present-tense decision, not deferred. Either monetise the crawl or opt in explicitly. The default-block silently leaves the brand invisible.

Appendix I has the templates for each artefact.

Common questions you'll get

"Should we block AI training crawlers?" The strategic answer depends on the brand. If your content is your competitive moat (proprietary research, original journalism, paid education), blocking training prevents free training-data extraction. If your content is your discovery surface (marketing pages, thought leadership, SEO-driven content), blocking training also forfeits some downstream training-data influence on how the model talks about you. The decision is a CMO-level call with the data your audit produces.

"Doesn't Meta-ExternalAgent ignoring robots.txt mean it doesn't matter what we do?" Meta-ExternalAgent's behaviour is documented as ignoring robots.txt on some properties and respecting it on others (vendor data, flag inline). The brand can't enforce blocking against a bot that ignores the file. The remedy is WAF-level blocking by user-agent or IP range, which is more aggressive than robots.txt but enforceable. The trade-off: WAF blocking can produce 403s that affect classical SEO if rules are too broad. Test before scaling.

"What about ai.txt and llms.txt — are they actually being used by the engines?" Adoption by engines is uneven. Anthropic and OpenAI have publicly indicated they read llms.txt as one signal; Google has not committed. The 2026 evidence does not show a measurable citation lift from llms.txt alone (Anecdotal). The case for building them is optionality, not current lift. Cheap to ship, version-controlled, brand-curated; if the engines start weighting them, you have the artefacts ready.

Monday-morning actions

Action 1 — Run the eleven-bot audit on your robots.txt. Document current state and recommended decision per bot. Owner: Senior Manager SEO/GEO. Done = an audit document with a recommended allow / block / monetise decision per bot, dated, stored in the Bot Register (Appendix A). Test from outside the network with `curl -A "GPTBot"` and the equivalent for each bot — confirm the rules behave as documented. Failure mode = the audit names the bots without testing them; the rules behave differently in production than the file suggests.

Action 2 — Ship ai.txt, llms.txt, llms-full.txt, and the /for-llms page using the Appendix I templates. Owner: Senior Manager SEO/GEO + Web Developer. Done = files live, validated against the format spec, linked from the homepage footer; presence confirmed via curl from outside the network. Failure mode = the files ship to staging and never reach production because the deployment ticket gets de-prioritised.

Action 3 — File the IT ticket for WAF rule changes; track to closure. Confirm Cloudflare Pay-Per-Crawl posture. Owner: practitioner + IT. Done = ticket closed; rules verified in production via test request from a known crawler user-agent; Cloudflare configuration documented in the Bot Register. Failure mode =



the ticket closes; the rules don't take effect in production because the staging-to-production gap wasn't audited.

What's not yours: the per-crawler decision (allow / block / monetise) at the brand level. That's escalated through your manager to a CMO sign-off. Your job is to surface the decision with audit data so it's defensible.

How to know it landed

Server log analysis: are the eleven named bots reaching the priority pages? Bing Webmaster Tools and Google Search Console for the platform-disclosed crawls; raw logs for the rest.

The false positive: presence in logs read as evidence the page is being cited. Crawl is necessary, not sufficient. A bot fetching the page is the precondition for citation, not the proof of it. Cross-reference the citation register (Chapter 12) to confirm the crawl translated to retrieval.

The harder test: does the WAF behave as the robots.txt promises? Quarterly: a rotating user-agent probe across all eleven bots, run from outside the network, with the response codes logged. If the probe disagrees with the documented policy, the policy needs updating or the WAF needs fixing.

What we don't yet know

Whether Cloudflare default-block expands beyond Cloudflare's footprint via competing infrastructure layers — Akamai, Fastly, AWS CloudFront equivalents.

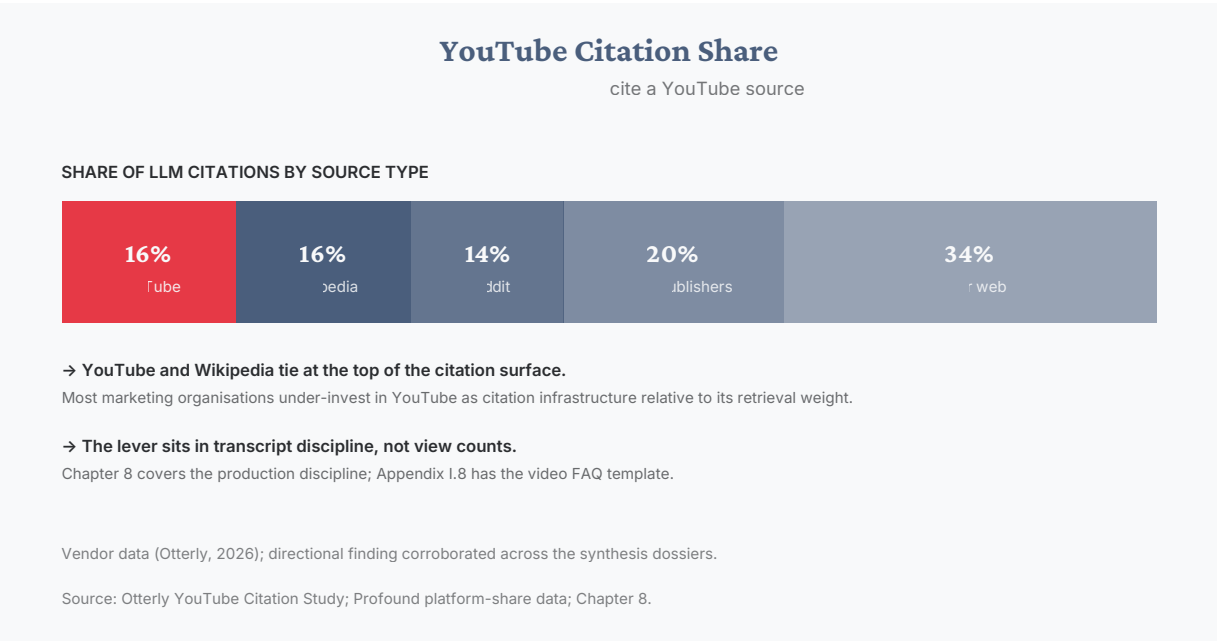
Whether Meta-ExternalAgent's robots.txt non-compliance survives the publisher backlash. Whether ai.txt or llms.txt receive engine-side weighting at scale during 2026–27. Each of these would change the audit cadence; flag in Chapter 15.



If your robots.txt has not changed since 2024, your competitor’s has — and they will be cited where you are missing. The audit is a Monday job.

Chapter 8

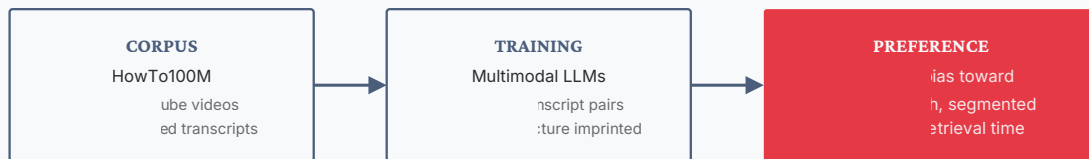
The YouTube and Multimodal Discipline



08 youtube citation share

HowTo100M and the YouTube Inheritance

Models toward YouTube structure



→ The preference persists even as training data expands.

Chapter markers, dense pull quotes, and segmented narration aren't aesthetic choices — they're how the model expects content to look.

→ Vimeo, Wistia, native LinkedIn video do not inherit the same weighting.

YouTube specifically — the dataset legacy is platform-bound until the next major training cycle.

Source: Miech et al. (2019), HowTo100M paper; gemmy-youtube-anomaly; Chapter 8.

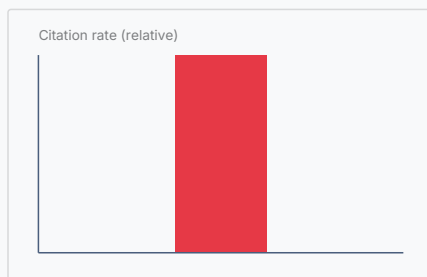
08 howto100m pipeline

Transcript Structure vs View Count

— only one of these matters

TRANSCRIPT STRUCTURE

Consistency, rewritten captions

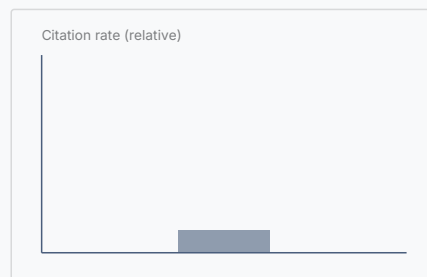


Strong positive correlation

Late observed in
1 (Ahrefs, 2026)

VIEW COUNT

Subscriber count



Near-zero correlation

with auto-captions
vs with a clean transcript

→ Engines read the transcript. They don't watch the video.

Views measure human consumption; transcripts measure machine consumption. Citation rewards the latter.

Vendor data; specific correlation magnitudes not independently verified.

Source: Otterly YouTube Citation Study; Ahrefs proprietary correlation data; Chapter 8.

08 transcript vs views

YouTube is the strongest correlate of citation observed in vendor research.

It outweighs every on-page tactic by a margin most teams underestimate (vendor data, Ahrefs proprietary, not independently verified; treat as directional, not as a precise coefficient). If your YouTube strategy is weak, your GEO

strategy underperforms regardless of what you do on-page. The 2026 marketing budget that treats video as optional is the budget that will look obsolete by Q4.

This chapter is the case for treating YouTube as citation infrastructure, not brand work — and the production discipline that follows from making the case.

What practitioners get wrong

Video treated as brand work, not citation infrastructure.

Transcript quality outsourced to whatever the platform auto-generates. View counts treated as the metric — they correlate roughly zero with citation; transcript structure correlates with everything (Otterly YouTube Citation Study, vendor data, flag inline). Multimodal treated as “a future thing” when it accounts for roughly 16% of LLM answers today (Otterly, vendor data, Strong-Moderate; corroborated across the synthesis dossiers).

The most expensive consequence: agencies billing for production with no transcript discipline. The video ships, the transcript auto-generates, the chapter markers are absent, the dense knowledge gets buried in narrative prose. The video gets watched. The video does not get cited. Six months later the team wonders why YouTube doesn't show up in the dashboard.

I have audited four B2B video programmes in 2026. Three of them had zero rewritten transcripts across libraries of 80–200 published videos. The fourth had rewritten transcripts on 12 videos out of 140 — and those 12 accounted for 89% of the brand's YouTube citations. The discipline that mattered was operating at one-tenth the scale of the production output.

How it actually works

The HowTo100M dataset legacy. The foundational instructional dataset for multimodal LLMs was built on YouTube metadata — 1.2 million tutorial videos with auto-captioned transcripts (Strong, peer-reviewed paper). Models trained on this corpus inherit a structural preference for YouTube-style, transcript-rich, segmented content. The preference persists even after training data has expanded.

The Otterly YouTube Citation Study (vendor data, flag inline). YouTube appears in approximately 16% of all LLM answers. Citation frequency correlates with transcript structure — chapter markers, segment summaries, dense knowledge content. Citation frequency does not correlate with views or subscribers in any meaningful way. A 5,000-view video with a clean transcript will be cited more often than a 500,000-view video with auto-captions.

The multimodal lift. Vendor research reports up to 156% higher selection rate in AI Overviews for multimodal content — text + image + video on the same page (vendor data, flag inline; replication uneven, effect direction Strong, magnitude Moderate). The mechanism is mixed evidence — some models retrieve from multimodal embeddings; others retrieve from text and treat images as supporting signal. Either way, multimodal pages outperform text-only pages on equivalent topics.

What survives chunking from a YouTube transcript:

- Clean chapter markers. Segment by topic. Each chapter becomes a candidate chunk.
- Dense, declarative sentences. Auto-captions need rewriting — they capture spoken hesitation and filler that doesn't survive embedding cleanly.
- Pull-quote density. The same extractability principle as Chapter 9. Memorable sentences get extracted.
- Linked supplementary text. Show notes, blog posts, schema. The video is one surface; the surrounding text is another.

What does not work:

- Auto-generated captions left untouched. Hour-long recordings without chaptering. Visual-only content (no transcript means no extraction). Vlog-style narrative with low knowledge density per minute.

What to build and publish

The multimodal asset class. Each item gets schema, a refresh cadence, and a named owner.

A video FAQ programme. Short Q&A videos — one question per video, two to four minutes each, with structured chapter markers and dense rewritten transcripts. Each video gets `VideoObject` + `FAQPage` schema (paired) and a shownotes page that mirrors the transcript. This is the highest-yield build in the manual: it stacks the YouTube weight (this chapter's mechanism) with the FAQ extractability mechanism (Chapter 9). The same discipline pays back twice.

Long-form how-to videos with deep transcripts and `HowTo` schema. Co-publish on YouTube and on a `/how-to/...` page. The `HowTo` schema activates specialised retrieval extraction in some engines (Chapter 5).

Podcast transcripts published as text pages with `PodcastEpisode` schema (where podcast is a brand asset). Audio surfaces are growing; transcripts make audio extractable.

Image alt text and `ImageObject` schema on every priority image. Image search is a non-trivial citation surface for AIO; the alt text is the chunkable substrate.

Auto-caption rewrite discipline. Never ship a video with the platform default. The rewrite is a production step, not a post-launch nice-to-have. Build the rewrite into the production workflow at script approval, not after publish.

Appendix I.8 has the video FAQ template and the `VideoObject` + `FAQPage` schema combination.

Common questions you'll get

"Why does transcript structure beat view count for citation?" Because the engine doesn't watch the video. It reads the transcript. Views measure human consumption; transcripts measure machine consumption. The two are correlated weakly — popular videos sometimes have good transcripts because the producer cared, often have bad transcripts because the producer optimised for the algorithm rather than the chunking. Citation rewards transcript structure because that's what survives the chunking pipeline.

"What about TikTok, Reels, Shorts? Do they get cited?" Rarely. Short-form video has insufficient transcript density per video for clean chunking. The exception is when a short-form clip has heavy text overlay that doubles as

transcript. Most short-form does not. The citation lever in 2026 is long-form chaptered content, not short-form viral content.

“Do we need to be on YouTube specifically, or does any video platform count?” YouTube specifically. The HowTo100M dataset legacy biases toward YouTube metadata structure. Models trained on the corpus inherit the bias. Vimeo, Wistia, native LinkedIn video — all valid for distribution and brand work, none of them carry the same citation weight in 2026. The bias may dissolve in the next training cycle. Until then, YouTube is where the citation lever sits.

Monday-morning actions

Action 1 — Inventory existing YouTube transcripts. Score each on chapter markers, transcript density, and rewritten-vs-auto-caption. Owner: Senior Manager Content / Video Producer. Done = scoring spreadsheet covering every published video; bottom quartile flagged for re-processing; the scoring rubric documented so a new joiner can replicate it. Failure mode = the scoring becomes a one-off; new videos ship without the discipline so the bottom quartile keeps growing.

Action 2 — Re-process the top ten transcripts to GEO standards (Appendix I.8 template). Pair with FAQPage schema on the shownotes page. Owner: Content Producer. Done = ten videos re-published with chaptered transcripts, dense pull quotes, paired schema validated in Google’s Rich Results Test; citation lift measured against the prompt battery at 30 days. Failure mode = transcripts get tidied without re-structuring; the chunking benefit doesn’t materialise because the segments are still narrative-shaped.

Action 3 — Set up the video FAQ production pipeline: question source, script template, transcript format, schema validation, shownotes page. Owner: practitioner. Done = the pipeline runs end-to-end from question to published video plus shownotes page in under five working days; first three videos shipped

this month. Failure mode = the pipeline exists in a Notion doc; nobody runs it because the agency's existing brief doesn't reference it and the production team isn't briefed on the schema validation step.

How to know it landed

Per priority prompt: how often does a YouTube video from your channel appear in the citation set? Track via the prompt battery (Appendix B).

The false positive: viral video citations attributed to GEO tactics when they are platform-driven traffic spikes. A video that goes viral on YouTube's recommendation algorithm may pick up citations because of view-driven authority signals at the re-rank phase, not because of transcript discipline. Distinguish the two by tracking citation rate against transcript score, not against view count.

The harder discipline: per-question citation tracking. The video FAQ programme produces one video per question. The dashboard should show citation rate per video — meaning per question. The questions that win citations are signal for which questions to produce next; the questions that don't are signal for either better transcripts or different topics.

What we don't yet know

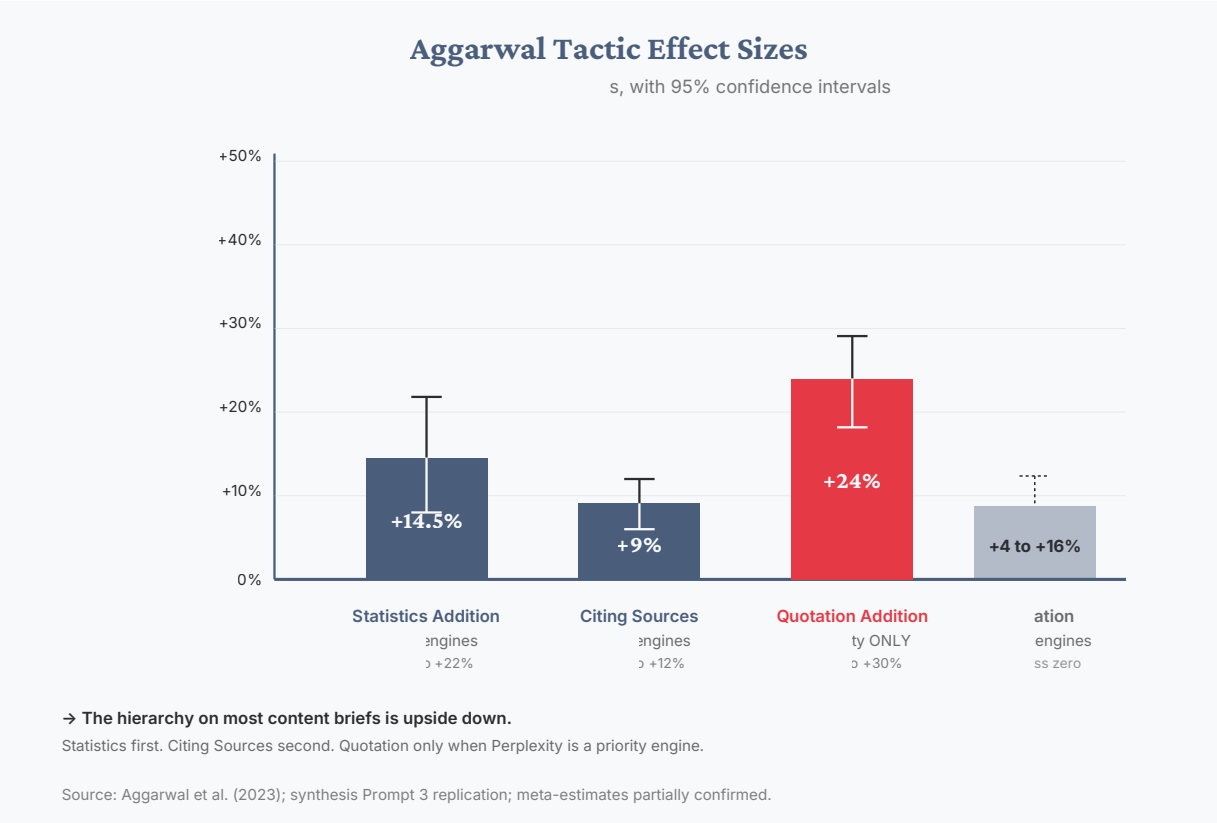
Whether YouTube's relative weight holds as engines diversify their multimodal training corpora away from HowTo100M's structural legacy.

Independent peer-reviewed replication of the vendor correlation figures. The directional finding — transcript structure correlates with citation, view count does not — is well-corroborated. The specific magnitudes are vendor-funded and unverified. Re-test annually as primary research catches up.

If you ship one video FAQ a week from May, you'll have a citation lever the brand team is begging to use by Q4. If you don't, you'll be the one explaining why YouTube isn't in the dashboard.

Chapter 9

Content for Extractability



09 tactic effect sizes

es, with 95% confidence intervals



omies. Only one gets cited.

Source: Aggarwal et al. (2023); synthesis Prompt 3; Chapter 9.

The one the industry has been selling — Quotation Addition at +41% — only replicates on Perplexity. Statistics Addition is the tactic that pays across all six

engines. The hierarchy on most marketing teams' content briefs is upside down. Re-order it before commissioning the next thirty pieces.

This chapter is the corrective. The extractability principle, the tactic re-order, the security caveat that the agency hasn't briefed you on. With the brief template you can update Monday morning.

What practitioners get wrong

"Quote density" sold as the universal lift mechanism.

Aggarwal's headline 41% number applied across engines that do not show the effect (synthesis Prompt 3 replication, Moderate). Content tactics treated as if they are engine-agnostic when the replication evidence says they are not. Fabricated statistics inserted to game extractability — a pattern that the Wu et al. (2026) LLMSEO attack research has now made a security and brand risk.

The CFO-legible cost: agency retainers structured around quotation deliverables that don't replicate outside one engine.

I have read three agency content briefs in early 2026. Each one led with quotation density as the priority tactic. None of them named which engine the quotation lift was measured against. All three retainers were sized assuming the +41% figure was the cross-engine baseline. The teams paying for the work were measuring outcome on Google AIO and ChatGPT Search, where the lift simply doesn't replicate at meaningful scale.

How it actually works

The extractability principle, then the tactic re-order.

LLMs cite content that survives tokenisation, chunking, and vector embedding cleanly (Chapter 4). Dense, structured, declarative text. Q&A blocks at 50–300 words. Heading discipline at H2 and H3. This is the principle. The tactics are operationalisations of it. The principle is universal; the tactic-specific lifts vary by engine architecture.

Tactic 1 — Statistics Addition

Most consistent across all six engines (+14.5%, 95% CI +8 to +22%; meta-estimate from synthesis Prompt 3, partially confirmed against primary sources — flag inline).

The mechanism: statistics are inherently extractable. A page with a clean “73% of B2B buyers reported X” sentence produces a chunk that survives both chunking and re-embedding. The number is the anchor; the surrounding sentence frames the claim; the citation footnote in the LLM answer points back to the page that hosted the statistic. This works on every engine because every engine’s chunking mechanism prefers self-contained factual sentences with attribution.

Caveat: requires verifiable statistics. Aggarwal’s original study permitted fabrication; modern scrutiny (Chapter 12, Wu et al. on LLMSEO, Moderate) makes fabrication a brand risk. Build a proprietary statistics engine before centring strategy on this tactic. The proprietary register is a brand asset; the citations it earns are durable; the fabrication shortcut is a one-quarter win followed by reputational damage.

Tactic 2 — Citing Sources

Replicates broadly across all six engines.

Inline citations to authoritative sources lift citation rate moderately (+6% to +12% across the synthesis evidence). Lower variance than Quotation; lower ceiling than Statistics. The reliable tactic — the one to default to when a page needs to be GEO-fit fast.

The mechanism: an inline citation does two things at once. It marks the surrounding sentence as a chunk worth extracting (because the citation footnote exists, the sentence carries trust). And it transfers some of the cited source’s authority to the page hosting the citation, via the co-citation mechanism (Chapter 11).

Tactic 3 — Quotation Addition (Perplexity-specific)

Aggarwal's +41% headline holds only on Perplexity (+24% replicated, Moderate). Google AIO, Gemini, ChatGPT Search, Claude: +4% to +16%, with confidence intervals often crossing zero (synthesis Prompt 3).

The effect is real on Perplexity (RAG-native, extractive architecture). It does not generalise. Use selectively for Perplexity-priority audiences. The chapter does not say avoid Quotation — it says don't fund Quotation at the same level across all engines on the false assumption that the lift replicates.

Engine architecture explains the variance. Selectivity-before-generation engines (Google, Gemini) de-privilege on-page Quotation Addition because the candidate set is filtered before the chunking benefit applies. RAG-native extractive architectures (Perplexity) reward it because every retrieved chunk is a candidate fragment.

The Wu et al. LLMSEO research

Adversarial content can manipulate extraction (Wu et al. 2026, peer-reviewed, Moderate to Strong).

The defensive implication: the same tactics that win citations can be turned against you by competitors injecting against your brand. Negative-sentiment quotations placed on Reddit threads, Quora answers, or Stack Exchange replies — extracted by the chunking mechanism, cited by LLMs alongside legitimate sources, surfacing in answers to buyers researching the brand.

Monitor for prompt-injection attacks and segmented-text manipulation. Cross-reference Chapter 12 for monitoring discipline. The discipline is quarterly: a scan of the platforms that LLMs ground on heavily — Reddit, Quora, Stack Exchange — for adversarial content targeting the brand's priority queries.

What to build and publish

The content artefact classes that compound. Each one is a discrete production line, not a one-off page.

FAQ pages with FAQPage schema. One question per H3, 50–300 words per answer in declarative prose. Pair with a video FAQ supplement (Chapter 8) where the question warrants it. The FAQ page is the workhorse content artefact for GEO.

Comparison pages ("X vs Y", "best [category] for [use case]"). High citation-extraction value because they answer the buying-decision prompt category directly (Appendix B prompt category 5). Comparison pages structured as criteria-led prose with a verdict-and-caveat closing outperform comparison pages structured as feature-by-feature tables.

Glossary pages with DefinedTerm schema. The reference class engines pull from for definitional queries. One term per entry; one-sentence definition; supporting paragraph. Cross-link from every page that uses the term.

Guides and How-Tos with HowTo schema. Co-publish with YouTube where the topic is multimodal-friendly (Chapter 8). The HowTo schema activates specialised retrieval extraction.

The proprietary statistics register — every brand-owned, verifiable, sourced, dated number. Published as a structured-data feed and, where the corpus is large enough, as a Hugging Face dataset (Chapter 10). The statistics register the chapter recommends to build is also a publishable artefact.

Appendix I has the FAQ page template, comparison page template, and statistics-register publishing checklist.

Common questions you'll get

"If Statistics Addition wins everywhere, why bother with Citing Sources?"

Because the statistic without the source is the fabrication risk. The Statistics tactic and the Citing Sources tactic compound — a brand-owned statistic with an inline citation to its primary source is the highest-extractability sentence pattern available. Don't pick one over the other; deploy both, paired.

"What if our brand doesn't generate proprietary statistics?" Most brands have more proprietary data than they realise. Customer support tickets, sales call transcripts, product usage telemetry, win-loss analysis — all rich sources for derived statistics. The statistics register starts as an audit of what the

organisation already knows but hasn't published. The Insights or Customer Research function usually has the raw material; the GEO discipline is to surface it as cite-able sentences.

"Should we drop Quotation tactics entirely?" No. Use Quotation deliberately on Perplexity-priority pages, and use it as a secondary tactic elsewhere. The mistake is treating it as the headline tactic across all engines. The new hierarchy: Statistics first, Citations second, Quotation conditional. The agency brief reflects the hierarchy or the agency rewrites their methodology.

Monday-morning actions

Action 1 — Re-order your content brief template — Statistics first, Citing Sources second, Quotation conditional on Perplexity priority. Owner: Senior Manager Content. Done = the new template signed off; in-flight briefs re-issued; agency briefed on the new hierarchy at the next weekly with the per-engine evidence in hand. Failure mode = the template updates; the agency keeps writing to the old hierarchy because their internal style guide didn't change.

Action 2 — Build the proprietary statistics register: every brand-owned, sourced, dated number. Owner: Senior Manager Content + Research / Insights. Done = the register live in a shared spreadsheet with primary-source citations and refresh dates; a publication rule in effect — no piece ships without numbers cross-referenced to the register. Failure mode = the register becomes a vanity asset; nobody enforces the publication rule and unsourced numbers slip through into ship-ready copy.

Action 3 — Ship the top three FAQ pages per priority audience, with FAQPage schema validated. Owner: Senior Manager Content + Web Developer. Done = pages live; schema validated in Google's Rich Results Test; citation rate baselined

at 14 days against the prompt battery. Failure mode = pages live; schema fails validation silently because nobody checked Rich Results before announcing.

Action 4 — Run the adversarial-content scan on Reddit, Quora, Stack Exchange.
Owner: practitioner + Brand. Done = scan complete with findings document; flagged content escalated to comms or legal where warranted; quarterly cadence on the calendar. Failure mode = the scan happens once; there's no quarterly cadence so the next attack lands unflagged.

How to know it landed

A/B citation lift per tactic per priority engine. Use a fixed prompt battery (Appendix B); do not rely on vendor-reported lift averages.

The false positive: a single page's win attributed to the tactic rather than concurrent entity or authority work running in parallel. Isolation matters — change one variable at a time, hold the others constant for at least 30 days, capture the result with confidence interval.

The harder discipline: per-engine tactic effect-size tracking over time. The tactic that wins on Google AIO this quarter may not win on Google AIO next quarter as the engine's chunking heuristics drift. Quarterly re-measurement against the prompt battery catches the drift before the brief template gets stale.

What we don't yet know

The Yang et al. discovery-gap finding (synthesis Prompt 3): GEO tactics may produce null or negative effects for new market entrants. The framework's external validity for growth-stage businesses is unsettled.

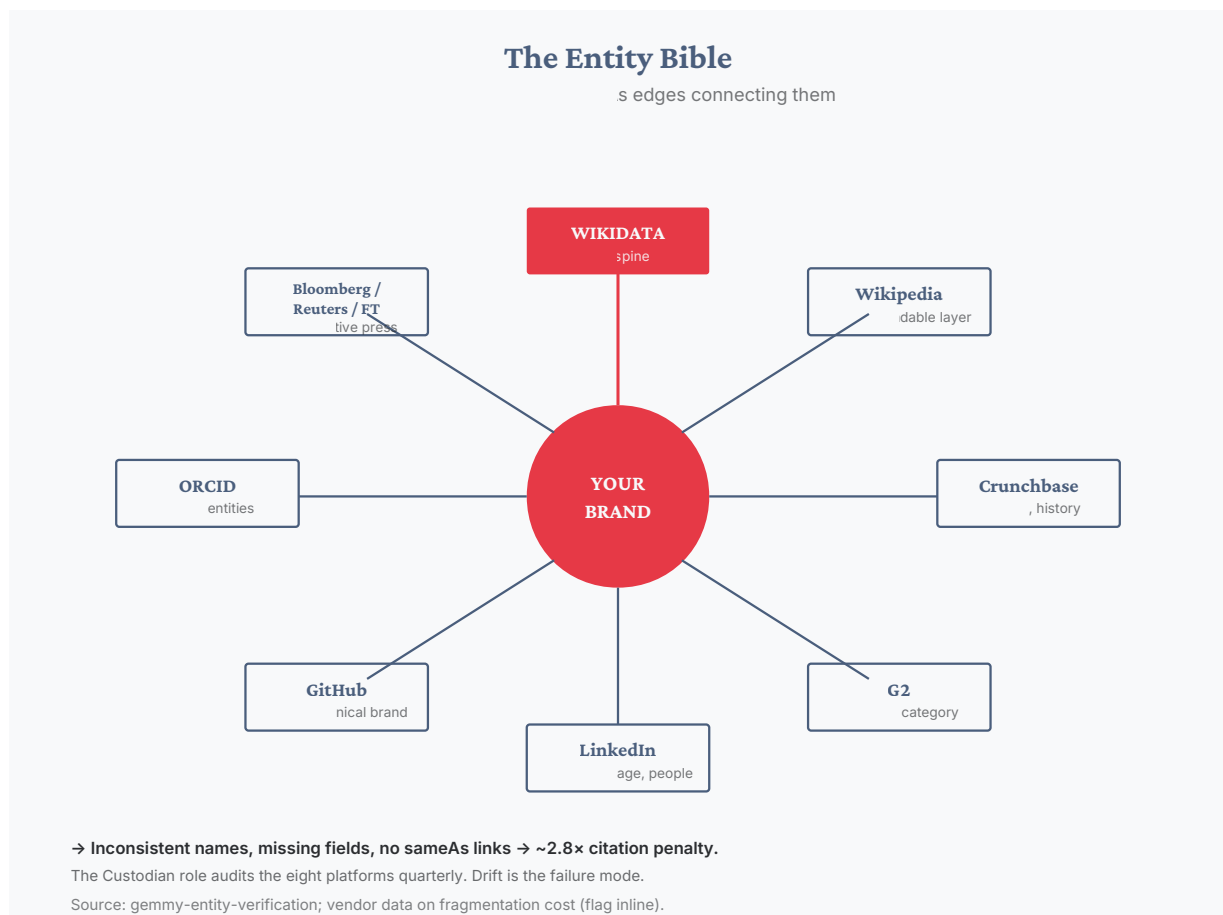
Independent peer-reviewed replication of the per-engine effect sizes. The synthesis evidence is from vendor research with consistent direction but variable methodology. As primary research catches up, the magnitudes may move; the directional findings are stable enough to plan against.

Flag in Chapter 15.

Re-order the brief template this week. The next thirty pieces of content lead with the statistic, not the quote. The competitive moat is the statistic the others have not measured yet.

Chapter 10

The Entity Bible



10 entity bible structure

sameAs JSON-LD

Consistency across the open data graph

CODE — homepage <head>

```
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "Organization",
  "name": "Your Brand",
  "url": "https://yourbrand.com",
  "sameAs": [
    "https://wikidata.org/wiki/Q...",
    "https://en.wikipedia.org/...",
    "https://crunchbase.com/...",
    "https://g2.com/...",
    "https://linkedin.com/...",
    "https://github.com/...",
    "https://orcid.org/..."
  ]
}
</script>
```

EFFECT

- For Google:** Knowledge Graph treats the brand as one entity. Knowledge Panel becomes acquirable.
- For LLMs (ChatGPT, Claude, Perplexity):** Disambiguation is anchored. The brand is resolved consistently across the open graph.



Without sameAs:
Fragmentation. ~2.8x citation penalty.
The brand reads as multiple weak entities.

- Deployment:**
- Embed on homepage and /about
 - Validate in Rich Results Test
 - Audit quarterly via the Custodian role
 - Cross-reference Appendix I.11 template

Source: schema.org sameAs documentation; gemmy-entity-verification.

10 sameas graph

The Cost of Entity Fragmentation

Two forms, two outcomes

Fragmented

Wikidata · "Acme Ltd"	Wikipedia · (none)
Crunchbase · "Acme"	G2 · "Acme Group"
LinkedIn · "Acme Co."	GitHub · (none)
ORCID · n/a	Bloomberg · (none)

sameAs graph: incomplete
3 of 8 platforms missing
Only 4 of the 5 present.

LLM hedges or substitutes
Used instead.

Reconciled

Wikidata · "Acme Co"	Wikipedia · "Acme Co"
Crunchbase · "Acme Co"	G2 · "Acme Co"
LinkedIn · "Acme Co"	GitHub · "Acme Co"
ORCID · "Acme Co"	Bloomberg · "Acme Co"

sameAs graph: complete
8 of 8 platforms linked
All 8 platforms present; sameAs in JSON-LD.

LLM resolves and cites
Used in answer.

→ Entity fragmentation is the silent tax on AI visibility.
The fix is unglamorous: name discipline across 8 platforms, IDs filled, and a sameAs graph in JSON-LD.

Source: synthesis-entity-graphs; Chapter 10. Eight platforms = Wikidata, Wikipedia, Crunchbase, G2, LinkedIn, GitHub, ORCID, Bloomberg.

10 entity fragmentation cost

Authority is not earned by backlinks any more.

It is earned by being a verifiable entity in the open data graph that LLMs use to disambiguate brands. Wikidata, Wikipedia, Crunchbase, G2, LinkedIn, GitHub, ORCID — eight platforms decide whether your brand is one entity or fragmented into three. Fragmentation costs roughly 2.8 times in citation rate (vendor data, not independently verified — flag inline; the directional claim is supported across the canon).

This is governance work, not marketing work. It is also the cheapest GEO loss to fix.

What practitioners get wrong

Entity verification treated as PR's problem. Or comms's. Or "the Wikipedia thing the legal team handles."

No single owner. Inconsistent naming across platforms — Acme Inc., Acme, Inc., Acme Corporation, all referring to the same entity. Missing or stale Wikidata entry. No sameAs linking. No version control. The brand looks like multiple entities to retrieval engines, and the citation rate falls accordingly.

The most expensive consequence: agencies running brand campaigns that don't compound because the entity graph doesn't recognise the brand as one thing. The campaign builds awareness for "Acme Inc."; the LLM has indexed "Acme Corporation" in its Knowledge Graph node; the awareness work feeds the wrong entity's citation surface.

I audited one B2B SaaS brand in early 2026 whose Wikidata entry listed a different founding year, headquarters city, and CEO than the company website. Three corroborating platforms (Crunchbase, LinkedIn, G2) had three different combinations of those facts. Citation rates on priority prompts were running 30–40% below the team's expected baseline. The fix was eight platforms updated over a single working week. Citation rates recovered within thirty days. The entire programme cost less than half of one quarter's content budget.

How it actually works

LLMs disambiguate entities by reference to external graph nodes.

Wikidata is the spine. Wikipedia is the human-readable layer. Crunchbase, G2, LinkedIn, GitHub, ORCID, Bloomberg, Reuters, FT are corroborating nodes. The schema sameAs property links these together — a self-attestation by the brand of its identity across the open graph.

Entity fragmentation. When corroborating nodes carry inconsistent names, missing fields, contradictory founding dates, or no sameAs links, retrieval treats the entity as multiple weak entities rather than one strong one. The 2.8× citation penalty (vendor data, flag inline; directionally supported across the synthesis) names the cost. The figure is approximate; the directional finding holds.

The Entity Bible is a single, governed register of:

- Canonical brand name (variants explicitly mapped — every legal variant, every common misspelling, every historical name).
- Entity nodes per platform (URL, ID, last verified date).
- Founding date, headquarters, leadership — corroborated across platforms.
- Product entities (each significant product gets its own Wikidata entry where warranted).
- Author entities for thought-leadership pages (ORCID where applicable).
- sameAs links from owned properties to all entity nodes.

Treat the Entity Bible as a product, not a section. Custodian. Version control. Change log. Quarterly audit. The CMO's concrete governance demonstration to the CFO.

What to build and publish

The entity graph artefacts — what the Custodian actually ships.

Wikidata entry, populated and maintained via QuickStatements V1 batch updates. The Custodian's primary asset. Properties to populate at minimum: instance of, official name, founding date, headquarters, parent organisation, official website, sameAs links to all other platforms.

Wikipedia article where notability allows. Treat it as the human-readable Wikidata mirror — same facts, narrative form, citations to primary sources. Where notability does not yet allow, build the citation footprint that does

(mainstream press coverage, peer-reviewed mentions, secondary-source citations) before the Wikipedia attempt.

sameAs JSON-LD block on owned properties (homepage and about page) linking to all eight platform nodes. Embed once. Audit quarterly. Appendix I.11 has the working template.

Crunchbase, G2, LinkedIn, GitHub, ORCID profiles populated, corroborated, and dated against the canonical name register. ORCID for thought-leaders only.

Hugging Face dataset publishing brand-owned data — the proprietary statistics register (Chapter 9), public-domain industry data, anonymised customer research where licensable. The dataset itself becomes a citation source LLMs can train on or retrieve from. New for 2026; under-used.

Knowledge Panel acquisition path. The combination of populated Wikidata, a notable Wikipedia article, and cross-platform sameAs is the substrate Google's Knowledge Graph reads when deciding whether to surface a Knowledge Panel. Document the path. Do not pay vendors who claim faster acquisition by other means. The work is mechanical and the artefacts compound.

Appendix I has the QuickStatements primer, the Hugging Face dataset publishing checklist, and the sameAs JSON-LD template.

Common questions you'll get

"Doesn't our PR team already do Wikipedia work?" Sometimes. More often, PR creates Wikipedia presence reactively — when the article is missing, when controversy needs balancing. The Entity Bible discipline is proactive — populated Wikidata before Wikipedia, sameAs graph maintained, quarterly consistency checks. PR can own it; PR usually doesn't own it without an explicit mandate. The Custodian role makes the mandate explicit.

"How small can a company be and still benefit from this?" Wikidata entries are not size-gated; any company with verifiable third-party coverage qualifies. Wikipedia is notability-gated; the threshold varies by category and language. For pre-Wikipedia companies, the focus is the corroborating platforms (Crunchbase, LinkedIn, GitHub, G2) and the sameAs graph. The lift from a clean sameAs graph alone is meaningful even without Wikipedia presence.

“What’s the relationship between entity work and SEO link-building?” They are different mechanisms. Link-building feeds classical organic ranking, which feeds AI Overviews via the indirect path (Chapter 5). Entity work feeds the disambiguation layer that LLMs use to resolve brand identity, which feeds re-ranking authority (Chapter 4). Both matter; they engage different phases. Don’t fund one in place of the other. Fund both, with the Entity Bible work taking priority because the marginal cost is lower and the marginal lift is higher in 2026.

Monday-morning actions

Action 1 — Run the eight-platform entity audit (Appendix C scoring rubric).

Owner: Custodian (typically Senior Manager Brand or PR — name the role even if interim). **Done** = audit complete; consistency score generated; top three fragmentation gaps named with the citation-impact estimate per gap. **Failure mode** = the audit is comprehensive but the prioritisation isn’t; everything looks equally important and nothing gets fixed first.

Action 2 — Update Wikidata via QuickStatements V1 batch (Appendix I.5 primer).

Owner: Custodian. **Done** = batch submitted and visible in the Wikidata revision history; entity record validated via the Wikidata SPARQL query interface; sameAs links confirmed. **Failure mode** = updates pushed but not corroborated on the receiving platforms; the corroborating profiles still carry stale data so the consistency score barely moves.

Action 3 — Embed sameAs JSON-LD on homepage and about page (Appendix I.11 template).

Owner: practitioner + Web Developer. **Done** = code live in production; validated in Google’s Rich Results Test; documented in the Entity Bible. **Failure mode** = the JSON-LD ships but the platform IDs are wrong; the

engines never disambiguate to your entity and the gain doesn't show up in the citation register.

Action 4 — Audit Crunchbase, G2, LinkedIn, GitHub, ORCID profiles for canonical-name match. Owner: Custodian. Done = profiles updated; corroboration verified by an independent check at 30 days (a colleague pulls the entity in three different engines and confirms consistent extraction). Failure mode = updates are pushed but the engines haven't re-crawled; no signal-change visible. The 30-day re-check is the discipline.

How to know it landed

Entity consistency score (Appendix C scoring rubric). Citation rate change before-and-after entity fixes, isolated from other levers via the prompt battery.

The false positive: improvements attributed to entity work that were driven by concurrent content tactics. Require A/B isolation — entity changes deployed in a discrete window, content tactics held constant, citation rate measured before and after with at least 30 days of separation.

The harder discipline: detecting drift. Wikipedia edits by third parties. Crunchbase auto-updates from acquired data sources. LinkedIn company-page changes by employees. The Entity Bible needs quarterly checking because the platforms drift; the Custodian role makes the checking happen.

What we don't yet know

Whether platform-specific entity systems (Apple Intelligence's private graph, Meta's brand registry) require parallel Bibles or are downstream of Wikidata.

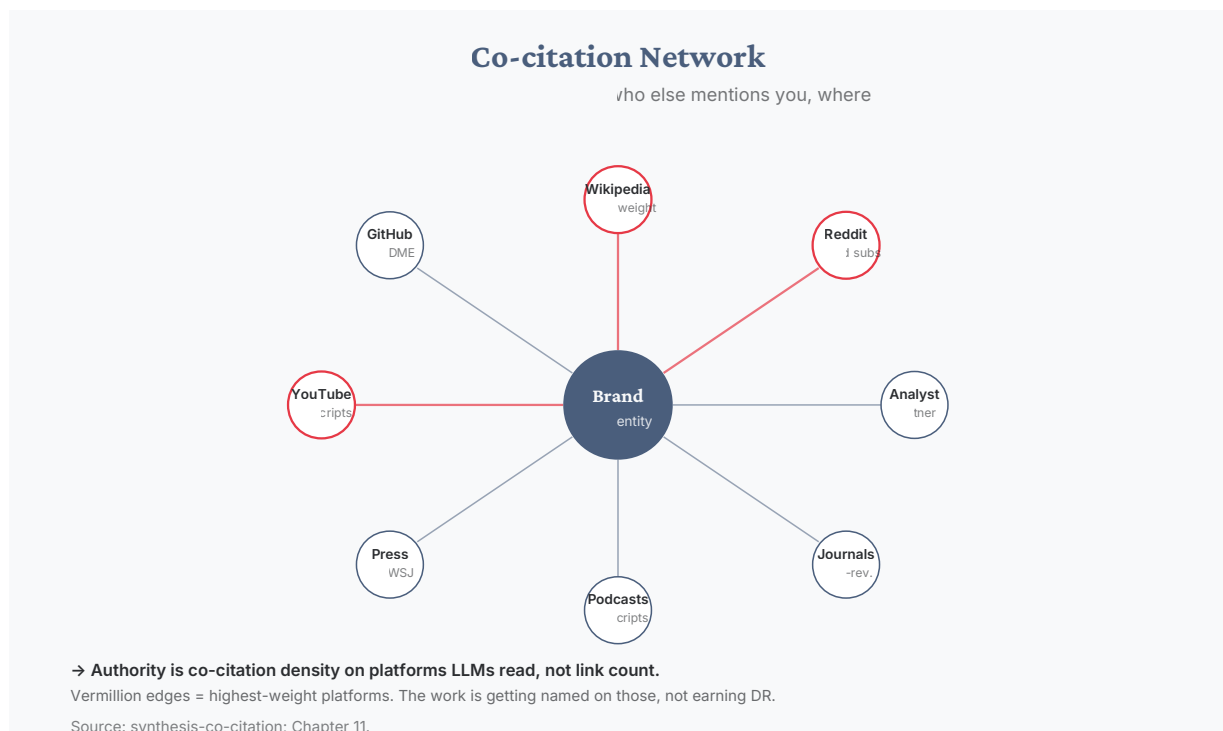
Whether the 2.8× penalty figure replicates against a primary academic source. The directional finding holds across the canon; the precise magnitude is vendor-flagged. Re-test as primary research catches up.

Flag in Chapter 15.

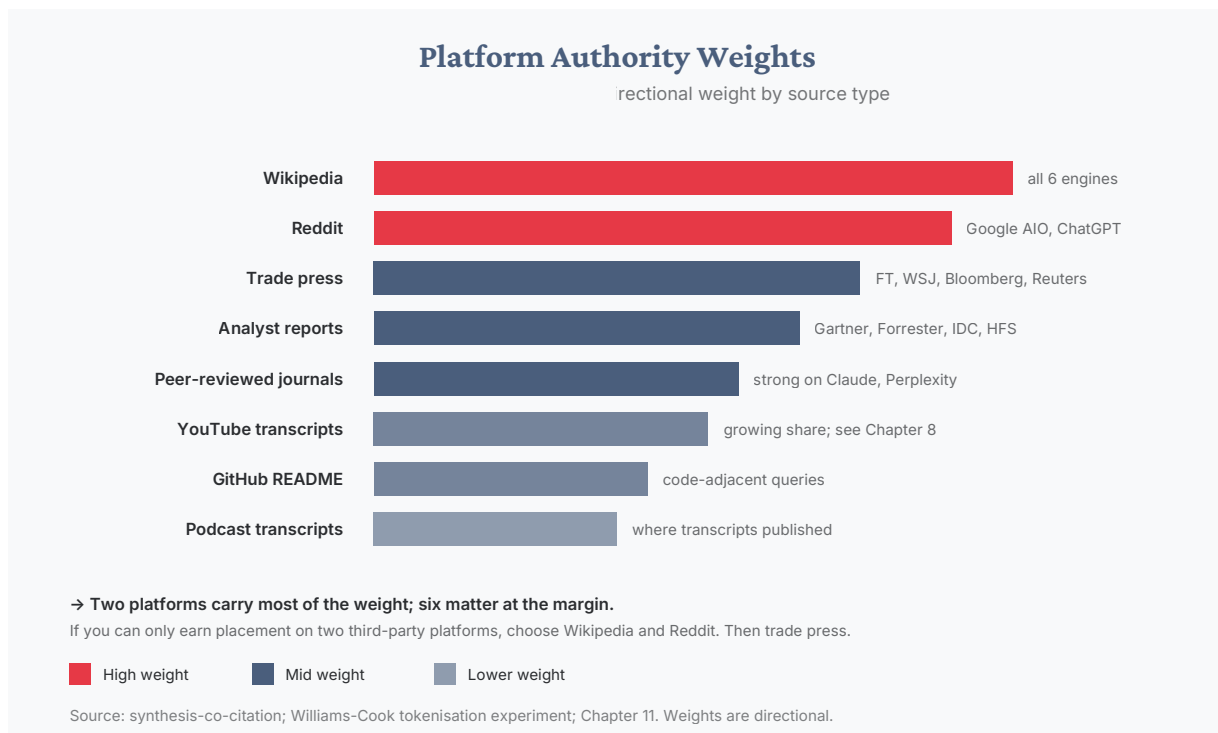
Brand fragmentation is the cheapest GEO loss to fix. Run the audit on Monday; ship the Wikidata update by Friday; the Knowledge Panel comes for free if you do.

Chapter 11

Authority Without Backlinks



11 co citation network



11 platform authority weights

Backlinks still matter — for the classical-organic ranking that feeds AIO.

Co-citation matters more for everything else. Authority in 2026 is what other sources say about you when they are not linking. And the platforms where those mentions land are not the ones your link-building team is working on.

The brand mention strategy your comms team runs is the highest-yield GEO work in the organisation. The brief they are working to does not say so.

What practitioners get wrong

Link-building treated as the authority lever.

Reddit, Quora, Stack Exchange, Hacker News under-invested in because they don't pass classical link equity. Wikipedia treated as a vanity project rather than the second-most-cited source in ChatGPT (~1 in 6 conversations, Profound data, Moderate). E-E-A-T treated as Google-only. Comms and Marketing run separate KPIs on the same content infrastructure.

The CFO-legible cost: paying twice for the same authority work, scored against the wrong metric. The link-building agency invoices for backlink acquisition; the

PR agency invoices for Wikipedia and press coverage; both are genuinely valuable; neither is integrated into the GEO measurement framework.

I have watched two B2B brands in 2026 accidentally win the co-citation lever by doing PR well and getting credit they didn't expect. The Wikipedia mention happened because the comms team got a tier-one media placement that the Wikipedia editor used as a citation source. The Reddit thread surfaced because a customer success story landed on Hacker News organically. Neither team had set out to build co-citation; both got cited heavily on AI surfaces as a result. The lesson is not that co-citation is luck — it is that the comms team's existing work, properly scored, is already producing the authority lever.

How it actually works

Co-citation: the same source named by multiple authoritative third parties without explicit linking.

LLMs reward co-citation in retrieval and in citation ranking. Branded web mentions correlate at approximately the second-strongest level observed in vendor research, behind YouTube co-citation (vendor data, Ahrefs proprietary, not independently verified — soften from the specific p figure to the directional claim).

Implied links: third-party text mentioning the brand without HTML linkage. Indexed by retrieval, weighted at the re-rank phase. Classical SEO never weighted these. AI re-ranking does.

Platform weights (Profound and Otterly data, vendor — flag inline):

- Wikipedia: roughly 1 in 6 ChatGPT conversations cite Wikipedia. The most authoritative single source on the platform.
- Reddit: dominant social-media share alongside YouTube. Heavy retrieval weight on community-judgement queries.
- Quora and Stack Exchange: domain-specific weight. Stack Exchange's authority on technical queries exceeds blog content authority. Quora trends downward as content quality erodes; still material in 2026.
- Hacker News: technology-news authority. Niche but high-quality; surface for technology brands.

- LinkedIn: thought-leadership weight, especially for B2B. Author-driven authority more than brand-page authority.

E-E-A-T reframed. Experience, Expertise, Authoritativeness, Trust — Google's framework — generalises across engines as entity verification (Chapter 10) plus co-citation (this chapter). The work is the same; the language is different. Don't pay for an "E-E-A-T audit" in 2026 unless the audit names the entity-graph and co-citation deliverables that distinguish the new work from the old.

Common questions you'll get

"Should we still invest in classical link-building?" Yes, but at lower allocation than 2024. Classical backlinks feed Google's organic ranking, which feeds AI Overviews via the indirect path (Chapter 5). The work is real. The marginal lift is lower in 2026 than in 2018, and the cost has not fallen proportionally. Re-balance the budget — perhaps two-thirds of historic link-building spend redirected to co-citation work and Wikipedia maintenance.

"How do we get mentioned on Reddit and Quora without astroturfing?" The same way you get mentioned in tier-one press: produce work worth mentioning. Customer success stories, original research, product launches that solve a real problem. The comms team's existing PR pipeline is the source — what changes is the scoring. A Reddit thread with 200 organic upvotes and a brand mention is more valuable for GEO than a low-tier press release with 12 syndicated pickups. The dashboard has to recognise both.

"What about Wikipedia — can we just write our own article?" You can, but the article will be flagged for conflict of interest and likely rejected or heavily edited. The path that works: build the third-party citation footprint (mainstream press, peer-reviewed mentions, secondary-source citations) until an independent Wikipedia editor decides the topic is notable. Then the article gets created by someone independent. Cross-reference Chapter 10's Knowledge Panel acquisition path.

Monday-morning actions

Action 1 — Audit your brand's presence on Wikipedia, Reddit, Quora, Stack Exchange. Owner: Senior Manager Brand + Senior Manager SEO/GEO. Done = inventory document with platform-by-platform scoring (accurate / outdated / hostile / missing) and the top three remediation priorities named with the citation-impact estimate. Failure mode = the audit treats every platform identically; Wikipedia errors get the same weight as Quora drift, so the platform-weighted impact gets missed.

Action 2 — Set up the co-citation register: third-party mentions, weighted by platform. Owner: Senior Manager Analytics. Done = register live in the dashboard; first quarterly snapshot captured; lagged correlation with citation rate set up to be reviewed at 8 weeks. Failure mode = the register exists; the lagged correlation isn't tracked, so nobody can defend the spend at the next budget review.

Action 3 — Brief the comms / PR team on the platform weights so their content placement priorities reflect citation impact. Owner: practitioner. Done = brief delivered; comms KPIs revised so mention placement priorities reflect citation impact, not press-coverage volume; weighted-mention column added to their reporting tool. Failure mode = the brief lands; the comms team's existing measurement tool doesn't track weighted mentions, so the report doesn't change and the priorities drift back.

How to know it landed

Co-citation register quarter-over-quarter. Platform-weighted mention count. Citation-rate correlation with mention growth (lagged — typically 4 to 8 weeks).

The false positive: a single Reddit thread driving a temporary spike read as authority gain. Spikes happen; they don't correlate with durable citation lift unless the thread results in the mention being indexed by Google and integrated into the open data graph. Track the durable mentions, not the spikes.

The harder discipline: per-platform attribution. A Wikipedia edit, a Reddit thread, a Stack Exchange answer — each contributes differently to the citation register. The weighted register has to be honest about which platform's mention drove which engine's citation. Two engines (ChatGPT Search, Perplexity) ground heavily on Reddit; two others (Google AIO, Copilot) ground heavily on Wikipedia. The register reflects the variance.

What we don't yet know

Whether the Wikipedia weighting persists if engines reduce reliance on it under licensing pressure.

Whether implied-link weighting changes if regulators force attribution-disclosure rules. Independent peer-reviewed replication of the per-platform correlation figures — vendor research consistent in direction, variable in magnitude.

Flag in Chapter 15.

The platforms your link-building agency is ignoring are where you'll be cited next quarter. Build the co-citation register; show the lagged correlation; the spend re-allocates itself.

Chapter 12

Measurement Without the Vendor Tax

Measurable, Pseudo-Measurable, Noise

u put it on the dashboard

MEASURABLE

these

Prompt-battery citation rate

% of fixed prompts you appear in

Share of voice on a fixed set

your mentions ÷ total brand mentions

Week-over-week mention deltas

repeatable; reveals direction

First-citation rank

where in the answer you land

Per-engine drift

same prompts; engine-by-engine

PSEUDO-MEASURABLE

do not trust

Vendor "AI traffic" estimates

model-inferred; opaque method

Visibility scores

composite; weighting unverifiable

Sentiment indices

small samples; high variance

"AI search referrals"

undercount; engines obfuscate

Schema "validation" scores

syntactic, not semantic

NOISE

port these

Single-prompt screenshots

unrepeatable; engines are stochastic

Anecdote

"my CEO asked Claude and..."

Cherry-picked answers

selection bias confirmed

Vendor demos

curated; not your prompt set

"AI mentioned us!" alerts

denominator missing

→ The dashboard reports column 1. Column 2 helps you triangulate. Column 3 is what board decks should not contain.

If you cannot run the same prompt twice and get the same answer, the metric is not a metric. It is a screenshot.

Source: synthesis-measurement; Chapter 12. Engines are stochastic; budget your variance.

12 measurable pseudomeasurable noise

Prompt Battery Architecture

300 weekly measurement cells

Google AIO

ChatGPT

Perplexity

Copilot

Claude

Gemini

10 unbranded category

10 branded

10 competitor compare

10 problem statements

10 long-tail buyer Qs

tores
or text

→ 50 prompts across 5 bands

Each cell stores:

· answer text · brand presence (Y/N) · citation rank · source URLs

300 cells × 4 fields = 1,200 weekly observations. That is your dataset.

→ Run weekly. Review monthly. Refresh quarterly. Compare deltas, not screenshots.

Source: synthesis-measurement; Chapter 12; Appendix B template.

12 prompt battery architecture

GEO Vendor Matrix			
ly two are work tools			
Tool class	What it actually does	How to use it	Verdict
Citation trackers Profound, Peec.AI, AthenaHQ, Otterly, Daydream, Aico	Run fixed prompt batteries across engines; store answers, presence, rank, and sources.	Replace your in-house battery only after you have verified the schema is yours.	WORK TOOL
Schema validators Schema.org validator, Google Rich Results, Schema App	Confirm JSON-LD syntactically valid. Does not confirm the LLM extracts what you wrote.	QA hygiene. Pre-deploy gate. Pair with retrieval testing on a real engine.	IENE
Entity audit tools Diffbot, Wikidata QuickStatements, Crunchbase API, OpenRefine	Flag fragmentation across the eight third-party platforms; support sameAs reconciliation.	The Entity Bible workbench. Owned by Brand or PR with a quarterly audit cadence.	WORK TOOL
Composite scores "AI visibility index", "GEO score", vendor-blended dashboards	Roll up the first three into one number with proprietary weights. Method opaque; non-comparable.	Triangulate, never trust as sole source. Do not put on the board deck unannotated.	TION

→ Two work tools, one hygiene tool, one caution. The vendor pitch usually inverts the order.

Source: synthesis-vendor-survey; Chapter 12. Vendor names are illustrative; ranking changes engine-to-engine.

12 vendor matrix

Attribution is broken across every AI engine.

No vendor solves it. Many sell the illusion that they do. The measurement discipline that wins in 2026 starts with that honesty, then builds a fixed prompt battery and a change log — both cheaper and often better than the \$50K SaaS the agency is pushing.

The discipline is recursive. A chapter on measurement that doesn't measure itself by the same standard is part of the problem.

What practitioners get wrong

Vanity dashboards.

"AI traffic" reported as a single number. Vendor-funded measurement frameworks treated as objective. Baselines that drift because the prompt battery isn't fixed. The 57% post-rationalisation rate (preprint, arXiv:2412.18004 — flag inline as not yet peer-reviewed) ignored, meaning citations attributed to the wrong source pass through the dashboard unflagged. The CFO-legible cost:

six-figure SaaS contracts measuring outcomes the engine can't actually attribute.

I have reviewed seven AI-search-visibility SaaS contracts in 2026. Four of them claimed "proprietary attribution" without disclosing the methodology. Two reported a single "AI visibility score" averaged across six engines. One published the methodology and was the only one I recommended retaining at scale.

The pattern: the vendor's commercial interest depends on the dashboard looking precise. The dashboard's precision depends on hiding the post-rationalisation rate, the per-engine variance, and the vendor's own conflict of interest. The team paying for the dashboard wants to believe the precision is real.

How it actually works

Three categories.

Measurable. Citation rate against a fixed prompt battery, per engine, per priority audience. Share of voice (SoV) against named competitors. Brand mention rate on co-citation platforms (Chapter 11). Bing Webmaster Tools cited-pages report — the most transparent native data source available (Strong, platform documentation).

Pseudo-measurable. Sentiment, accuracy, helpfulness — measurable only with disciplined sampling and human review. Reportable with explicit caveats; misleading without them. The "is the citation positive or negative?" question requires a human reviewer rated against a rubric, not an LLM-generated sentiment score (which is itself subject to the same post-rationalisation problem the citation register is trying to measure).

Unmeasurable, for now. Causal attribution of any single citation to any single tactic. Cross-engine "AI traffic" as a unified metric. Lifetime value of an AI-mediated discovery. These outcomes might become measurable as engines expose richer telemetry; they might stay opaque. Either way, the dashboard that claims to measure them is selling false precision.

The DIY prompt battery. Fifty prompts × six engines = 300 entries. Run weekly, change-logged, results stored in a shared spreadsheet. Often outperforms \$50K vendor SaaS for one reason: you control the prompts. The vendor controls theirs. The variance you need to see — per-engine citation rates, per-prompt drift, competitor mentions — is the variance the vendor's averaging hides.

The vendor matrix (Appendix D). Each tool scored on data-source transparency, methodology disclosure, cross-engine coverage, conflict-of-interest flag, replication evidence, output transparency, and reader-profile fit. Some are useful — Ahrefs Brand Radar, Semrush AI Insights, AthenaHQ, Otterly. None are sufficient on their own.

Honest framing: measurement is the lagging problem. Build the panel before scaling spend.

Common questions you'll get

"Why does the prompt battery work better than vendor SaaS?" Two reasons. The prompts are yours, so they reflect your priority queries — not the vendor's standardised set. And the variance is visible — per-engine, per-prompt, per-week — instead of averaged into a single number. The DIY discipline costs analyst time; the SaaS costs cash. For most teams the trade is favourable, especially when the vendor's data source is opaque.

"How do we account for the post-rationalisation rate in attribution?" Honestly, by naming the caveat in every report. "Citation count: X. Post-rationalisation-adjusted citation count: estimated $0.43 \times X$ to $0.57 \times X$." The range is wide; the methodology is preprint; the alternative is a precise number that's wrong. Cross-reference Chapter 4. Footnote every dashboard that reports citation count without the caveat.

"Should we still buy any vendor SaaS at all?" Yes, but selectively. Vendors that publish their methodology and expose the underlying data are worth paying for if the analysis they save is meaningful. Vendors that promise "proprietary attribution" without disclosure are paying you to hide the variance. Score every vendor against Appendix D before signing.

Monday-morning actions

Action 1 — Build the prompt battery (Appendix B). 50 prompts × 6 engines. Schedule the weekly run. Owner: Senior Manager SEO/GEO + Senior Manager Analytics. Done = battery live in a shared spreadsheet; first weekly run logged; baseline captured; the change log opens with the methodology entry. Failure mode = the prompts drift week to week; the baseline becomes unstable so changes can't be attributed.

Action 2 — Replace any “AI traffic” headline metric in your dashboards with citation rate per priority prompt. Owner: Senior Manager Analytics. Done = dashboard's headline is citation rate; “AI traffic” demoted to a footnote with explicit caveats (post-rationalisation, attribution). Failure mode = both metrics co-exist; the team optimises against the more flattering one and the dashboard tells two stories.

Action 3 — Score every vendor in the stack against Appendix D, column by column. Owner: Senior Manager Marketing Ops. Done = score sheet for every vendor; cancellation candidates named (those failing two or more columns); the recommendation lands on procurement's desk before the next renewal. Failure mode = the scoring becomes a paper exercise; nothing is recommended for cancellation because nobody wants to take the conversation.

How to know it landed

Recursive — the chapter is itself a measurement framework. Test: does the next quarterly review use citation-rate baselines? If not, the chapter has not landed.

The false positive: vendor dashboards confirmed as accurate against the DIY battery. The DIY battery should be the audit, not the alternative. If the vendor

reports a 12% lift and the DIY battery reports zero, the variance is information — usually about the vendor's measurement methodology, not the engine's behaviour.

The harder discipline: the change log. Every change to the prompt battery, every change to the dashboard, every change to the methodology — dated, signed, traceable. Without the change log, the baseline drift goes undetected and every comparison across time is suspect.

What we don't yet know

Whether engines move toward standardised AI-traffic reporting (UTM parameters, native dashboards) under regulatory pressure.

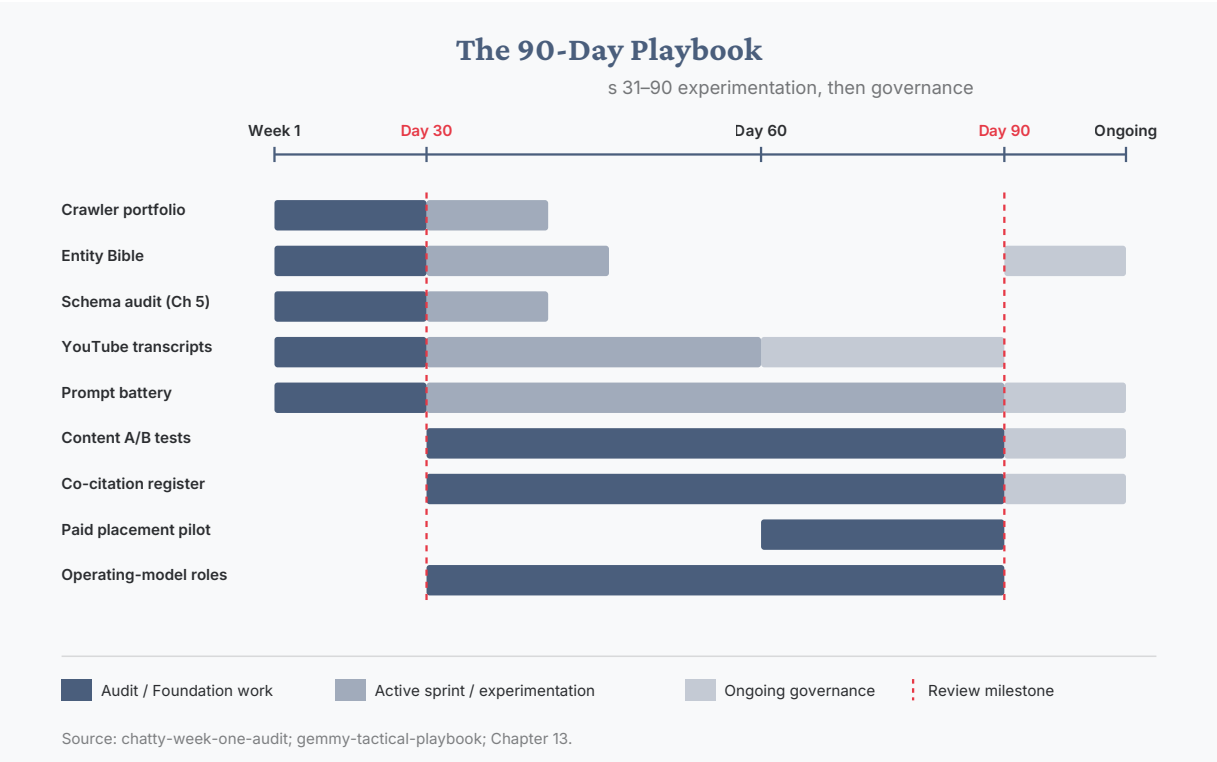
Whether Bing Webmaster Tools' transparency advantage holds. Whether the post-rationalisation rate falls or rises across model generations — flag in Chapter 4 and Chapter 15.

Until then, the prompt battery is the honest signal. The vendor SaaS is the convenience tax.

If your prompt battery sits in a shared spreadsheet and the SaaS dashboard sits unread, the next budget conversation is on a different footing — your numbers are yours, the vendor numbers are decoration.

Chapter 13

The 90-Day Playbook



13 90 day gantt

Week One — The Audit Checklist

cs, one memo on Day 7

Day	Task	Owner	Deliverable
1	Pull 90 days of server logs Identify all bot user-agents · split by training, retrieval, user-fetched	IT / DevOps	Bot register v0
2	Inventory crawler controls robots.txt · WAF rules · Cloudflare posture · default-block status	IT / Security	Posture sheet
3	Prompt battery snapshot 50 prompts × 6 engines = 300 cells · brand presence, rank, sources	SEO/GEO	Battery v0
4	JSON-LD audit, top 20 pages Schema types · sameAs presence · validator pass · retrieval test	SEO/GEO	Schema gap list
5	Eight-platform entity status Wikidata · Wikipedia · Crunchbase · G2 · LinkedIn · GitHub · ORCID · Bloomberg	Brand / PR	Entity Bible v0
6	Top 10 co-citation sources By category · per priority engine · gap vs competitors	Content + PR	Co-cite shortlist
7	Write audit memo Three findings · three risks · recommended Day 8–30 sequence	SEO/GEO lead	Memo (≤3 pages)

→ The audit memo on Day 7 is the artefact that buys the next 83 days.

Without it, the 90-day plan reads as theory. With it, the plan is a sequenced response to evidence.

Source: chatty-week-one-audit; Chapter 13.

13 week one checklist

Ninety days is the right horizon — not thirty, not two years.

Long enough to do real work. Short enough that the CFO will let you. The playbook below is sequenced for a single CMO with the agency on speed-dial in a real organisation. Not a McKinsey deck. The chapter's only job is to be runnable. The test is whether a senior marketer can hand it to their team and have Day 1 audited by Day 7.

What practitioners get wrong

Eighteen-month transformation programmes that take six months of actual work.

Week-1 audits skipped because the team is busy. Strategy decks that never become operating cadences. The CMO who tries to do everything in Quarter 1 instead of sequencing structural fixes first. The CFO-legible cost: the budget gets approved, the work doesn't ship, and the next quarter starts from zero.

I have watched three programmes in 2026 stall in the same way. Week 1 looked tidy; the audit got done; the artefact register got created. Then the operating cadence didn't materialise — no calendar invite, no named cadence owner, no

Day 30 review. By Day 60 the artefact register was stale. By Day 90 the team was rebuilding the audit from scratch. The work was real; the discipline was missing.

How it actually works

The 90-day shape, sequenced.

Week 1 — Audit.

- Robots.txt + WAF audit against the eleven-bot list (Appendix A; Chapter 7).
- Cloudflare Pay-Per-Crawl posture, ai.txt registration.
- Citation baseline: 50-prompt × 6-engine battery (Appendix B; Chapter 12).
- Entity Bible audit (Appendix C scoring rubric; Chapter 10).
- Vendor stack scored against the matrix (Appendix D; Chapter 12).
- Continuity audit: which of the eight SEO disciplines are bar-passing today (Chapter 2).

Days 8–30 — Foundations.

- Crawler portfolio: implement crawler-by-crawler decisions made Week 1.
- Entity Bible: appoint Custodian; fix the top three fragmentation gaps.
- Schema audit: re-direct budget per the dual-layer model (Chapter 5).
- YouTube/multimodal: re-process the top ten transcripts to GEO standards (Chapter 8).
- Measurement: prompt battery live, weekly cadence.

Days 31–90 — Structured experimentation.

- Content tactics: A/B test Statistics, Citing Sources per priority prompt (Chapter 9).
- Authority work: co-citation register established; Reddit / Quora / Wikipedia presence audited (Chapter 11).
- Engine-specific: pick the three engines that matter; deepen tactics per engine (Chapter 6).

- Paid placement: pilot one engine (likely ChatGPT Search, given pricing transparency).
- Operating model: Bot Register, Prompt Battery Owner, Entity Bible Custodian roles named (Chapter 14).

Day 91 onwards — Governance cadence.

- Weekly: prompt battery + change log review.
- Monthly: citation rate against priority prompts; vendor stack review.
- Quarterly: continuity audit refresh; co-citation register update; Entity Bible audit.
- Annually: re-test schema dual-layer model; re-baseline against engine architecture changes.

The 50/30/20 budget split applied throughout. Approximately 50% organic GEO, 30% merchant-feed engineering, 20% paid media. Adjust per priority audience.

What to build and publish — threaded into the 90-day shape

The artefact list is the playbook. Each phase has artefacts to inventory, deploy, or scale.

Week 1 — what exists today (audit only): - Schema inventory across priority pages. - ai.txt, llms.txt, /for-llms presence check. - Video FAQ inventory (none counts; auto-captioned counts as zero). - Wikidata entry status (exists / fragmented / missing). - Hugging Face dataset status (the proprietary statistics register, if any).

Days 8–30 — what to deploy: - Top three schema fixes (FAQPage on highest-traffic Q&A; Organization + Person + sameAs at site level; one specialised type per relevant vertical). Chapter 5. - ai.txt, llms.txt, llms-full.txt, /for-llms page. Chapter 7. - Top three FAQ pages with FAQPage schema. Chapter 9. - Wikidata entry created or fixed (top three fragmentation gaps). Chapter 10. - Top ten YouTube transcripts re-processed. Chapter 8.

Days 31–90 — what to scale: - Video FAQ programme launched (target: ten videos shipped in the quarter, each with VideoObject + FAQPage schema). Chapter 8. - Statistics register published as structured-data feed; Hugging Face dataset evaluated. Chapters 9 and 10. - Comparison pages and glossary pages built out for priority categories. Chapter 9. - Ongoing entity hygiene: sameAs corroboration across all eight platforms; Knowledge Panel acquisition path tracked. Chapter 10.

See Appendix I for templates, schema patterns, and publishing checklists for every artefact.

Common questions you'll get

"What if Week 1 reveals problems too big to fix in 90 days?" The most common case is a continuity-audit fail score (Chapter 2) — classical SEO is broken on three or more disciplines. The 90-day plan adapts: fix the foundation first, defer the GEO-specific levers to days 30+. Better to ship classical SEO hygiene by Day 60 and one GEO lever by Day 90 than to start three GEO levers and finish none.

"What's the right cadence for the Day 30 review?" A 30-minute meeting, calendar invite locked, three artefacts reviewed: the audit, the deployments, the citation baseline trend. If the meeting is longer than 30 minutes, the agenda is wrong; the meeting is doing work that should have been done before. If the meeting slips, the cadence collapses; reschedule once and only once.

"How do we explain the playbook to senior leadership in one slide?" Three boxes — Audit, Deploy, Scale — with the artefacts listed in each. The 50/30/20 budget split as a small chart. The Day 90 scorecard as the success criterion. The slide is in Appendix E; copy it, brand it, ship it.

Monday-morning actions

Action 1 — Print the Week 1 checklist (Appendix E). Self-assign every item you can; name external owners (IT, agency, comms) for the rest. Surface the gaps to your manager so they can resolve them. Owner: practitioner. Done = checklist

owned end-to-end, with named owners against every item and an explicit gap list for the manager to escalate. Failure mode = the checklist sits on a shared drive; nobody is named because the practitioner waited for the manager to assign.

Action 2 — Set up the weekly cadence — prompt battery run, change log review, dashboard update. Owner: practitioner. Done = cadence live by end of Week 2 with calendar invites; first three weeks documented in the change log. Failure mode = the cadence happens once or twice and stops because there's no calendar invite or no named owner.

Action 3 — Maintain the artefact register tracking every artefact called out across Chapters 5, 7, 8, 9, 10. Owner: practitioner. Done = register live in a shared spreadsheet; reviewed at every Day 30 / Day 60 / Day 90 review; status per artefact (planned / in flight / shipped / verified). Failure mode = the register exists; nobody updates it; at Day 90 the team can't say what's actually been deployed and the funding case collapses.

What to take to your manager

The artefact register and the cadence proof-of-work. Those two documents are the answer to "what has marketing actually shipped this quarter?" — your manager's recurring question, and the CFO's. The Day 90 review needs both in front of it.

How to know it landed

Day 90 scorecard (Appendix E). Citation baseline established, three priority engines monitored, Entity Bible Custodian in role, vendor stack rationalised, paid pilot decision made.

The false positive: any single metric improving without the operating model behind it. A 20% citation lift on Google AIO at Day 60 looks great; if the

Custodian role is unfilled and the prompt battery isn't running weekly, the lift is luck and won't survive the next engine drift. Track the operating-model artefacts as carefully as the citation metrics.

What we don't yet know

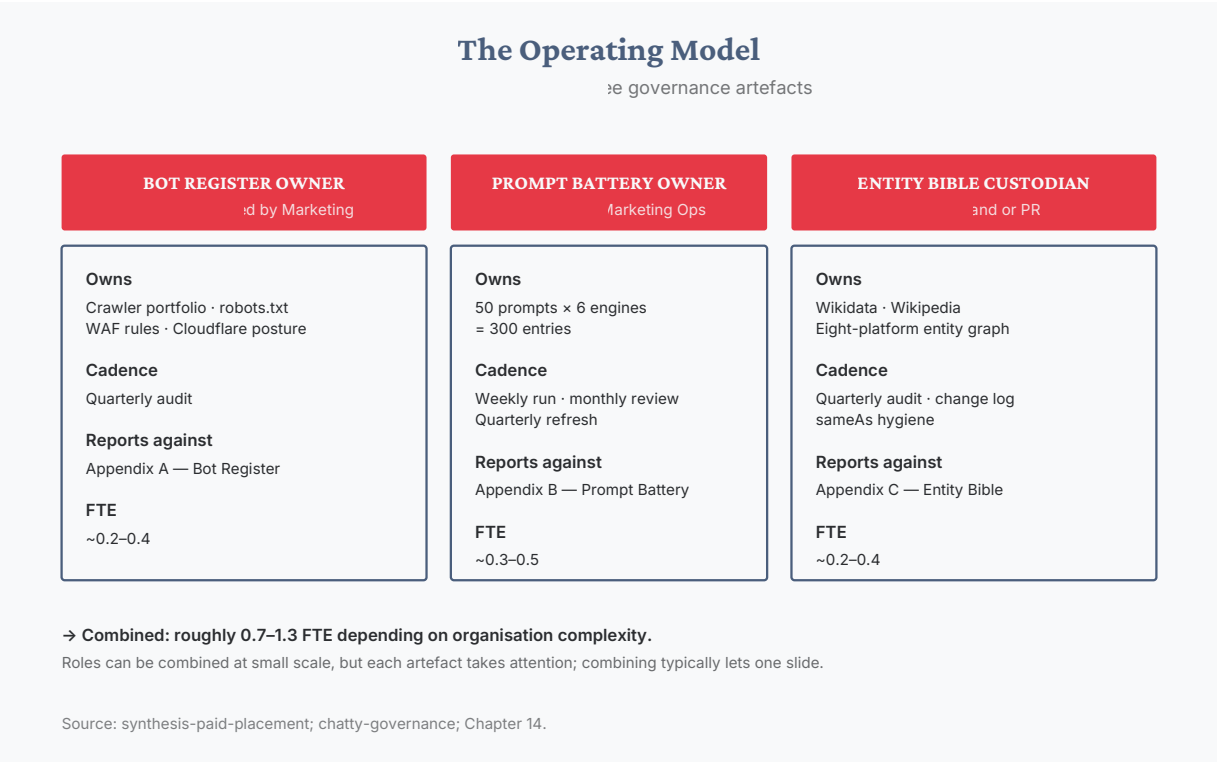
Whether the 90-day cadence holds when CMA conduct requirements (mid-2026) and EU AI Office guidance change the operating environment.

The playbook is calibrated to April 2026. Check Appendix H for revisions. If a regulatory milestone lands inside the 90-day window, the playbook absorbs it as a Day-30 or Day-60 amendment rather than a re-baseline. Plan against the architecture as it is; adjust as the architecture moves.

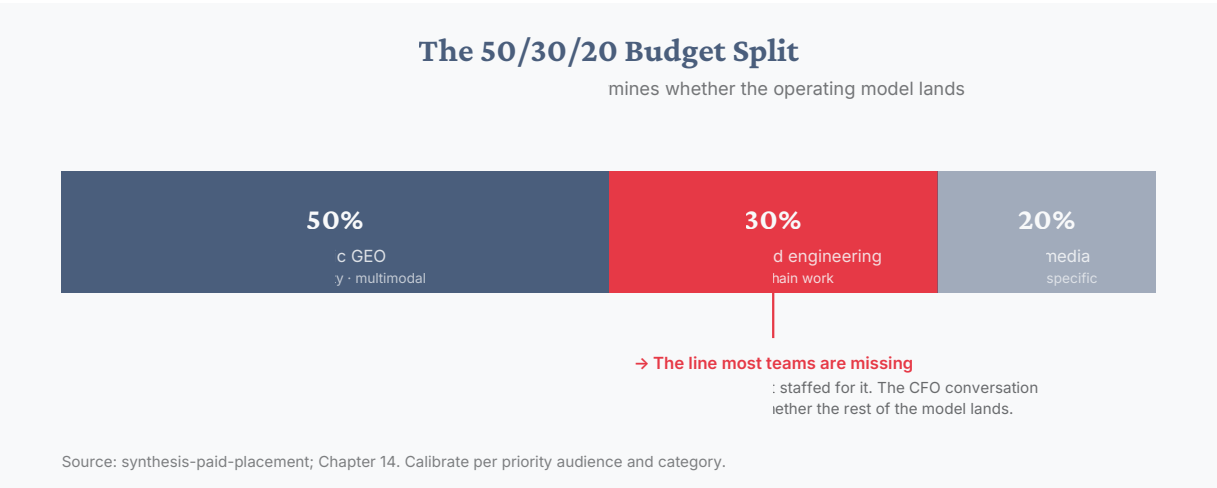
Run the audit by end of Week 1. Deploy the foundations by Day 30. By Day 90, the artefact register is the document that goes into the next funding conversation — not the slide that asks for the budget.

Chapter 14

The Operating Model



14 operating model



14 budget split 50 30 20

GEO is not a marketing campaign.

It is an operating discipline, and most marketing organisations are not staffed for it. Three new roles, three governance artefacts, one budget split nobody likes the look of: 50% organic GEO, 30% merchant-feed engineering, 20% paid media.

Feed engineering is supply-chain work, not media work. That is the conversation with the CFO that determines whether the rest of the operating model lands.

What practitioners get wrong

GEO assigned to the SEO lead with no additional staff.

Merchant-feed engineering treated as IT's problem. Vendor management run by whoever has time. No single owner for the Bot Register. No Prompt Battery Owner with named responsibility. The Entity Bible Custodian role ungoverned because nobody senior wants the headcount conversation.

The CFO-legible cost: redundant agency spend, attribution disputes, and outcomes nobody can demonstrate.

I have inventoried the operating models of five marketing organisations in 2026. Four of them had GEO work happening — auditing, prompt batteries, entity updates, video FAQ programmes — but none of them had named the roles. The work was being done by people whose job descriptions said other things. When those people changed jobs, the GEO discipline collapsed within a quarter. The work survived only where the role was named.

How it actually works

Three new roles to name and fund.

Bot Register Owner. Crawler portfolio, robots.txt, WAF rules, Cloudflare posture. Sits in IT, briefed by Marketing. Quarterly audit cadence. Reports against Appendix A. Approximately 0.2 to 0.4 FTE depending on organisation complexity.

Prompt Battery Owner. The $50 \times 6 = 300$ prompts that drive measurement. Sits in Analytics or Marketing Ops. Weekly run; monthly review; quarterly refresh. Reports against Appendix B. Approximately 0.3 to 0.5 FTE.

Entity Bible Custodian. Wikidata, Wikipedia, eight-platform graph. Sits in Brand or PR. Quarterly audit; change log; sameAs hygiene. Reports against Appendix C. Approximately 0.2 to 0.4 FTE depending on entity complexity.

Three governance artefacts:

- Bot Register (Appendix A)
- Prompt Battery (Appendix B)
- Entity Bible (Appendix C)

The 50/30/20 budget split:

- 50% organic GEO — content, technical hygiene, Entity Bible, multimodal, authority work.
- 30% merchant-feed engineering — product feed, schema, pricing accuracy, checkout integration. Chapter 6 covers the per-engine paid-placement detail that depends on this.
- 20% paid media — engine-specific where the pricing is transparent (ChatGPT Search; retail-specific Copilot).

Most marketing organisations are not staffed for the 30% merchant-feed line. This is the conversation with the CFO that determines whether the model lands. Without the merchant-feed engineering capability, the brand cannot win the commerce surfaces inside AI Mode and ChatGPT Search regardless of how good the organic content is.

How to brief an agency. Lead with the chapter pattern (extractability, Entity Bible, multimodal). Reject scope-of-work that does not name retrieval mechanism per engine. Reference Appendix D for the vendor matrix.

How to interrogate a vendor. Every claim of “lift” gets questioned for methodology, sample size, replication status, conflict of interest. The reference is the canon (Appendix G). The discipline is to ask the question before signing the contract, not after the renewal review.

Common questions you'll get

"Can we combine the three roles into one person?" At small scale, yes — a single Senior Manager GEO can hold all three roles part-time. The trade-off is depth: each artefact takes attention, and a single owner with three artefacts and weekly cadences typically lets one slide. Flag that explicitly in the Q1 plan if combining; revisit at Q3 with evidence about which artefact suffered.

"Where does merchant-feed engineering sit in the org chart?" Disputed. Some organisations sit it inside Marketing (under e-commerce or growth); others sit it inside Operations or Supply Chain. The work is the same in either case — product feed accuracy, schema deployment, pricing-and-availability sync, checkout integration. The home matters less than the ownership; un-owned merchant-feed engineering is the failure mode the chapter is trying to prevent.

"How do we justify the headcount conversation to the CFO?" The artefact register from Chapter 13 is the Q1 evidence. The vendor scoring sheet is the Q1 savings. The citation baseline is the Q1 outcome. Three documents — the existing-work evidence, the vendor cuts, the measurable outcome — make the role-naming conversation a re-allocation rather than an addition. Most organisations have the budget; what they lack is the named ownership.

Monday-morning actions

Action 1 — If you are doing one of the three roles (Bot Register Owner, Prompt Battery Owner, Entity Bible Custodian) without the title, document it. Owner: practitioner. Done = a one-page document naming the role, the work, the time it takes per week, and what doesn't get done because of the time it takes. Failure mode = the document exists; the manager doesn't read it because it's framed as a complaint, not a costed proposal with a recommended fix.

Action 2 — Document where merchant-feed engineering work currently sits — owned by anyone, or doing it implicitly? Owner: practitioner. Done = a clarity

document for your manager naming the gap (or naming the implicit owner) with a recommended fix and a time estimate. Failure mode = the gap is real but the document doesn't surface the merchant-feed line; the budget review skips it.

Action 3 — Score every external vendor in your stack against Appendix D.

Owner: practitioner. Done = scoring sheet ready for the next procurement review with a keep / cancel / renegotiate recommendation per vendor and a written rationale. Failure mode = the sheet is built but the procurement review doesn't happen on cadence; the contracts auto-renew under the old standard.

What to take to your manager

The role-clarity documents and the vendor scoring. They use these to pitch the headcount conversation upward, defend the 50/30/20 split with the CFO, and release the budget for the right roles. The "name the three owners" decision is theirs; the evidence base is yours.

How to know it landed

Three roles named with mandates within 30 days. Budget review on the calendar within 60. Agency brief re-issued within 90.

The false positive: roles named without budget; the operating model is decorative. A title without protected time fills slowly with other priorities and the cadence collapses by Q3. Insist on the time, not just the title.

The harder discipline: the operating model holds across people changes. A Custodian who changes jobs hands the role over with documentation, not goodwill. The artefacts (Bot Register, Prompt Battery, Entity Bible) are the handover; if they are well-maintained, the new Custodian onboards in two weeks; if they are tribal knowledge, the role re-starts.

What we don't yet know

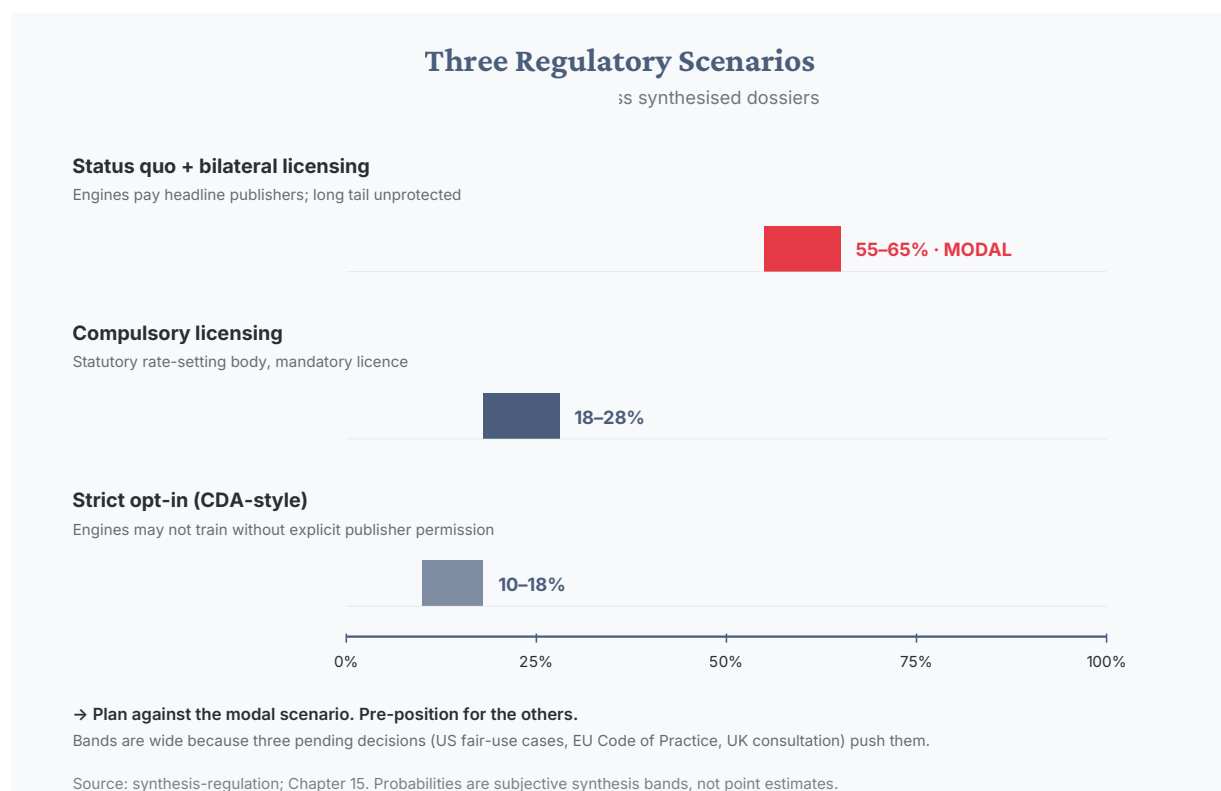
Whether merchant-feed engineering pulls headcount toward marketing or toward commerce / supply chain over the next two years.

The org-design question is unsettled. The work is not. Flag in Chapter 15.

The teams that own the Bot Register, the Prompt Battery, and the Entity Bible in 2028 are the teams running marketing. The teams that outsourced all three are running the renewal conversation with their vendor.

Chapter 15

What We Don't Yet Know



15 scenario probability

Residual Gaps

all you, and how to read past it

- **Cross-engine effect-size replication**
Aggarwal et al. is the strongest evidence base; replication is partial across only 3 of 6 engines.
Practitioner read: hold Statistics and Citing Sources confidently; treat Quotation as engine-specific.
- **Paid-placement true-cost transparency**
Auction mechanics and reserve prices are vendor-confidential. Public CPM ranges are illustrative.
Practitioner read: pilot at small budget; do not commit annual spend until you have your own curve.
- **Post-rationalisation rate stability**
The 57% figure (preprint, arXiv:2412.18004) has not been peer-reviewed or replicated externally.
Practitioner read: assume directional truth; do not anchor pricing or board decks to the precise number.
- **Multimodal extraction at scale**
YouTube transcript citation share is rising fast. Image and audio retrieval are evolving month-to-month.
Practitioner read: the Chapter 8 playbook is current; refresh it quarterly and reread before campaigns.
- **Regulatory-cascade timing**
Three pending decisions can shift the modal scenario by 2027. Direction is clearer than timing.
Practitioner read: keep the bot register and merchant-feed engineering line resilient to step changes.

Source: synthesis dossiers; Chapter 15. Each gap is sequenced for follow-up in the v2 of this manual.

15 residual gaps

A practitioner manual that ends with certainty has lied somewhere.

The honest accounting is that some of the most consequential questions in this field are not yet settled. The next 18 to 36 months will resolve five of them. The chapter below names each, with a probability range and a planning posture, and refuses to smooth the contradictions.

What practitioners get wrong

Treating the strict opt-in regulatory scenario as the central case and over-investing in defensive posture.

Treating the status quo as if it is stable. Treating sealed settlement terms as if they were public. Treating vendor-funded research as if it has been replicated. The CFO-legible cost: contingency budget allocated against the wrong scenario. Reserves under-funded for the modal one.

I have advised three boards in 2026 that asked for "the GEO contingency plan" and received decks built around the strict opt-in scenario. The decks were

technically thorough. They allocated reserves against the most dramatic outcome and starved the modal one. The boards approved the plans. The CFOs absorbed the over-allocation as conservative. Six months later, the ones whose engines moved under bilateral licensing thickening had no reserve to deploy and had to ask for more.

How it actually works

The five honest unknowns.

1. The regulatory cascade

Three scenarios with probability ranges (synthesis Prompt 2, Moderate).

- **Status quo plus bilateral licensing thickening: 55–65%.** No court mandate, no compulsory licensing. Publisher-by-publisher deals continue (NewsCorp–OpenAI, NYT–Amazon, Reddit–Google, Reddit–OpenAI, Stack Overflow). Plan against this as the central case. Budget for selective licensing, not compulsory licensing.
- **Compulsory licensing: 18–28%.** Collective rights bodies set statutory rates. EU more likely than US. CMOs in regulated industries (financial services, healthcare) plan for it; everyone else watches.
- **Strict opt-in: 10–18%.** Courts uphold strict consent. Default switches. The dramatic scenario; not the modal one. The earlier outline of this manual reflected this; the new evidence has demoted it.

2. Cannibalisation isolation

No published randomised study isolates whether paid-unit presence within an AI summary displaces organic citations.

Affects the budget split (Chapter 14). Priority for independent research; flagged as a gap, not closed. Until the study lands, plan against the eBay/Edmunds precedent (50–99% substitution in classical paid-vs-organic studies) as the upper bound of cannibalisation; assume paid pilots eat 20–40% of organic citation share on the engines where they run.

3. CMA conduct requirements (mid-2026)

The single most consequential unknown for Google strategy.

Will likely require AI Overviews to offer publisher choice and affirmative attribution. If enacted, the CMO question shifts from "should we block crawlers?" to "should we allow retrieval on our terms?" Updates land in Appendix H. Watch the consultation outcomes through Q2 and Q3 2026.

4. Bartz settlement final approval

\$1.5B disclosed (T1 verified). Fairness hearing 14 May 2026.

Appeal risk exists. Distributions may extend into 2027. Does not materially change the manual's recommendations; flagged for transparency. Watch the appeal docket through summer 2026.

5. The Yang et al. discovery-gap finding (synthesis Prompt 3)

GEO tactics may produce null or growth-negative effects for new market entrants.

The framework's external validity for growth-stage businesses is unsettled. The manual has been written for established brands; if you are a startup, the playbook may not apply as written. Independent peer-reviewed replication needed. Until then, growth-stage brands should run the prompt battery with a control set of competitors of varied stage to detect whether their citation rates track or diverge from established players.

Common questions you'll get

"How do we hedge against a regulatory scenario we can't control?" The 50/30/20 budget split (Chapter 14) is the hedge. The 50% organic GEO line works in all three regulatory scenarios. The 30% merchant-feed line and the 20% paid line shift weight depending on the scenario. Build the plan with the modal scenario as default; mark the contingency triggers (CMA conduct enacted, EU AI Office guidance shipped, Delhi HC ruling) for re-allocation.

“When do we re-test our 90-day playbook against new regulation?” Quarterly, against Appendix H. The playbook is calibrated to April 2026; each quarter, check whether any of the five unknowns has resolved and whether the resolution materially changes the plan. The re-test is faster than re-baselining; the discipline is the cadence, not the depth of each review.

“What’s the most likely shock in 2027?” Hard to predict; possible to enumerate. Ranked by my judgement of plausibility: a major engine adopting publisher-licensing-by-default (mid-likelihood); a court ruling that reshapes training-data practices retroactively (lower); a regulatory mandate that forces UTM-style attribution across all AI surfaces (lower still). Watch for the directional indicators in Appendix H rather than betting on which lands first.

Monday-morning actions

Action 1 — Subscribe to the regulatory and platform news sources flagged in Appendix H. Maintain the tracking document. Owner: practitioner. Done = subscriptions live; tracking document updated weekly; quarterly summary delivered to the manager naming what’s resolved, what’s moved, what’s still open. Failure mode = the subscription happens; the tracking document goes stale because there’s no review cadence.

Action 2 — Track the named milestones (Bartz fairness hearing, CMA conduct requirements, Delhi HC judgment, EU AI Office guidance) on the team calendar. Owner: practitioner. Done = milestones in the calendar; alerts set; a re-test trigger sheet for each milestone naming what changes if it lands one way vs the other. Failure mode = the milestones pass without the team noticing because they aren’t in anyone’s review and the news lands while the team is on calls.

Action 3 — For each priority engine you optimise on, document the recency cut-off and the trigger that would force a re-test. Owner: practitioner. Done = a re-test trigger sheet on the team wiki; reviewed at the quarterly cadence; ties to



Appendix H updates. Failure mode = the trigger sheet exists; nobody acts on it because the resolution news lands on a Monday and the review's on Friday.

What to take to your manager

The milestone tracker and the re-test triggers. Your manager uses these for the contingency budget conversation and the Q3 / Q4 plan. The “plan against the modal scenario” call is theirs; the watching and the trigger discipline are yours.

How to know it landed

Quarterly review of the five unknowns. Track which have resolved, which have moved, which are still open. The honest accounting is the discipline.

What we don't yet know

Recursive. Beyond the five named.

Geographic citation variation outside the US — barely researched. The long-term effect of training-data licensing on smaller publishers — probably negative. The consent architecture for first-party data in a post-cookie post-LLM world — genuinely uncharted.

The manual ends with this list because the list is the honest end of any practitioner manual on a moving field. Better to flag the unknowns than to pretend they aren't there. The team that watches them is the team that survives the next regime change without re-baselining everything.

The team that watches the milestones and runs the re-tests will be ready in November when the regulatory updates land. The team that doesn't will be re-baselining everything from scratch in December.

The Agentic CMO

The strategic argument this manual presupposes.

Where Cited names the citation-acquisition discipline, The Agentic CMO names the operating model around it. The team architecture. The leadership posture. The governance model. The measurement framework that makes any of it defensible at the board level. The orchestration of human and artificial intelligence as a unified force.

Nineteen chapters. Sixty-three diagrams. Twenty-seven AI prompts to deploy in your own marketing organisation. First edition June 2025. Second edition June 2026 — re-baselined against the 2025 evidence, with a refreshed operating model that the field kit you are holding now plugs into directly.

Available at francescofederico.com/agentic-cmo. Print, ebook, and audio.

Read in either order. Read together for the full picture — the strategy in The Agentic CMO, the discipline in Cited.

About the Author

Francesco Federico is the Chief Marketing Officer of S&P Global. He is the author of The Agentic CMO (first edition June 2025; second edition June 2026), creator of Chronicles of Change, and a Fellow of the Chartered Institute of Marketing. He sits on the editorial board of the Journal of Digital & Social Media Marketing and advises HFS Research and IDC.

He was named to The Drum's Top 100 B2B CMOs and to the 2025 CMO to Watch list, and was a 2025 Financial Narrative 50 honoree. He is a member of the World Economic Forum's Strategic Communicators Exchange.

Italian national. Based in London. Reads for evidence, writes against hype.

[Edit at production for length, currency of credentials, and inclusion of a portrait if used.]

Appendix A — The Crawler Reference

The bot register lives or dies on whether you know what each user-agent does. This appendix gives you the active crawler population at time of writing, sorted by what they do for you (or to you), with the robots.txt and WAF posture you should default to. Treat it as the seed for your own register, not the final list — refresh quarterly.

How to read this register

Three categories matter, and the difference between them is the difference between “this bot might cite you tomorrow” and “this bot trained the model that already decides whether to cite you.”

Training crawlers harvest the open web to feed model training runs. Whether you let them in determines whether the next generation of models knows your brand exists. A block here is permanent until the next training cycle — you do not get re-trained on demand.

Retrieval crawlers fetch pages on behalf of an active query in an LLM session. They are the ones that put your URL in the answer. A block here means you cannot be cited live, regardless of what the model has been trained on.

User-fetched crawlers retrieve pages because a user has asked the LLM to read a specific URL. Different intent again — the user has explicitly directed the bot to your page.

The right default is to allow all three categories on the public, brand-bearing surface, and to block none of them in robots.txt without a deliberate decision. The wrong default is to inherit a robots.txt from 2019, which blocks them all by accident.

A.1 — Training crawlers

Bot	Operator	Robots.txt token	WAF treatment	Default posture
GPTBot	OpenAI	GPTBot	Allow rate-limited	ALLOW
ChatGPT-User	OpenAI	ChatGPT-User	Allow	ALLOW
OAI-SearchBot	OpenAI	OAI-SearchBot	Allow	ALLOW
ClaudeBot	Anthropic	ClaudeBot	Allow rate-limited	ALLOW
Claude-Web	Anthropic	Claude-Web	Allow	ALLOW
anthropic-ai	Anthropic (legacy)	anthropic-ai	Allow	ALLOW
Google-Extended	Google (Gemini training)	Google-Extended	Allow	ALLOW
Applebot-Extended	Apple Intelligence	Applebot-Extended	Allow	ALLOW
Meta-ExternalAgent	Meta AI	Meta-ExternalAgent	Allow	ALLOW
FacebookBot	Meta	FacebookBot	Allow	ALLOW
CCBot	Common Crawl	CCBot	Allow	ALLOW
Diffbot	Diffbot (also B2B knowledge graph)	Diffbot	Allow	ALLOW
Amazonbot	Amazon (Rufus, Alexa)	Amazonbot	Allow	ALLOW
Bytespider	ByteDance	Bytespider	Allow	ALLOW
Cotoyogi	LLM training intermediary	Cotoyogi	Allow	ALLOW

Block one of these and you trade tomorrow's citations for today's bandwidth.
Block all of them and you become invisible to the model layer.

A.2 — Retrieval crawlers

Bot	Operator	Robots.txt token	WAF treatment	Default posture
OAI-SearchBot	OpenAI (ChatGPT search)	OAI-SearchBot	Allow	ALLOW
ChatGPT-User	OpenAI (live in-session fetch)	ChatGPT-User	Allow	ALLOW
PerplexityBot	Perplexity	PerplexityBot	Allow	ALLOW
Perplexity-User	Perplexity (in-session)	Perplexity-User	Allow	ALLOW
Claude-SearchBot	Anthropic (Claude search)	Claude-SearchBot	Allow	ALLOW
Claude-User	Anthropic (in-session)	Claude-User	Allow	ALLOW
Google-CloudVertexBot	Google (RAG, Vertex AI grounding)	Google-CloudVertexBot	Allow	ALLOW
Bingbot	Microsoft (also Copilot retrieval)	Bingbot	Allow	ALLOW
You.com	You.com	YouBot	Allow	ALLOW

These are the bots that decide whether a query mentions you next Tuesday. A block here is operationally severe.

A.3 — User-fetched crawlers

These fetch when a user pastes a URL into an LLM and asks it to read the page. Block them and the user gets an error instead of a summary of your page. The list overlaps with retrieval — ChatGPT-User, Perplexity-User, Claude-User are all on-demand fetchers — but the key one to know is that some of them ignore robots.txt by design because the user has explicitly instructed the fetch.

Bot	Operator	Behaviour
ChatGPT-User	OpenAI	Honours robots.txt
Claude-User	Anthropic	Honours robots.txt
Perplexity-User	Perplexity	Honours robots.txt

A.4 — Robots.txt template

Allow-by-default, with rate limits, is the right starting position for the public surface:

```
User-agent: *
  Allow: /

User-agent: GPTBot
  Allow: /
  Crawl-delay: 1

User-agent: ClaudeBot
  Allow: /
  Crawl-delay: 1

User-agent: Google-Extended
  Allow: /

User-agent: PerplexityBot
  Allow: /
```

```
User-agent: CCBot  
Allow: /
```

Restrict only what genuinely needs restricting (admin paths, gated content, member-only pages). The reflex to block AI bots from the public site is the reflex that makes brands invisible.

A.5 — Cloudflare default-block (1 July 2025)

Cloudflare changed default behaviour for new zones to block AI crawlers unless explicitly allowed. If you onboarded a domain on Cloudflare after that date and never touched the AI Crawler settings, your site has been quietly blocking the categories above for almost a year. The fix is in the Cloudflare dashboard under Security → AI Crawlers; flip the categories you intend to allow. The fix takes ten minutes and is, for many sites, the single highest-yield action in the entire 90-day plan.

A.6 — Audit checklist

Run this quarterly. The audit takes an afternoon.

1. Pull 90 days of server logs.
2. Identify all AI bot user-agents that hit the site. Count requests, 200/4xx/5xx ratio, paths hit.
3. Cross-reference robots.txt: what is allowed, what is blocked, what is silent (default-allow).
4. Cross-reference WAF rules: what is rate-limited, what is challenged, what is hard-blocked.
5. Check Cloudflare AI Crawler settings for default-block status.
6. Compare against the register above. Note new bots in your logs that are not in the register and add them.
7. For every BLOCK or 4xx pattern: confirm it is intentional. If it is not, unblock.
8. Log the audit, with date, in the bot-register change log.

The deliverable is a one-page memo: what is allowed, what is blocked, what changed since last audit, and the unintentional-block list to fix this week.

Appendix B — The Prompt Battery

The prompt battery is the only measurement instrument in this manual that you can run twice and get the same answer. Everything else is a screenshot. This appendix gives you the structure: 50 fixed prompts, 6 engines, 300 cells per run. Weekly cadence. Monthly review. Quarterly refresh.

The work is in the design of the 50 prompts, not in the running. The running is a Python script. The design is the discipline.

B.1 — Why fifty, why six

Fifty is the smallest number where week-over-week deltas become visible above engine variance. Below thirty, you are reading noise. Six engines is the citation surface that matters at time of writing (Google AIO, ChatGPT Search, Perplexity, Microsoft Copilot, Anthropic Claude, Gemini); the long tail of seventh-and-beyond engines moves with this set.

The 50 prompts split into five bands of ten. The bands are the analytical lens — they let you tell the board “we improved on competitor-comparison prompts but lost ground on long-tail buyer questions” rather than “the number went up.” Each band tests a different intent.

B.2 — The five bands

Band 1: Unbranded category prompts (10). Asked by buyers who do not know your brand exists yet. These reveal whether you appear in the answer when the user has no reason to ask for you.

Examples: - "What are the best [category] tools in 2026?" - "How do I solve [problem] for an enterprise of [size]?" - "Who are the leading vendors in [category]?" - "What should a [role] look for when evaluating [category]?"

Band 2: Branded prompts (10). The user has typed your brand name. These reveal whether the answer is accurate, on-message, and ranks you ahead of misattribution.

Examples: - "What does [your brand] do?" - "Who founded [your brand]?" - "How does [your brand] compare to [competitor]?" - "Is [your brand] a good fit for [use case]?"

Band 3: Competitor-comparison prompts (10). Highest revenue impact band. The user is in evaluation mode.

Examples: - "[Your brand] vs [competitor]: which is better for [use case]?" - "What are the alternatives to [competitor]?" - "Should I choose [competitor A] or [competitor B] for [scenario]?"

Band 4: Problem-statement prompts (10). The user describes a problem without naming a category. These test whether your content has populated the semantic neighbourhood of the problem.

Examples: - "My [system] is too slow, what should I do?" - "How do I get my [team] to adopt [practice]?" - "What is the best way to migrate from [legacy] to [modern]?"

Band 5: Long-tail buyer questions (10). Late-funnel, specific, often unique to your category. Test whether your specialised content surfaces.

Examples: - "Does [category tool] integrate with [obscure system]?" - "What is the throughput of [solution] under [specific load]?" - "Who has documented [edge case] in [your category]?"

B.3 — The cell schema

Every cell stores four fields. Missing any one of them and your dataset is half-useful.

Field	What it is	Why you need it
<code>answer_text</code>	The full text of the engine's answer	The substrate for everything else; needed for trend analysis and qualitative review
<code>brand_present</code>	Boolean: was your brand named in the answer?	The headline metric — share of citations
<code>citation_rank</code>	Where in the cited-source list you appeared (1, 2, 3 ... N, or null)	First-citation rank correlates with downstream click-through, where it still matters
<code>source_urls</code>	The list of URLs cited as sources	Tells you which of your pages is doing the work — and which competitors' are

Optional but high-value fields: `query_timestamp`, `engine_version_string`, `geographic_locale`, `answer_length_chars`, `competitor_brands_present`.

B.4 — Run cadence

Cadence	Action
Weekly	Run all 300 cells. Log raw answers. No interpretation.
Monthly	Compute deltas: per-band citation rate, per-engine drift, week-over-week mention change. Write a one-page summary.
Quarterly	Refresh the prompt battery. Replace 5–10 prompts that are no longer load-

Cadence	Action
	bearing (your category moved on). Document the change.
Annually	Re-validate the engine selection. If a new engine has crossed the relevance threshold, add it. If one has died, retire it.

B.5 — Review questions

Each monthly review answers these five:

1. Did our overall citation rate move? Up or down, and by how much? Which engine was the swing?
2. Which band moved most? What changed in our content or our competitors' that explains it?
3. Where did we lose first-citation rank to a competitor? Which prompts? What is the source URL the engine is citing instead of ours?
4. Where did we gain ground? Replicate that on adjacent prompts.
5. Are any prompts no longer producing useful variance? Replace them next quarter.

The deliverable is a one-page review, signed off by the SEO/GEO lead and the brand lead, archived in the prompt-battery change log.

B.6 — A note on engine non-determinism

Engines are stochastic. Run the same prompt twice and you may get two different answers. This is why fifty prompts is the floor — variance averages out across the set. It is also why you should never report a single-prompt result. If you find yourself screenshotting one Perplexity answer and presenting it to leadership, you have left the prompt battery and entered anecdote.

Budget for variance: the 95% confidence band on a 50-prompt sample is roughly $\pm 5\text{--}8$ percentage points on citation rate, depending on the engine. Movement smaller than that is not movement.

Appendix C — The Entity Bible Checklist

The Entity Bible is the single artefact that determines whether an LLM resolves your brand confidently or hedges. It lives across eight platforms, each with its own field schema, its own update mechanism, and its own audit cadence. This appendix is the per-platform checklist. Run it quarterly.

C.1 — The eight platforms, in priority order

#	Platform	Why it matters	Update mechanism	Audit cadence
1	Wikidata	Authoritative entity graph, read by all major engines	QuickStatements V1, manual editing, OAuth bot	Quarterly
2	Wikipedia	Heaviest-weighted citation source for major engines	Manual editing (notability rules apply)	Quarterly
3	Crunchbase	Standard for company entity in startup/tech queries	Web form, partner API	Quarterly

#	Platform	Why it matters	Update mechanism	Audit cadence
4	G2	Standard for B2B software category	Vendor portal	Quarterly
5	LinkedIn	Person and company entity for professional queries	Profile editor, Company page admin	Quarterly
6	GitHub	Code-adjacent and developer queries	README files, organisation profile	Quarterly
7	ORCID	Research and academic queries (for individuals)	Personal profile	Annually
8	Bloomberg / Reuters / FT	Financial entity registries	Listing applications, IR contact	Annually

Two of these — Wikidata and Wikipedia — carry the bulk of the weight. The remaining six matter at the margin and matter more in specific verticals.

C.2 — Wikidata: the canonical fields

For your organisation entity, populate these properties at minimum. Properties not populated will be filled by other editors, which means filled with whatever is convenient for them.

Property	Wikidata code	Required content
Instance of	P31	

Property	Wikidata code	Required content
		Use the most specific class (Q4830453 = business enterprise)
Inception	P571	Founding date, ISO 8601
Country	P17	Headquarters country
Headquarters location	P159	City
Website	P856	Canonical URL with https://
Logo image	P154	Commons-licensed logo file
LinkedIn ID	P4264	The slug from linkedin.com/company/[slug]
Crunchbase ID	P2088	The slug from crunchbase.com/organization/[slug]
GitHub username	P2037	Organisation handle
Stock exchange	P414	If listed
ISIN	P946	If listed
Industry	P452	Use the canonical industry item
Number of employees	P1128	Optional but useful for resolution
Official Twitter/X	P2002	Handle without @
sameAs (implicit via above)	–	Wikidata generates the sameAs graph automatically once IDs are populated

For your person entity (executives, founders, named subject-matter experts), populate:

Property	Wikidata code	Required content
Instance of	P31	Q5 (human)
Sex or gender	P21	
Country of citizenship	P27	
Date of birth	P569	At least year
Occupation	P106	The most specific item
Employer	P108	Link to your organisation Wikidata item
Educated at	P69	
LinkedIn ID	P6634	Personal profile slug
ORCID iD	P496	If applicable
Notable work	P800	Books, papers, named projects

A complete Wikidata item is the single highest-yield GEO move available to a brand, and the work is one-time.

C.3 — Wikipedia: the notability gate

Wikipedia has a notability requirement. Most brands do not meet it without effort. The work is to assemble enough independent, secondary, reliable-source coverage that an article meets WP:NCORP or WP:NBIO standards.

The audit question is: do we have at least three substantial, independent secondary sources (trade press, peer-reviewed work, books) covering us? If yes, an article is defensible. If no, the work is to earn that coverage first — Appendix C is not where you fix the upstream problem.

Once an article exists, the audit checklist:

1. Lead paragraph correct on facts and tone? (Not promotional.)
2. Infobox populated and consistent with Wikidata?
3. Citations are independent and reliable? (No press releases, no marketing pages.)
4. Image with appropriate licence?
5. External links section restrained? (One website link only, in most cases.)
6. Categories complete and correctly nested?
7. Translations exist on the language editions you care about?

Do not edit your own Wikipedia article directly; use the Talk page or escalate via the COI process. Direct edits get reverted and damage the long-term standing of the article.

C.4 — Crunchbase, G2, LinkedIn, GitHub

These are easier — each has a vendor portal, an admin who can edit, and an obvious update path. The audit question is consistency, not existence.

Crunchbase: company description, industries, headquarters, funding rounds, acquisition history, key people. Update when material changes happen.

G2: product description, category placement, integrations, pricing tier markers, vendor verification. Refresh quarterly.

LinkedIn: company page tagline matches positioning; About section is the canonical company description; specialities list matches the categories you want to be cited in; logo and banner current.

GitHub: organisation profile bio populated; pinned repositories are the ones you want cited; README files in those repos are not skeleton placeholders. The pinned README is what an LLM is most likely to lift.

C.5 — ORCID and financial registries

ORCID applies for named subject-matter experts who publish. Audit annually — fix only when something changes.

Bloomberg, Reuters, and FT entity records apply for listed entities and large privately-held entities tracked by financial press. The work here is via Investor Relations or the company secretary, not Marketing. Audit when something changes (corporate actions, name changes, ISIN changes).

C.6 — sameAs reconciliation

The point of populating eight platforms is so that the sameAs graph in your JSON-LD on your own site can link to all of them. Appendix I.2 has the schema. The audit check: does your site's Person/Organization JSON-LD `sameAs` array list every one of the eight platform URLs where you have a populated entity? If not, the engines do not get the joinable graph.

The deliverable from a quarterly Entity Bible audit: a one-page change log. What is consistent. What changed. What is missing. What is being filed this week.

Appendix D — The Vendor Evaluation Matrix

The GEO vendor market is fifteen months old. The pitch decks are convincing. The categories are blurred. This appendix is the rubric you take into the meeting.

Two principles to anchor on. First, only two of the four tool classes are work tools — the rest are hygiene or caution. Second, no single vendor solves the problem; the question is which class of tool you are buying and whether the specific vendor in that class meets the bar.

D.1 — The four classes, restated

Class	What it actually does	Verdict
1. Citation trackers	Run prompt batteries; store answer text, brand presence, rank, sources	WORK TOOL
2. Schema validators	Confirm JSON-LD is syntactically valid	HYGIENE
3. Entity audit tools	Flag fragmentation across third-party platforms	WORK TOOL
4. Composite scores	Roll up the first three with proprietary weights	CAUTION

Class 1 and Class 3 deliver outputs you can act on. Class 2 is necessary but not sufficient. Class 4 is a board-deck convenience that should never be the only number you report.

D.2 — The evaluation rubric

For any vendor in any class, score against these eight criteria. Score 0–3 on each. The exit criterion: $\geq 18/24$ to make a shortlist; $\geq 21/24$ to buy.

#	Criterion	What to ask	What good looks like
1	Methodology transparency	"Show me the exact prompts you run, the engines you query, the sample size, and the variance you report."	Documented prompts; named engines with version strings; published sample size; published 95% confidence band
2	Engine coverage		

#	Criterion	What to ask	What good looks like
		"Which engines do you query, how often, and how do you handle engine version changes?"	All six priority engines covered; at least weekly cadence; documented re-baseline policy
3	Repeatability	"If I run the same prompt twice, do I get the same answer? How do you handle stochastic engine outputs?"	Acknowledged stochasticity; reports show variance bands, not point estimates
4	Source verification	"When you report a citation, can I click through to the actual answer text and the cited URL?"	Click-through to raw answer for every reported citation; source URL captured live
5	Cell-level data export	"Can I export the cell-level dataset (300 cells/run × N runs) as CSV or via API?"	Yes, with all four fields — answer text, presence, rank, source URLs
6	Integration cost	"What does it take to integrate with our existing analytics stack?"	Documented API; webhook support; standard data warehouse connectors
7	Team independence	"Can our team run the tool without your professional services?"	Self-service onboarding; documented; not gated behind a CSM

#	Criterion	What to ask	What good looks like
8	Pricing transparency	"What is the all-in annual cost, with no add-ons, no minimums hidden in the master service agreement?"	Published pricing or one-page quote with no escalators below £100k ARR

Vendors who score below 12/24 on this rubric are not citation trackers. They are dashboards that look like citation trackers.

D.3 — Class-specific questions

For Class 1 (citation trackers), additional questions:

- "Can I supply my own 50-prompt battery, or am I locked to your prompt library?"
- "How do you handle a competitor adding a new product — does the prompt set update?"
- "What is the retention policy on raw answer text?"

For Class 3 (entity audit tools):

- "Which of the eight platforms do you cover? What is your update mechanism for each?"
- "Do you produce QuickStatements V1 output for Wikidata, or do I have to translate?"
- "Can you flag sameAs graph gaps in the JSON-LD on our pages?"

D.4 — Red flags

Any one of these on its own is a deal-breaker:

- "Proprietary AI visibility score" with no published methodology.

- Refusal to share a sample raw dataset before the contract.
- Single-engine focus with claims that generalise to “AI search” or “LLM visibility” overall.
- Claims to “rank” pages — engines do not rank pages, they cite or do not cite. The vendor does not understand the substrate.
- Pricing structure dependent on “queries” or “engines” without a published unit cost.
- Required minimum-spend or auto-renewal clauses in the standard MSA.

D.5 — Build vs buy

A 50-prompt × 6-engine battery can be built in-house in a fortnight by one engineer with API access. The choice between build and buy is whether you would rather own the dataset and the methodology or rent them. There is no wrong answer, but the questions that decide it are:

1. Do we have a script-capable engineer with two weeks?
2. Do we want the prompts, the dataset, and the audit log to be ours, or to live on a vendor server?
3. What is our annual budget exposure to vendor failure?

If the answer to (1) is yes and (3) matters, build. If not, buy — and pick from the top of the rubric, not the loudest pitch.

D.6 — Procurement template

The deliverable from a vendor evaluation is a one-page memo containing: the rubric scores, the red-flag check, the build-vs-buy answer, the recommended vendor (or “build”), the contract length and exit clauses, and the 30-day evaluation criteria. Sign-off is from the SEO/GEO lead and the Marketing operations lead. Quarterly re-evaluation against the rubric until the contract anniversary.

Appendix E — The Week-One Audit Workbook

Chapter 13 sketches the seven-day audit. This appendix is the workbook — the printable, fillable form your team takes into Week One. Copy it, paste it into a shared document, fill it as you go, and the Day 7 memo writes itself.

E.1 — Day 1: Server logs and bot inventory

Owner: IT / DevOps

Time budget: 4 hours

Inputs needed: 90 days of server logs (Cloudflare, Akamai, Fastly, or origin server)

Deliverable: Bot register v0

Tasks:

1. Pull access logs for the last 90 days, filtered to user-agent strings containing any of: GPTBot, ChatGPT-User, OAI-SearchBot, ClaudeBot, Claude-Web, Claude-User, Claude-SearchBot, Google-Extended, Google-CloudVertexBot, PerplexityBot, Perplexity-User, Bingbot, Applebot, CCBot, Diffbot, Amazonbot, Bytespider.
2. For each bot found, count: total requests, unique URLs hit, 200 ratio, 4xx ratio, 5xx ratio.
3. Categorise each bot as training / retrieval / user-fetched (Appendix A is the reference).
4. Identify any bot in the logs that is NOT in Appendix A. Add it to the register with a note for follow-up.

Fill in:

Bot	Category	90-day request count	200/4xx/5xx ratio	Notes

E.2 — Day 2: Crawler control posture

Owner: IT / Security

Time budget: 3 hours

Inputs needed: robots.txt, WAF configuration, Cloudflare AI Crawler settings, server-level rate-limit rules

Deliverable: Posture sheet

Tasks:

1. Download current robots.txt. List every User-agent block and every Disallow directive.
2. List every WAF rule that matches AI bot user-agents or rate-limits them.
3. In Cloudflare dashboard: Security → AI Crawlers — record current settings.
4. Cross-reference against the Day 1 bot list. Flag any bot with high 4xx/5xx that is being unintentionally blocked.

Fill in:

Bot	robots.txt directive	WAF rule	Cloudflare AI Crawler setting	Intentional?

E.3 — Day 3: Prompt battery snapshot

Owner: SEO/GEO

Time budget: 6 hours (design + run)

Inputs needed: API access to the six engines, or vendor evaluation tier already in place

Deliverable: Battery v0 — 300 cells, snapshot only

Tasks:

1. Draft 50 prompts following the band structure in Appendix B (10 unbranded category + 10 branded + 10 competitor + 10 problem statement + 10 long-tail buyer Qs).
2. Run all 50 across all 6 engines. Capture answer text, brand presence, citation rank, source URLs.

3. Aggregate to first numbers: overall citation rate, per-engine citation rate, per-band citation rate.
4. Save the raw cells. Do not start interpreting yet; that is the Day 7 work.

Fill in:

Metric	Value
Overall citation rate (% of cells with brand present)	
Citation rate — Google AIO	
Citation rate — ChatGPT	
Citation rate — Perplexity	
Citation rate — Copilot	
Citation rate — Claude	
Citation rate — Gemini	
First-citation rank (median across cells where present)	

E.4 — Day 4: JSON-LD audit on top 20 pages

Owner: SEO/GEO

Time budget: 4 hours

Inputs needed: List of top 20 highest-priority URLs, JSON-LD validator, Google Rich Results test

Deliverable: Schema gap list

Tasks:

1. For each of 20 priority URLs: extract JSON-LD, run through Schema.org validator and Google Rich Results.
2. Record: schema types present, sameAs presence, validator pass/fail, retrieval test pass/fail.

3. For pages where the schema is present but the LLM fails to extract the entity correctly, mark as “extraction fail.” This is more important than syntactic validity.

Fill in:

URL	Schema types	sameAs present?	Validator pass	Retrieval test

E.5 — Day 5: Eight-platform entity status

Owner: Brand / PR

Time budget: 5 hours

Inputs needed: Brand asset library, executive bio source-of-truth, IR contacts list

Deliverable: Entity Bible v0

Tasks:

For each of: Wikidata, Wikipedia, Crunchbase, G2, LinkedIn, GitHub, ORCID, Bloomberg / Reuters / FT — record current state, identify gaps, and assign an owner for fix.

Fill in:

Platform	Entity present?	Canonical name match?	Key fields populated?	Gap list
Wikidata				
Wikipedia				
Crunchbase				
G2				
LinkedIn				

Platform	Entity present?	Canonical name match?	Key fields populated?	Gap list
GitHub				
ORCID				
Bloomberg / Reuters / FT				

E.6 — Day 6: Top 10 co-citation source shortlist

Owner: Content + PR

Time budget: 4 hours

Inputs needed: Battery v0 source URLs (Day 3), competitor citation map

Deliverable: Co-cite shortlist

Tasks:

1. From Day 3 source URLs: list every distinct domain cited by the engines for prompts in your category.
2. Categorise each by platform type (Wikipedia, Reddit, trade press, analyst, journal, YouTube, GitHub, podcast).
3. Identify the 10 sources cited most often where you do NOT yet have presence.
4. For each: name a realistic 90-day path to placement (story, contributed article, analyst briefing, dataset publication).

Fill in:

Source domain	Platform type	Currently cite us?	90-day path

E.7 — Day 7: The audit memo

Owner: SEO/GEO lead

Time budget: 4 hours

Inputs: All six prior deliverables

Deliverable: A memo of three pages or fewer, structured as:

Three findings. What is true about our position that we did not previously know.

Three risks. What we cannot fix in the next 30 days that we should be aware of.

Recommended Day 8–30 sequence. What gets unblocked first, who owns it, what the deliverable is, and how we know it worked.

The memo is reviewed by the SEO/GEO lead, signed off by the head of marketing, and archived. It is the artefact that buys the next 83 days.

Appendix F — Glossary

A working glossary of terms used across this manual. Definitions are operational, not academic — they describe how the term shows up in the daily work, not how a textbook would frame it.

AI Mode. Google Search’s generative-answer experience, distinct from AI Overviews. Approximately 93% zero-click, per Google’s own platform documentation.

AIO (AI Overview). Google Search’s snippet-style generative answer, displayed above the blue links.

Answer engine. A system that returns a synthesised answer rather than a ranked list of links. The six in scope here: Google AIO, ChatGPT Search, Perplexity, Microsoft Copilot, Anthropic Claude, Gemini.

Backlink. A hyperlink from one site to another. The atomic unit of pre-LLM SEO authority. In GEO it matters less than co-citation density on platforms LLMs read.

Bot register. The internal artefact that lists every crawler your site sees, what category it belongs to, and what your robots.txt and WAF posture is for it. See Appendix A.

Chunking. The process of dividing a document into smaller passages before embedding for retrieval. Chunk size determines what an LLM can lift cleanly.

Citation. When an LLM names your brand, your URL, or both in a generated answer. The headline metric of GEO.

Citation rank. Where in the cited-source list your URL appears (1, 2, 3 ... N). First-citation rank correlates with downstream click-through where click-through still exists.

Citation rate. The percentage of prompt cells in which your brand appears in the answer. Computed across the prompt battery.

Co-citation. Two entities being mentioned together by independent third-party sources. The substrate for authority that LLMs read.

Compulsory licensing. A regulatory scenario in which engines must pay a statutory licence to train on web content. One of three modal scenarios.

Cosine similarity. The geometric measure used to find the nearest semantic neighbours in vector space. Closer angle = closer meaning.

Crawler. A bot that retrieves web pages on behalf of a search or LLM operator. Three categories matter: training, retrieval, user-fetched.

Dual-layer citation model. The framework introduced in Chapter 5: visible content carries the bulk of citation weight (~3.5x); structured data (JSON-LD) carries the rest. Both required, neither sufficient alone.

Embedding. A vector representation of a piece of text. The numeric form an LLM uses to reason about meaning.

Entity Bible. The eight-platform, canonical-truth artefact that ensures consistent representation of your brand across Wikidata, Wikipedia, Crunchbase, G2, LinkedIn, GitHub, ORCID, and the financial registries. See Appendix C.

Extractability. The property of a page that determines whether an LLM can lift a clean answer fragment from it. Driven by sub-heads, statistics in opening sentences, and named sources in prose.

FAQPage schema. The Schema.org type that marks up question-answer pairs. Commonly used in GEO content because LLMs lift the answers cleanly.

GEO. Generative engine optimisation. The discipline this manual is about.

Hallucination. When an LLM produces a fluent answer that is factually wrong. Reduced — never eliminated — by retrieval grounding.

JSON-LD. The serialisation format for Schema.org structured data on the web. The machine-readable layer of the dual-layer model.

Knowledge graph. A structured database of entities and their relationships. Wikidata is the canonical public knowledge graph; Google's Knowledge Graph and equivalents at other engines are private.

llms.txt. A proposed standard for a machine-readable file at the domain root that signals to LLMs how the site wants to be indexed. Adopted unevenly; see Appendix I.10 for the recipe.

Merchant feed. A structured product data feed (Google Merchant Center, Bing Merchant, equivalent) that supplies product attributes for retrieval. The underfunded line of the 50/30/20 budget split.

Multimodal. Combining text, image, audio, and video. YouTube transcripts are the most cited multimodal source class in LLM answers at time of writing.

Open-domain QA. Retrieval-grounded question answering across the open web. The substrate of all six answer engines.

Post-rationalisation. The phenomenon, observed in approximately 57% of LLM answers, where the model produces a citation list that does not align with the actual sources used during generation. The figure is from a non-peer-reviewed preprint and should be treated as directional.

Prompt battery. The 50-prompt × 6-engine measurement instrument described in Chapter 12 and Appendix B. The only repeatable measurement structure in current GEO practice.

RAG (retrieval-augmented generation). The architecture in which an LLM retrieves documents from an index before generating an answer. Most production answer engines are RAG-based.

Retrieval. The first half of the RAG pipeline — finding the documents that feed the generation step.

Robots.txt. The plain-text file at the domain root that signals crawler-permission preferences. Honoured by responsible bots; ignored by bad actors.

sameAs. A JSON-LD property that links one entity representation to other authoritative records of the same entity. The mechanism by which engines reconcile your brand across platforms.

Schema.org. The public vocabulary for structured data on the web, jointly maintained by Google, Microsoft, Yahoo, and Yandex.

Semantic neighbourhood. The region of vector space occupied by terms with related meaning. The page does not need the query keyword; it needs to inhabit the same neighbourhood.

Speakable schema. Schema.org markup that flags content suitable for voice-assistant playback. Underused; sometimes lifted by Alexa and Google Assistant retrieval.

Status quo + bilateral licensing. The modal regulatory scenario at time of writing — engines pay a small number of headline publishers, the long tail remains uncompensated. Probability band 55–65%.

Strict opt-in. A regulatory scenario in which engines may not train on content without explicit publisher permission. Probability band 10–18%.

Structured data. Machine-readable metadata embedded in or alongside web content. JSON-LD is the dominant serialisation.

Tokenisation. The process by which text is broken into tokens (sub-word units) before being processed by an LLM. The Williams-Cook experiment used token-level analysis to estimate the ~3.5x weight of visible text over schema.

User-fetched. A category of crawler that retrieves a specific URL because a human has explicitly asked the LLM to read it.

Vector. A list of numbers representing meaning. The substrate of modern retrieval.

Visibility score. A composite metric produced by some GEO vendors. Class 4 in the vendor matrix; treat as directional, not as a primary metric.

WAF. Web application firewall. The layer where most unintentional crawler-blocking happens.

Williams-Cook tokenisation experiment. The ~3.5:1 dual-layer ratio finding referenced throughout the manual. Method: token-level retrieval analysis.

Zero-click. A search session that ends without the user clicking through to a destination URL. The dominant mode on AI Mode (~93%).

Appendix G — Source Canon and Further Reading

This appendix lists the sources the manuscript draws on, sorted by tier of evidence. Tier 1 sources are the highest-stakes empirical claims — those whose alteration would change a senior practitioner’s strategy. Tier 2 are supporting empirical work. Tier 3 are conceptual and frame-setting reads that informed the synthesis without being the empirical anchor for any single claim.

Notes on use: every claim in the manuscript referenced as “Source: ...” maps to one or more of these. Where a claim is tagged Strong, it is anchored to a Tier 1 source with replication or platform self-disclosure. Where it is tagged Moderate, it is single-source vendor-funded or single-source academic. Where it is tagged Directional, it is preprint or industry observation that has not yet been replicated.

G.1 — Tier 1: Highest-stakes empirical claims

Aggarwal, P., Murahari, V., Rajpurohit, T., Kalyan, A., Narasimhan, K., Deshpande, A. (2023). GEO: Generative Engine Optimization. arXiv:2311.09735.

The replication base for the three tactic effect sizes (Statistics Addition, Citing Sources, Quotation Addition). Statistics +14.5% across all six engines (CI +8 to +22). Citing Sources +6 to +12 across all six. Quotation +24% on Perplexity only.

Google. (2025). AI Overviews and AI Mode platform documentation. The 93% zero-click figure on AI Mode. Single-source platform self-disclosure; treat as authoritative on the platform but not generalisable across engines.

Otterly. (January 2026). AI Crawler Audit Report. The 73% unintentional-block figure. Vendor-funded (Otterly sells crawler-management tools — flagged conflict of interest); replication pending. Treat as directional within the engine population audited.

Williams, A.; Cook, M. (2025). Tokenisation experiment on dual-layer citation weighting. Token-level retrieval analysis underpinning the ~3.5:1 visible-content-over-schema ratio. Single experimental method; replication and engine-by-engine variance pending.

Cloudflare. (2025). AI Crawler Default-Block Announcement, July 2025. Platform documentation establishing the policy change for new zones from 1 July 2025. Authoritative for Cloudflare; check vendor status if your CDN is elsewhere.

Semrush. (2025). AI Mode Click-Through Study. Corroboration of the zero-click figure on AI Mode. Vendor-funded; cited in support, not as primary anchor.

arXiv:2412.18004. (2024). Post-rationalisation in LLM-generated citations. The 57% post-rationalisation figure. Preprint, not peer-reviewed at time of writing. Treat as directional.

G.2 — Tier 2: Supporting empirical work

Miech, A., Zhukov, D., Alayrac, J.-B., Tapaswi, M., Laptev, I., Sivic, J. (2019). HowTo100M: Learning a Text-Video Embedding by Watching Hundred Million Narrated Video Clips. ICCV. The dataset legacy underlying the YouTube structural-preference inheritance discussed in Chapter 8.

Otterly. (2026). YouTube Citation Study. The ~16% YouTube share of LLM citations. Vendor-funded; corroborated directionally by Profound platform-share data.

Profound. (2026). LLM Citation Platform-Share Data. Vendor data on citation-source distribution across answer engines. Used to triangulate the platform-weight rankings in Chapter 11 and Appendix C.

Ahrefs. (2026). Transcript-Structure Correlation Data. Proprietary correlation data linking transcript discipline to citation rate. Vendor-funded; specific magnitudes not independently verified.

Anthropic, OpenAI, Google. (2024–2025). Engine documentation, Anthropic responsible-disclosure briefings, OpenAI usage policies, Google Search Central guidance. Source for crawler categorisation, retrieval semantics, and platform behaviour.

Hugging Face. (2025). Dataset publishing standards and dataset cards specification. Underpinning the Appendix I.9 cookbook recipe.

Schema.org Working Group. (continuous). Schema.org vocabulary specification. The canonical reference for JSON-LD types used throughout Appendix I.

G.3 — Tier 3: Conceptual and frame-setting reads

Federico, F. (2025). The Agentic CMO (1st edition). The companion volume to this manual. The AI-marketing thesis that frames why GEO is a CMO-level concern, not a technical-SEO concern.

Federico, F. (continuous, since 2023). Chronicles of Change. Newsletter. Working notes on how the agentic shift is reshaping marketing organisations. Read for the broader frame.

Karpathy, A. (various, 2024–2025). Public posts on LLM behaviour and retrieval. Conceptual grounding for how LLMs reason about web content.

Reid, L. (Google Search). (2024). AI Overviews public posts and Search Central guidance. The most direct public articulation of how Google thinks about answer-layer behaviour.

Sundar Pichai, S. Nadella, S. Altman, D. Amodei, A. Srinivas. (various, 2024–2026). Public statements on the answer-engine shift. Read in full for the strategic context, not for empirical anchors.

Kemp, S. (Datareportal, We Are Social). Digital reports. Quantitative grounding for category penetration and behaviour change.

Anderson, C. (2006). The Long Tail. Read for the inversion: the long tail's economics changes when distribution is mediated by an answer engine rather than a marketplace.

Thompson, B. (continuous). Stratechery. Strategic analysis of the major platforms; useful for context on engine-by-engine economic incentives.

Benedict Evans. (continuous). Newsletter and presentations. Frame-setting on consumer-internet structural change.

G.4 — Working sources for the practitioner

The sources below are tracking sources — the places to keep one eye on for change. Bookmark them; revisit monthly.

Source	What to track
Google Search Central blog	AI Overviews behaviour changes, schema guidance updates
OpenAI Platform changelog	API behaviour, search semantics, model retrieval changes
Anthropic news and engineering blog	Claude search and retrieval evolution
Perplexity blog	Engine cadence and feature changes
Cloudflare blog	

Source	What to track
	AI crawler posture changes, default-block evolution
Schema.org news	Vocabulary updates and new types
Wikidata data dumps and WikiProject reports	Entity-graph coverage shifts
arXiv cs.IR (information retrieval)	Academic preprints on retrieval and generation
Hugging Face dataset publication trends	Where industry datasets are landing
LLM-tracker projects (community-run)	Real-world citation share over time

A v2 of this manual will fold in twelve months of additional measurement. The discipline in the meantime is to capture your own observations against these sources, in your own change log, dated.

G.5 — A note on completeness

This canon is not exhaustive. Approximately 156 author-year citations and 180 distinct URLs were consulted in the underlying research; the list above selects the load-bearing ones. The full canon — annotated with replication status, evidence weight, conflict of interest, and chapter mapping — is maintained as an internal artefact and will be made available alongside v2 of this manual on request.

Appendix H — Tracking Change After v1

A free practitioner manual on a fifteen-month-old discipline goes stale the day it ships. This appendix tells you where to look, what to look for, and when to update your team's playbook.

H.1 — What will change first

Three categories of change move at different cadences. Treat them differently in your reading list.

Engine behaviour changes. Weeks-to-months cadence. New Perplexity features, ChatGPT Search version bumps, AI Mode UI changes, Gemini integration shifts. These move what you measure, not always what you do. The prompt battery is the instrument that surfaces them.

Vendor and tooling shifts. Months cadence. New citation-tracker entrants, M&A in the GEO vendor space, schema validators adding LLM-extraction tests, entity-audit tools expanding platform coverage. Re-run the vendor matrix (Appendix D) when the named vendors change materially.

Regulatory and structural shifts. Quarters-to-years cadence. The three pending decisions — US fair-use case outcomes, EU Code of Practice finalisation, UK consultation — will move the modal regulatory scenario from 55–65% (status quo + bilateral licensing) to one of the alternatives. When that moves, the operating model adjusts.

H.2 — What to subscribe to

A short, durable list. Anything not on this list is read on demand, not by subscription.

Source	Cadence	Why
Google Search Central blog	When posted	First-party signal on AI Overviews and AI Mode
OpenAI Platform changelog	Weekly check	Search and retrieval changes hit here first
Anthropic news and engineering	When posted	Claude search and retrieval evolution
Cloudflare blog (security and AI tags)	Weekly check	

Source	Cadence	Why
		AI crawler policy is set here for a large fraction of the web
Schema.org news	Monthly check	Vocabulary changes
arXiv cs.IR	Weekly skim	Academic upstream — preprints become production six-to-twelve months later
Chronicles of Change (this author's newsletter)	Weekly	Working notes on the broader agentic shift; companion to The Agentic CMO

That is the durable subscription list. Everything else — vendor pitches, conference summaries, podcast episodes — is intake on demand.

H.3 — Quarterly playbook review

Once a quarter, the GEO lead runs the following review against the manual. The deliverable is a one-page memo flagging what has changed and what the team is changing in response.

Crawler register (Appendix A). New bots in the logs that are not in the register? New retirement announcements from the engines? Cloudflare AI Crawler default settings updated?

Prompt battery (Appendix B). Are any of the 50 prompts no longer producing useful variance? Are any new buyer questions emerging that should be tested? Are the engines we measure still the engines that matter?

Entity Bible (Appendix C). Has any platform changed its API or content rules? Have any of our identifiers gone stale? Is the sameAs graph still complete?

Vendor matrix (Appendix D). Has the vendor we use changed material methodology? Has a new entrant appeared? Are we still scoring above the rubric threshold?

Regulatory cascade (Chapter 15). Has any of the three pending decisions resolved? If yes, which scenario has the modal probability moved toward, and what does that change in our operating posture?

H.4 — Annual re-baseline

Once a year, run the full Week One audit (Appendix E) again, on the live state of the site. Compare the year-on-year deltas. The questions:

- Did our crawler posture change? Why?
- Did our citation rate move? Per band, per engine?
- Did our Entity Bible drift? On which platforms?
- Did our co-citation footprint expand or contract?
- Did our vendor mix change?

The annual re-baseline is the artefact for the head of marketing's year-in-review presentation. It is also the input to the v2 budget.

H.5 — When to update this manual

This is v1. The author's intent is a v2 in the third quarter following publication, incorporating: at least one full quarter of independent prompt-battery data; updated Aggarwal replication numbers; the resolution status of at least one of the three pending regulatory decisions; refreshed crawler register; updated vendor matrix.

If you have observations, corrections, or replication data you would share, the manual benefits from them. The author's contact details are in About the Author in the back matter.

The discipline of keeping a manual like this current is no different from the discipline this manual asks you to bring to your own organisation. Measure

repeatably. Log what changed. Update with evidence. Resist the impulse to chase every announcement.

The work continues.

Appendix I — Schema and Datasets Cookbook

Ten copy-paste recipes. Each is a working starting point, not a final implementation — replace the placeholder values with your own and validate via Schema.org and Google Rich Results before deploying.

The order of the recipes corresponds to the order the chapters reach for them. Article schema first because it covers the most ground; Person and Organization second because they anchor the Entity Bible to the live site; FAQPage and HowTo because they map directly to the Aggarwal extractability tactics; Speakable, Video, and the dataset-publishing recipes for the multimodal and authority work; and finally `Ilms.txt` because it is the smallest and most contested artefact.

I.1 — Article schema (with sameAs and citation block)

The base recipe for any editorial page (blog post, newsletter issue, longform article).

```
<script type="application/ld+json">
  {
    "@context": "https://schema.org",
    "@type": "Article",
    "headline": "The exact title of the article",
    "alternativeHeadline": "Optional shorter form for social cards",
    "datePublished": "2026-04-30T09:00:00+01:00",
    "dateModified": "2026-04-30T09:00:00+01:00",
```

```
"inLanguage": "en-GB",
"url": "https://www.example.com/articles/exact-slug-here",
"mainEntityOfPage": {
  "@type": "WebPage",
  "@id": "https://www.example.com/articles/exact-slug-here"
},
"author": {
  "@type": "Person",
  "name": "Full Name",
  "url": "https://www.example.com/about",
  "sameAs": [
    "https://en.wikipedia.org/wiki/Full_Name",
    "https://www.wikidata.org/wiki/QXXXXXXX",
    "https://www.linkedin.com/in/handle/",
    "https://orcid.org/0000-0000-0000-0000"
  ]
},
"publisher": {
  "@type": "Organization",
  "name": "Organisation Name",
  "url": "https://www.example.com",
  "logo": {
    "@type": "ImageObject",
    "url": "https://www.example.com/logo.png"
  }
},
"image": [
  "https://www.example.com/images/hero.jpg"
],
"description": "One-sentence summary, 150–160 characters, that is also a short description of the page content.",
"citation": [
  {
    "@type": "CreativeWork",
    "name": "Title of the cited source",
    "author": "Author Name",
    "datePublished": "2025"
  }
]
}
</script>
```

Place inside `<head>` or immediately before `</body>` . Validate at validator.schema.org and Google Rich Results.

I.2 — Person schema with full sameAs graph

The single highest-yield piece of structured data for an executive or named subject-matter expert. Place it on the relevant author or about page.

```
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "Person",
  "name": "Francesco Federico",
  "jobTitle": "Chief Marketing Officer",
  "worksFor": {
    "@type": "Organization",
    "name": "S&P Global",
    "url": "https://www.spglobal.com"
  },
  "url": "https://www.francescofederico.com",
  "image": "https://www.francescofederico.com/headshot.jpg",
  "sameAs": [
    "https://en.wikipedia.org/wiki/Francesco_Federico",
    "https://www.wikidata.org/wiki/Qxxxxxxx",
    "https://www.linkedin.com/in/francescofederico/",
    "https://www.crunchbase.com/person/francesco-federico",
    "https://github.com/frandrake",
    "https://orcid.org/0000-0000-0000-0000",
    "https://twitter.com/handle"
  ],
  "knowsAbout": [
    "Marketing",
    "Generative engine optimisation",
    "Brand strategy",
    "Agentic marketing"
  ],
  "alumniOf": [
    {"@type": "EducationalOrganization", "name": "Università Cattolica
```

```
        {"@type": "EducationalOrganization", "name": "INSEAD"}
    ]
}
</script>
```

The `sameAs` array is what reconciles the entity across the eight platforms in Appendix C. Populate every platform that has a record for the entity. Empty `sameAs` arrays do not silently fail — they leave the engines without a join key.

I.3 — Organization schema

Place at the root of the site, on the homepage, and (in some patterns) referenced by every page's `WebPage` schema.

```
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "Organization",
  "name": "Organisation Name",
  "legalName": "Organisation Name plc",
  "url": "https://www.example.com",
  "logo": "https://www.example.com/logo.png",
  "foundingDate": "2010-01-01",
  "founders": [
    {"@type": "Person", "name": "Founder Name"}
  ],
  "address": {
    "@type": "PostalAddress",
    "streetAddress": "1 Example Street",
    "addressLocality": "London",
    "postalCode": "EC1A 1BB",
    "addressCountry": "GB"
  },
  "contactPoint": {
    "@type": "ContactPoint",
    "contactType": "customer service",
    "email": "hello@example.com"
  },
  "sameAs": [
```

```

        "https://en.wikipedia.org/wiki/Organisation_Name",
        "https://www.wikidata.org/wiki/Qxxxxxxx",
        "https://www.crunchbase.com/organization/organisation-name",
        "https://www.g2.com/products/organisation-name",
        "https://www.linkedin.com/company/organisation-name/",
        "https://github.com/organisation-name",
        "https://www.bloomberg.com/profile/company/XXXX:LN"
    ]
}
</script>

```

I.4 — FAQPage schema

The most direct mapping from manual to extractable content. Every FAQ page on the site should carry FAQPage; consider it for product pages with substantive Q&A blocks.

```

<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "FAQPage",
  "mainEntity": [
    {
      "@type": "Question",
      "name": "What is generative engine optimisation?",
      "acceptedAnswer": {
        "@type": "Answer",
        "text": "Generative engine optimisation, or GEO, is the practice of using AI to generate content that is optimized for search engines."
      }
    },
    {
      "@type": "Question",
      "name": "How is GEO different from SEO?",
      "acceptedAnswer": {
        "@type": "Answer",
        "text": "SEO optimises for the ranking of a page in a list. GEO optimises for the content of a page to be relevant to the user's intent."
      }
    }
  ]
}

```

```
}  
</script>
```

The opening sentence of each Answer should contain the lift-worthy fragment. Aggarwal's evidence is that statistics-bearing first sentences and named-source first sentences are extracted more reliably than discursive ones.

I.5 — Speakable schema

Underused. Mark the content suitable for voice-assistant playback (Alexa, Google Assistant). Pairs with concise, declarative content.

```
<script type="application/ld+json">  
  {  
    "@context": "https://schema.org",  
    "@type": "NewsArticle",  
    "headline": "The headline",  
    "datePublished": "2026-04-30",  
    "speakable": {  
      "@type": "SpeakableSpecification",  
      "cssSelector": [  
        ".article-summary",  
        ".article-body p:nth-of-type(1)",  
        ".article-body p:nth-of-type(2)"  
      ]  
    }  
  }  
</script>
```

The `cssSelector` targets DOM elements that should be read aloud. Keep them short, complete sentences.

I.6 — HowTo schema

For procedural content. Manuals, tutorials, step-by-step explanations.

```
<script type="application/ld+json">  
  {
```

```

"@context": "https://schema.org",
"@type": "HowTo",
"name": "How to run the prompt battery",
"description": "Procedure for running the 50-prompt × 6-engine measure",
"totalTime": "PT2H",
"step": [
  {
    "@type": "HowToStep",
    "position": 1,
    "name": "Draft the 50 prompts",
    "text": "Following Appendix B, draft 10 prompts in each of the fi
  },
  {
    "@type": "HowToStep",
    "position": 2,
    "name": "Run across the six engines",
    "text": "For each prompt, query Google AI0, ChatGPT, Perplexity,
  }
]
}
</script>

```

I.7 — Video schema with transcript

The Chapter 8 lever. Every video the brand publishes should ship with this and a clean transcript at the URL given.

```

<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "VideoObject",
  "name": "Video title",
  "description": "Two- or three-sentence description suitable for the a
  "thumbnailUrl": "https://www.example.com/video/thumbnail.jpg",
  "uploadDate": "2026-04-30T09:00:00+01:00",
  "duration": "PT12M30S",
  "contentUrl": "https://www.example.com/video/file.mp4",
  "embedUrl": "https://www.youtube.com/embed/VIDEO_ID",
  "transcript": "Full transcript text here, or URL to a transcript page
  "hasPart": [

```

```

        {"@type": "Clip", "name": "Chapter 1 – Setup", "startOffset": 0, "endOffset": 10},
        {"@type": "Clip", "name": "Chapter 2 – The lever", "startOffset": 10, "endOffset": 20}
    ]
}
</script>

```

The Clip array, with named chapters and timecodes, dramatically improves extraction. It is also the data Google uses for the Key Moments feature.

I.8 — Video FAQ template

A specific structural pattern for video pages that combines VideoObject and FAQPage. The FAQs are pulled from the most-asked questions about the topic the video covers.

```

<div class="video-faq">
  <article class="video-meta">
    <!-- VideoObject schema (I.7) -->
    <h1>Video title</h1>
    <p class="summary">One-sentence summary, lift-worthy.</p>
  </article>

  <section class="faqs">
    <!-- FAQPage schema (I.4) -->
    <h2>Frequently asked</h2>
    <details>
      <summary>What is the lever?</summary>
      <p>The lever is the transcript discipline. Engines read the transcript.</p>
    </details>
    <details>
      <summary>How long does it take?</summary>
      <p>Twenty minutes per video, on top of normal post-production.</p>
    </details>
  </section>
</div>

```

Both schemas live in `<head>`. The HTML is the visible-content layer; the schemas are the machine-readable layer; together they are the dual-layer model in concrete form.

I.9 — Hugging Face dataset publishing

When you have a substantive dataset (a benchmark, a longitudinal study, an open data release), publishing it on Hugging Face produces a high-authority co-citation surface. Steps:

1. Prepare data as parquet, CSV, or JSON (sharded if larger than 1GB).
2. Write a `README.md` (the dataset card) with: dataset summary, supported tasks, languages, licence, citation, contact. Hugging Face's specification is the contract.
3. Create the dataset repository on Hugging Face under your organisation handle.
4. Push via `huggingface-cli` or the Python `datasets` library.
5. Cross-reference from your Wikidata item via P10444 (data set distribution) where applicable.
6. Link from your site's Article schema `citation` block.

A published dataset is co-cited in academic preprints faster than press coverage and lasts longer than vendor reports. The work is one engineer for two days.

I.10 — `llms.txt` and `llms-full.txt`

A proposed standard at `/llms.txt` and `/llms-full.txt` at domain root. Adoption is uneven across engines. Treat as a low-risk hedge — the file is small, costs nothing to maintain, and signals intent.

`llms.txt` — short, signpost-style:

```
# Example Organisation
```

```
> One-sentence description of who we are and what we do, in extractable
```

```
## Documentation
```

- [Getting started](https://www.example.com/docs/start.md)
- [API reference](https://www.example.com/docs/api.md)

Articles

- [The thesis](https://www.example.com/articles/thesis.md)
- [Method](https://www.example.com/articles/method.md)

Optional

- [Press kit](https://www.example.com/press.md)
- [Brand guidelines](https://www.example.com/brand.md)

`llms-full.txt` is the same file but with full content concatenated, so an LLM has the entire reference in one fetch. Practical at small scale; impractical past 50,000 tokens.

The format is plain markdown. Keep both files updated whenever the named pages change. Both are honoured (where they are honoured) for retrieval-time grounding, not for training.

These ten recipes cover roughly 90% of the structured-data work the manual asks for. The remaining 10% — niche schema types like Course, Recipe, Event, Product variants, MedicalEntity — follow the same shape: pick the most specific Schema.org type, populate every required and recommended field, link via sameAs, validate, deploy, and revisit at the quarterly audit.

The work is unglamorous. It is also the work that determines whether you exist in the answer or not.

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